



清华大学 集成电路学院
School of Integrated Circuits, Tsinghua University

集成电路先进封装技术

蔡坚

2025-11-05

提纲



■ 问题

- 什么是集成电路封装?
- 什么是先进封装技术?

■ 业界关注的封装技术

- 倒装芯片技术 (Flip Chip)
- 硅通孔互连 (TSV) 三维封装
- 扇出型圆片级封装 (Fan-out WLP)
- 玻璃封装基板 (Glass Substrate)

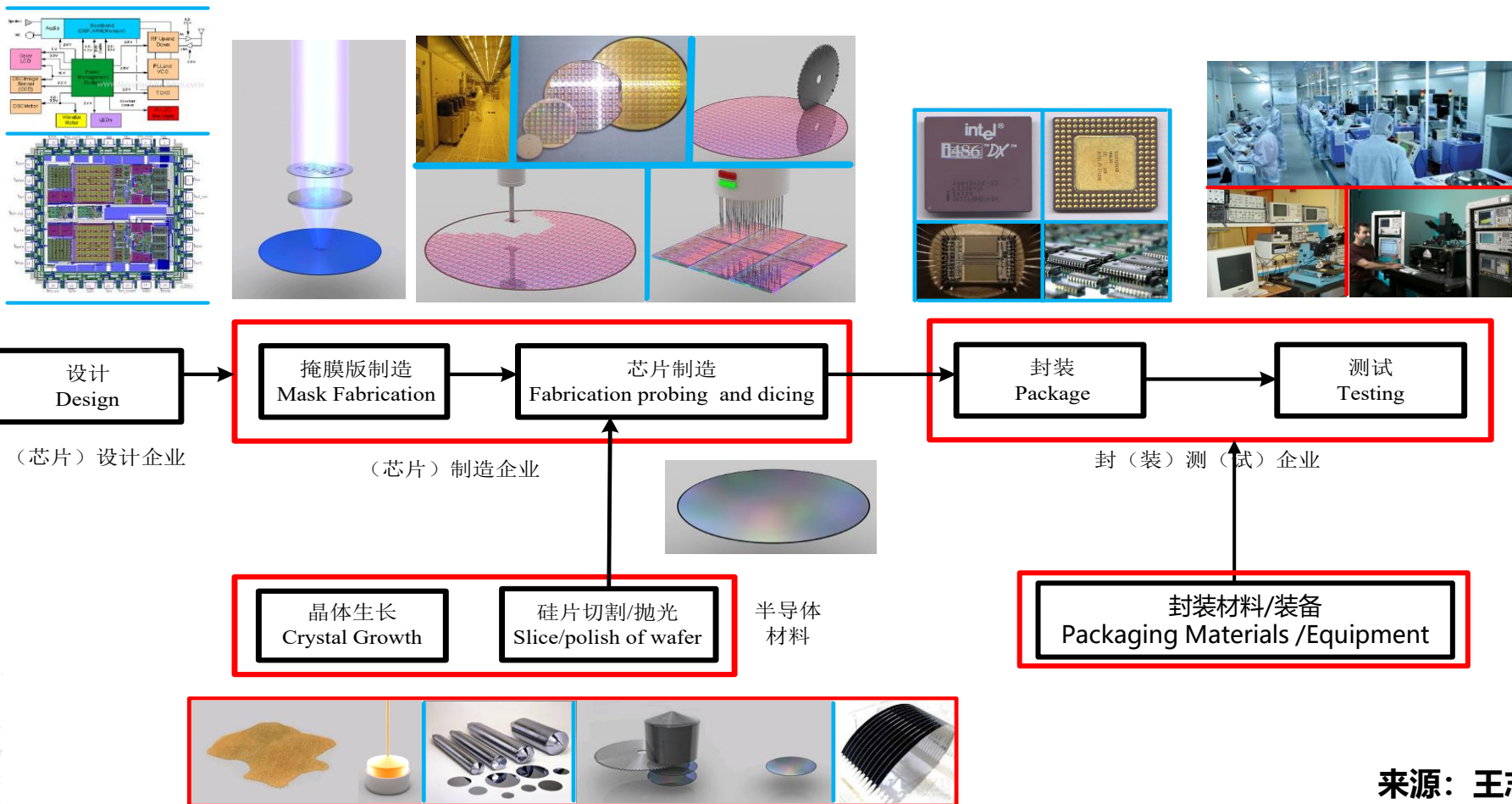
■ 先进封装案例

■ 先进封装技术发展的讨论



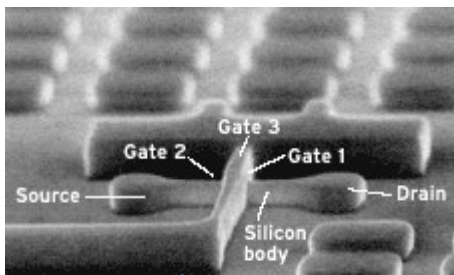
产业链中的封装

■ 不可或缺的一环

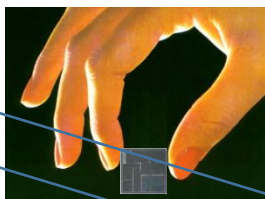
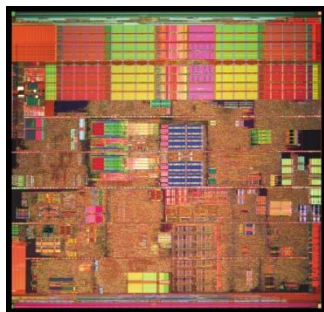


来源：王志华教授

封装是跨尺度的工程



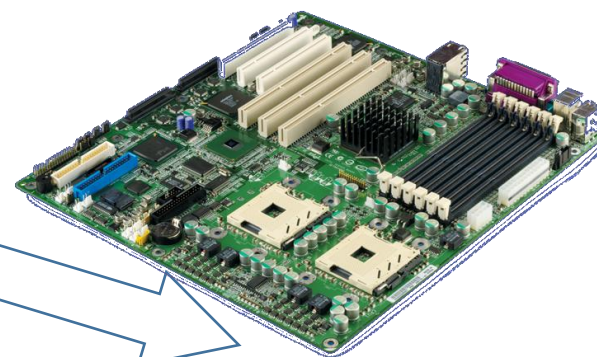
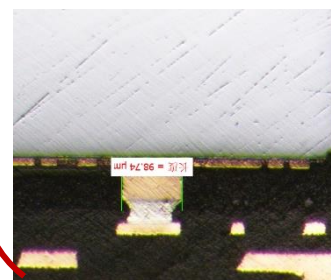
晶体管: 45nm工艺



集成电路: ~9mm*9mm
2.28亿个晶体管



封装: ~37.5mm*37.5mm



主板

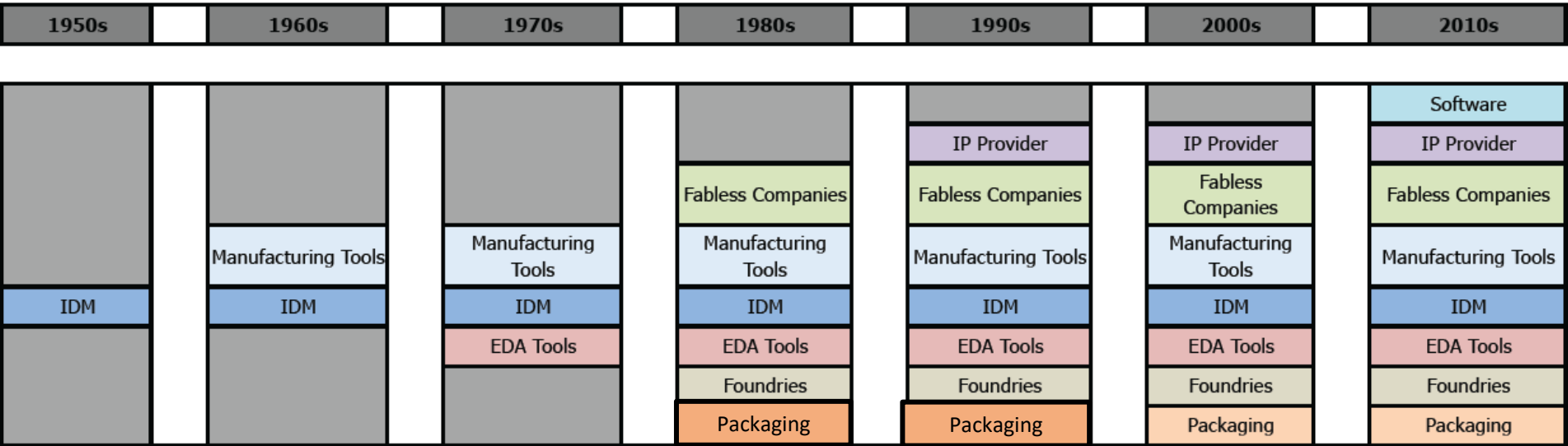
封测产业的基本情况

■ 两种商业模式并存

- IDM的封测厂
- OSAT（封测代工厂）



Functional Evolution of the Semiconductor Ecosystem (1950s–2010s)



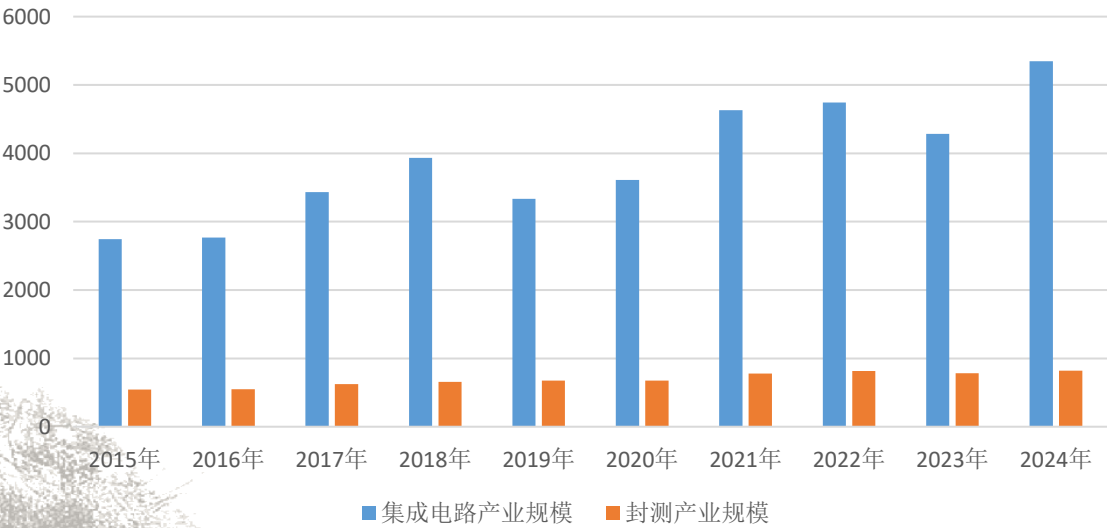


封测产业的基本情况

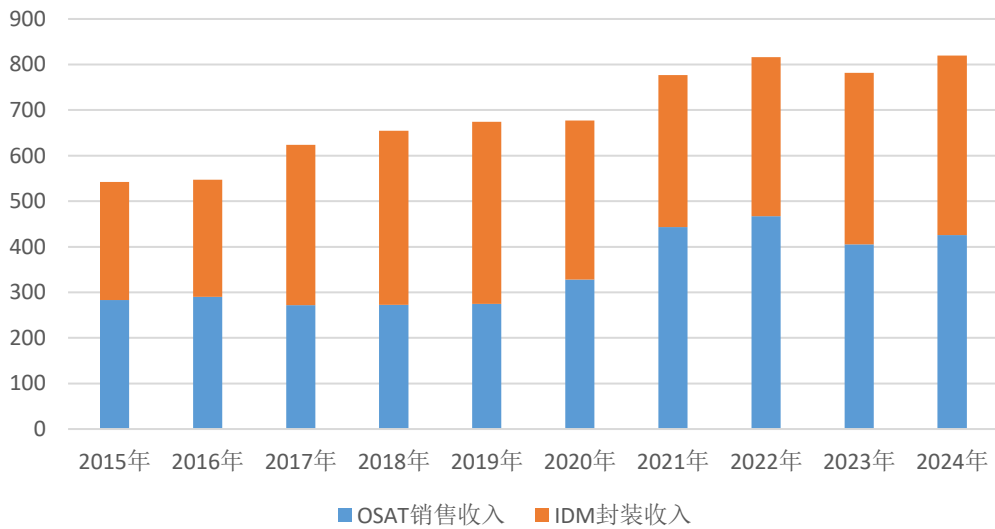
■ 产业规模

- 2024年：封测收入约为全产业收入的15.3%；OSAT收入约占封测收入52%

单位：亿美元



单位：亿美元



来源：集成电路产业发展研究报告



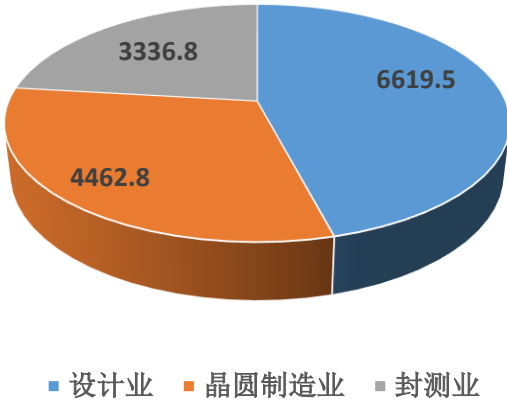
封测产业的基本情况

■ 前10大封测厂商营收占比约80%

● 中国大陆的封测产业占有重要规模

	公司名称	区域	2024年营收 (亿人民币)	全球占比 (%)
1	日月光 (ASE)	中国台湾	765	25.2
2	安靠科技 (Amkor)	美国	470	15.5
3	长电科技 (JCET)	中国大陆	346	11.4
4	通富微电 (TFME)	中国大陆	242	8.0
5	力成科技 (PTI)	中国台湾	172	5.7
6	华天科技 (HT)	中国大陆	143	4.7
7	智路封测 (Wise Road)	中国大陆	112	3.7
8	京元电子 (KYEC)	中国台湾	68	2.2
9	韩亚微 (HANA)	韩国	59	1.9
10	南茂科技 (ChipMOS)	中国台湾	53	1.7

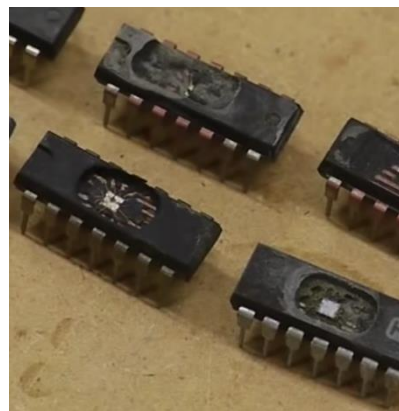
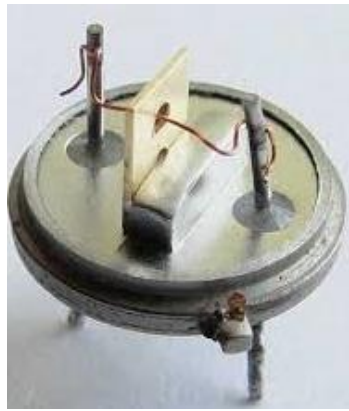
中国大陆设计、晶圆制造和封测业规模
(单位: 亿人民币)



什么是封装？

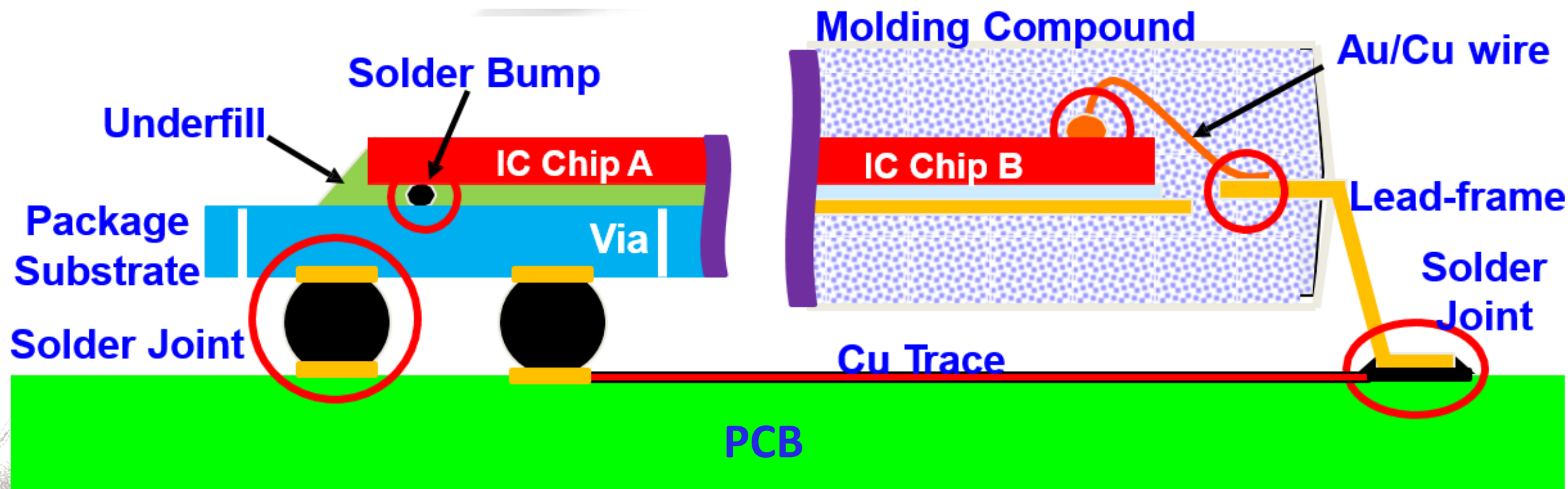
■ 将芯片（集成电路）**包封**在某一种**标准组件**中的方法，从而使之可以用于终端产品

- 基本功能：为芯片提供**电互连**、**机械支撑**、机械和**环境保护**及**导热**通道
- 扩尺度工程：实现空间尺度的变化
- 拓展摩尔：完成系统集成的功能



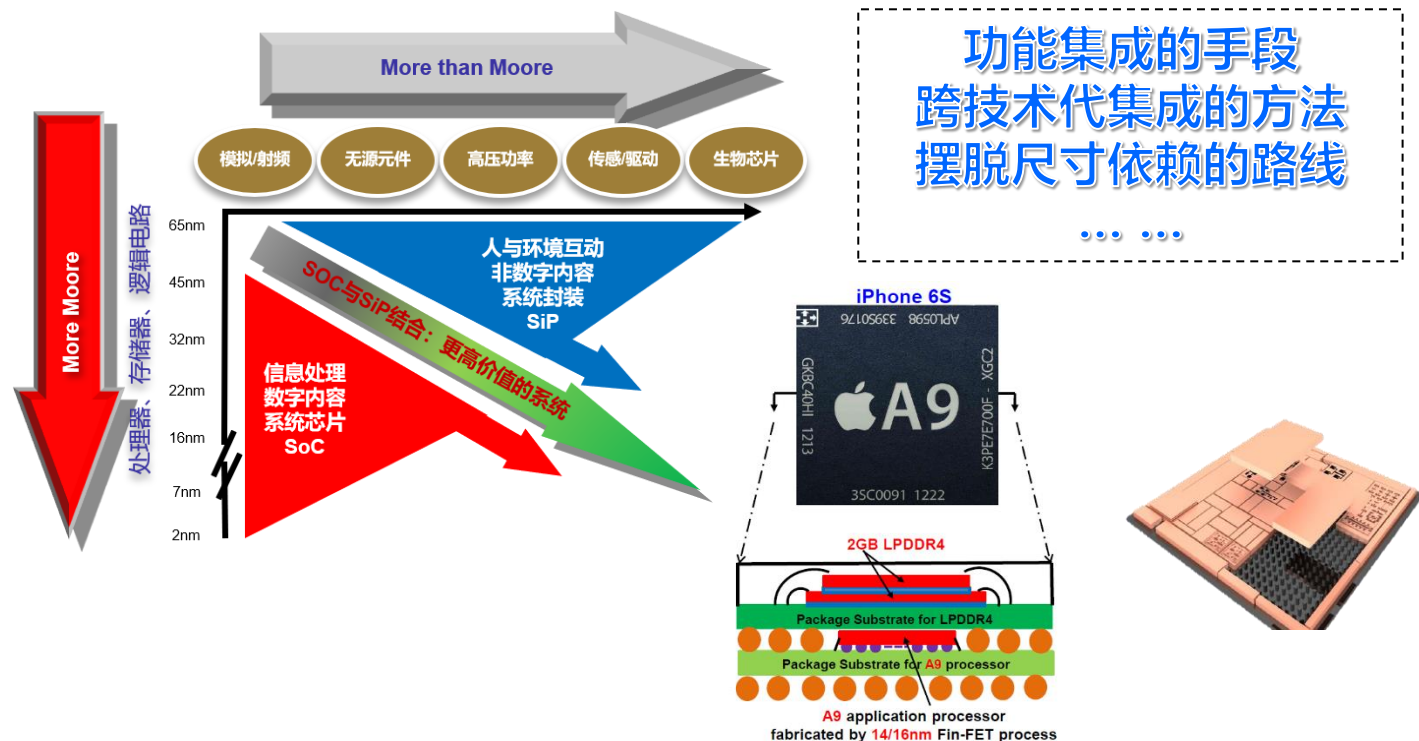
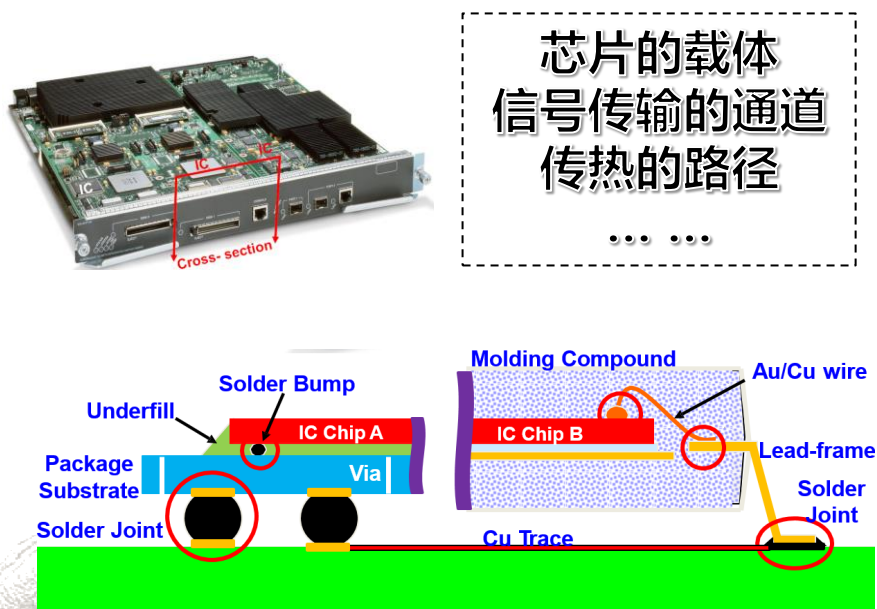
封装是什么？

■ 封装是构建系统产品的关键环节



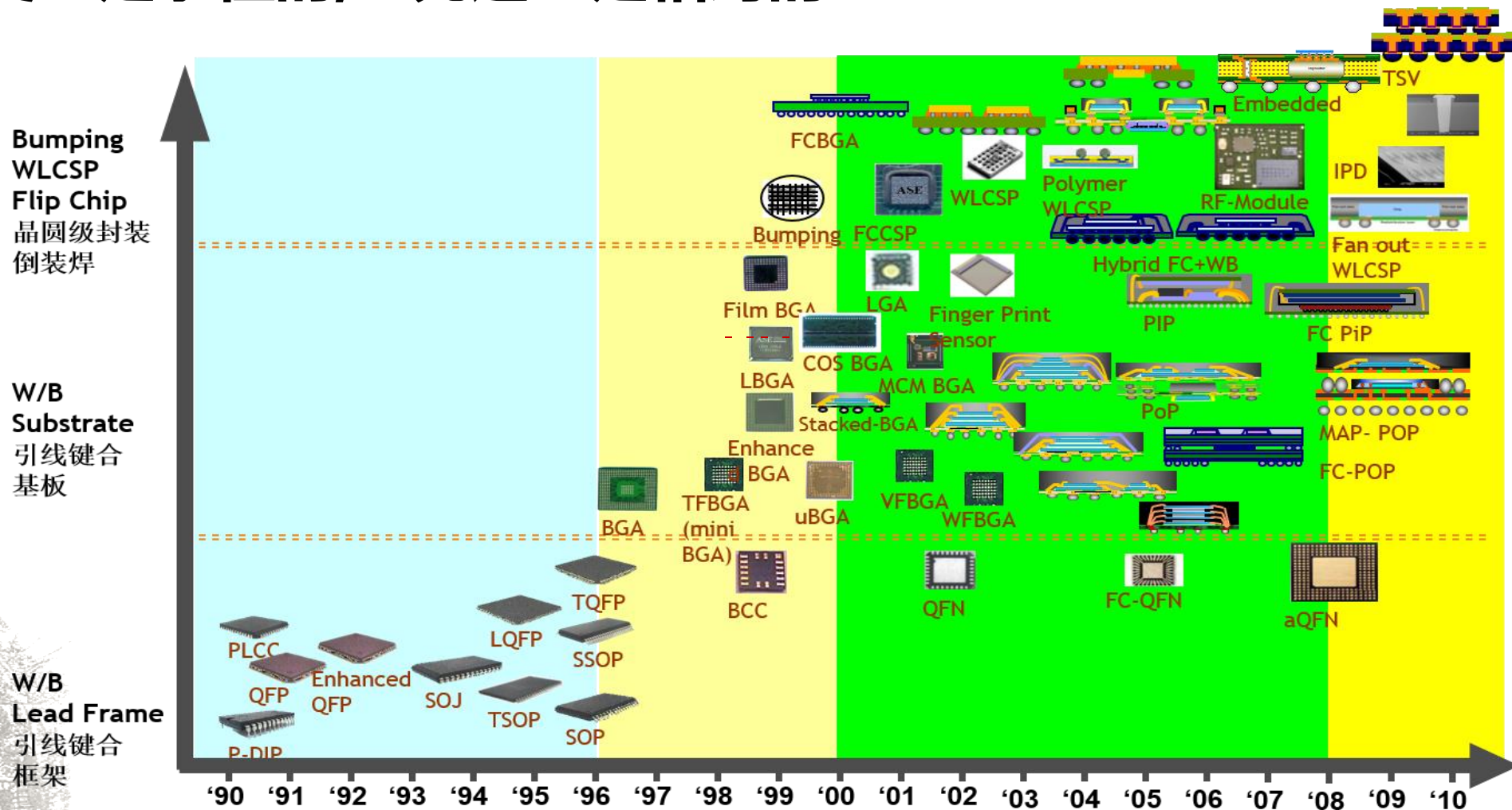
封装正在变得更为重要

- 异质集成复杂系统：A9采用的集成封装
- Chiplet可能带来的新变化：芯片模块化？



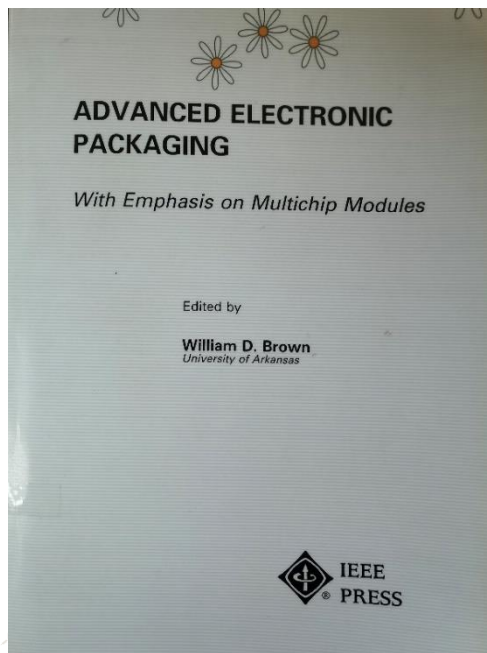
什么是先进封装技术？

■ “变”是永恒的, “先进”是相对的



什么是先进封装技术？

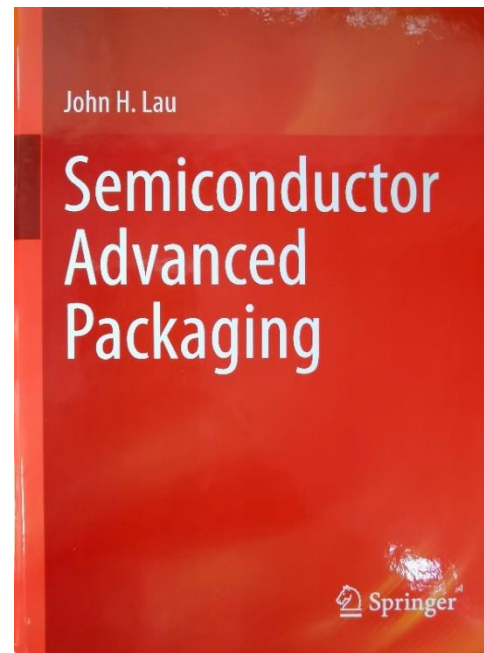
■ “变”是永恒的，“先进”是相对的



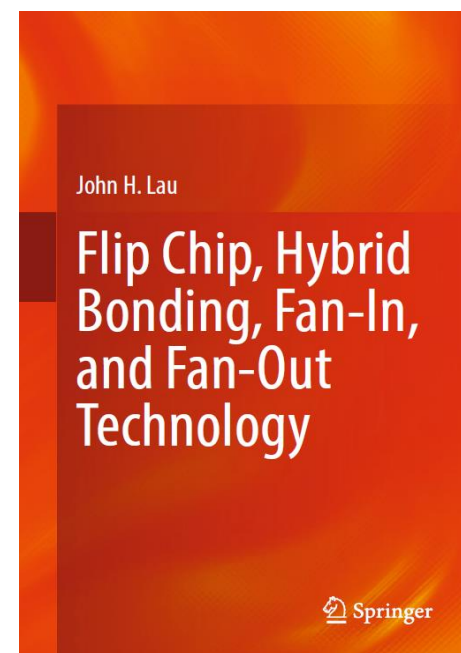
1998年



2013年



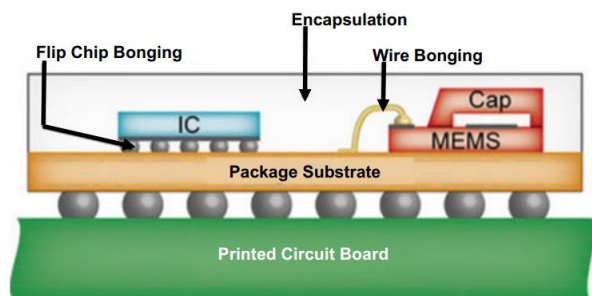
2021年



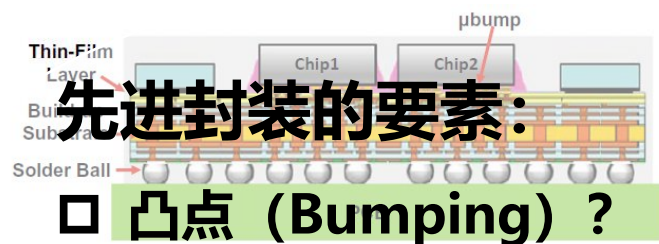
2024年

什么是先进封装技术？

■ 先进封装：从2D集成到3D集成



2D：引线键合+倒装芯片技术



2.1D：有机基板+高密度有机布线层

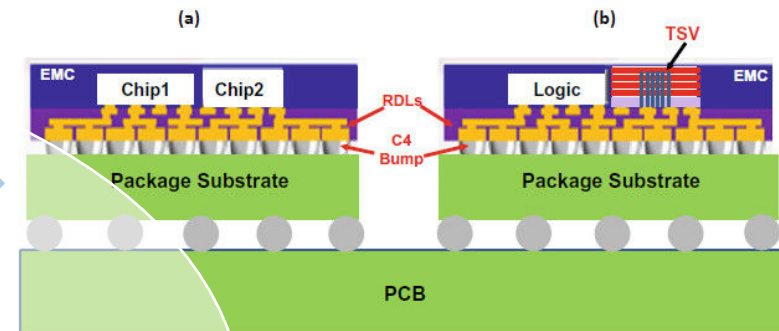
□ 凸点 (Bumping) ?

□ 再布线 (RDL) ?

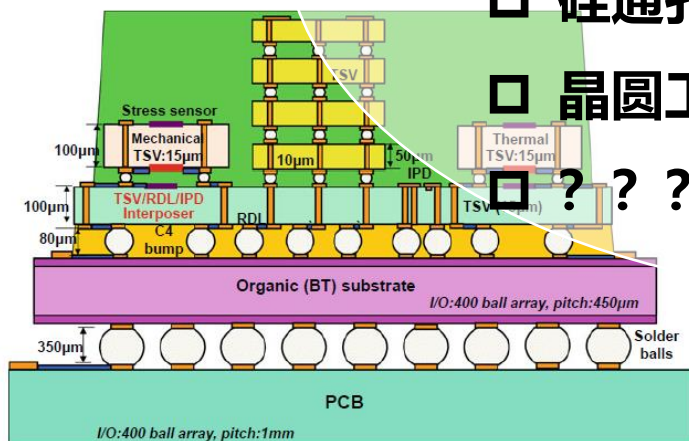
□ 硅通孔技术 (TSV) ?

□ 晶圆工艺 (Wafer Level) ?

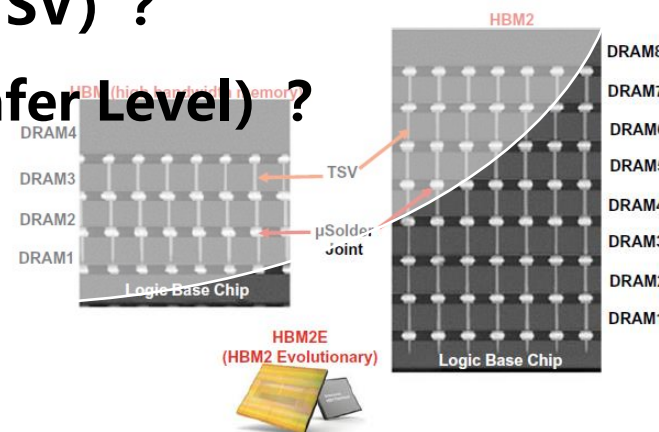
□ ???



2.3D：有机基板+有机RDL层



2.5D：有机基板+硅转接板



3D：堆叠芯片技术

来源：Semiconductor Advanced Packaging.
John Lau

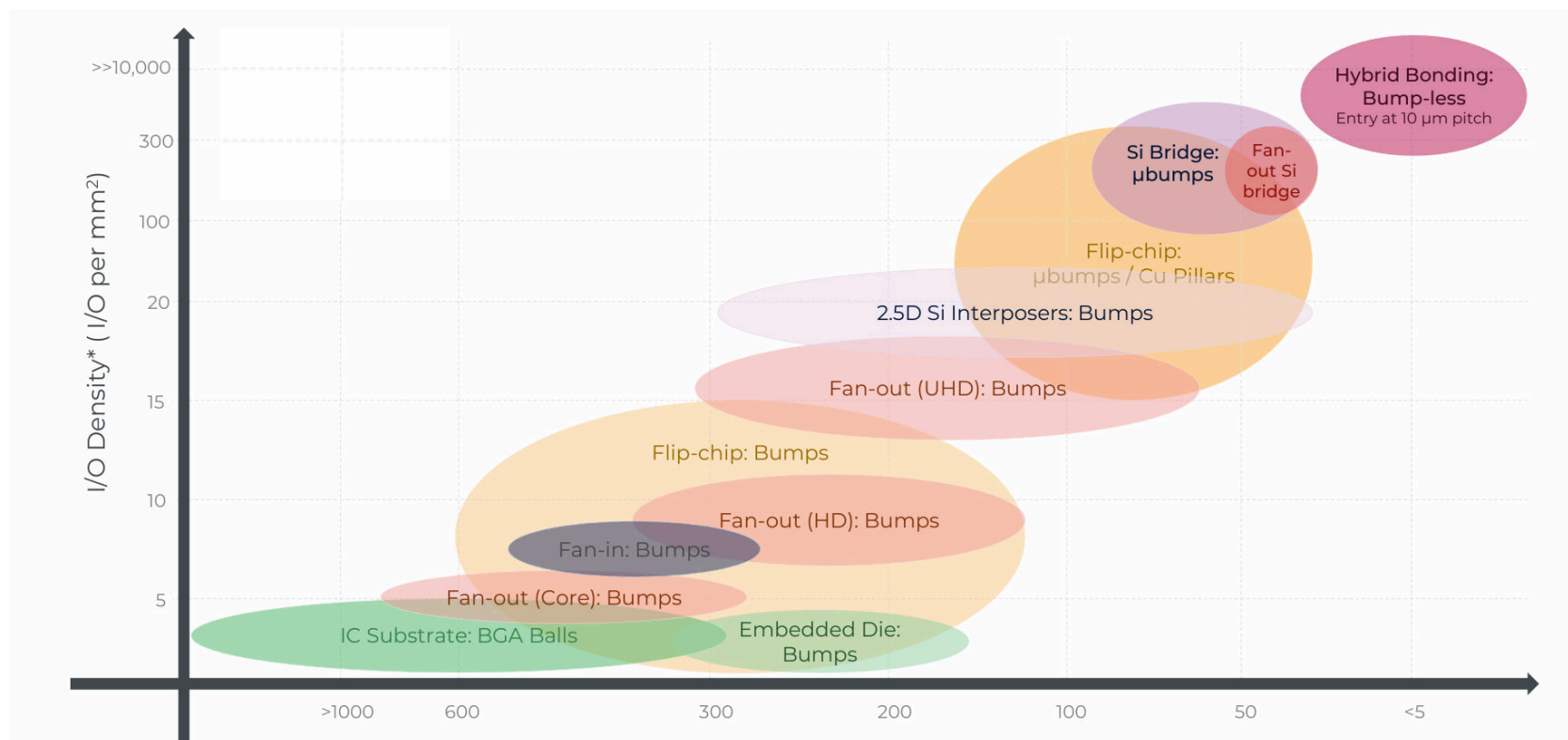
什么是先进封装技术？

- 高密度互连
- 晶圆级工艺
- 系统级封装
- 三维集成方法
- 异质异构集成

Flip Chip (C4 Bump) : $10^1/\text{mm}^2$

Flip Chip ($\mu\text{Bump}/\text{Cu Pillar}$) : $10^2/\text{mm}^2$

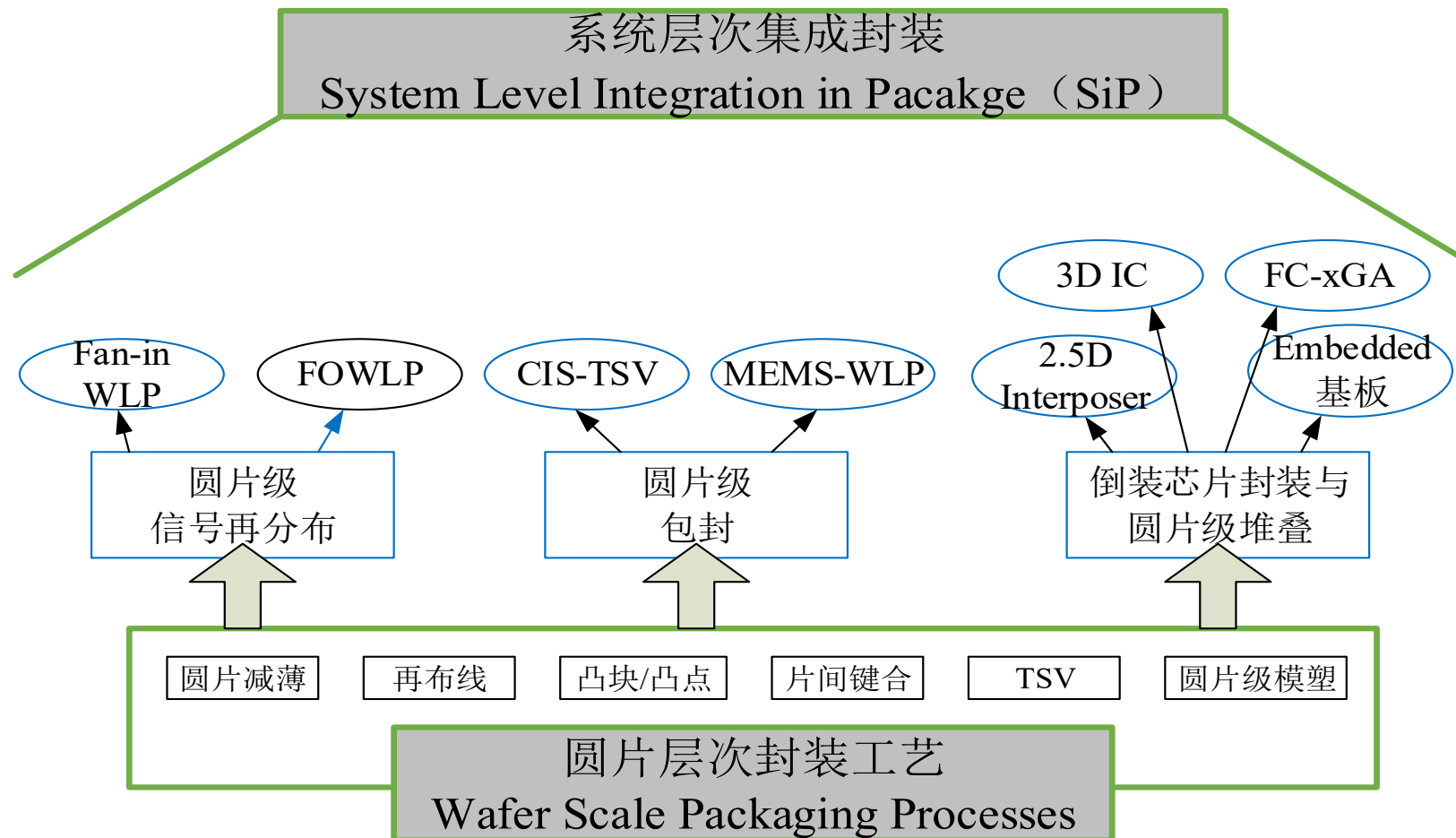
Hybrid Bonding : $10^4\text{-}10^6/\text{mm}^2$



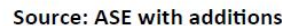
Status of the Advanced Packaging Industry 2023, Yole Development

什么是先进封装技术？

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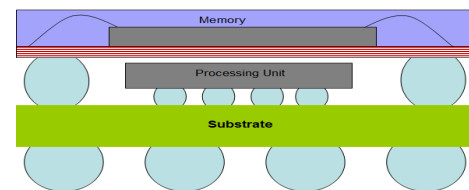


- ## 电子组装制造

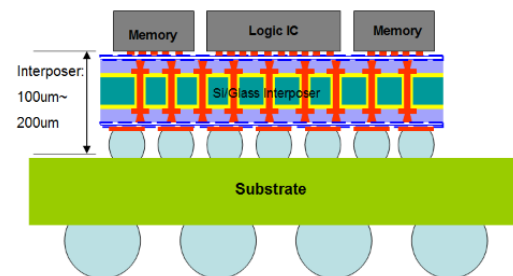


什么是先进封装技术？

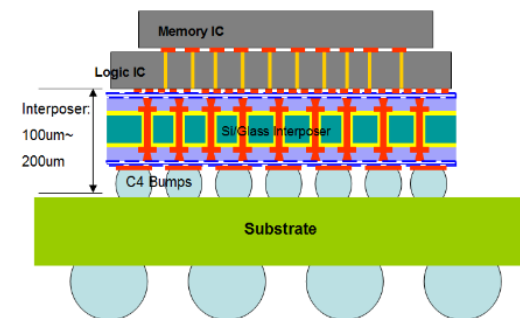
- 高密度互连
- 晶圆级工艺
- 系统级封装
- **三维集成方法**
- 异质异构集成



POP



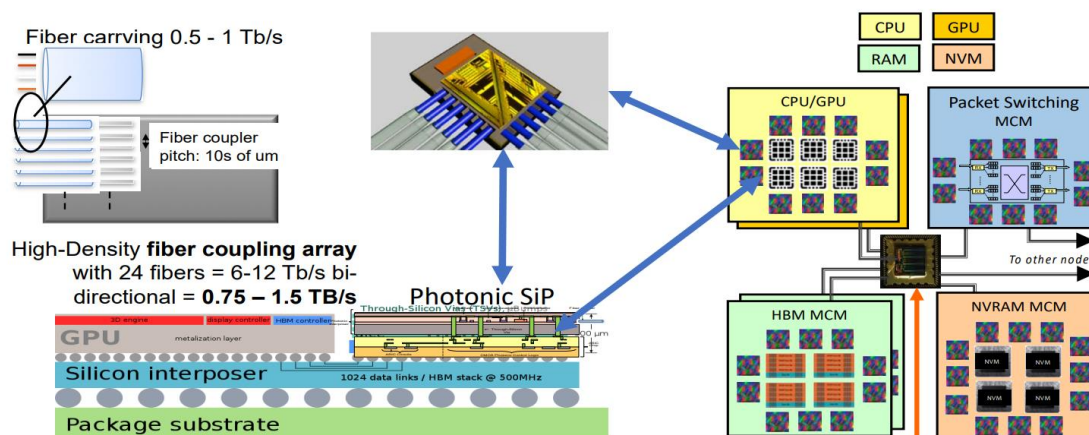
3D TSV



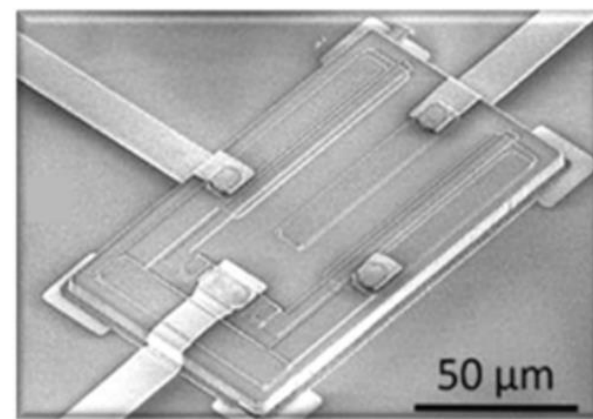
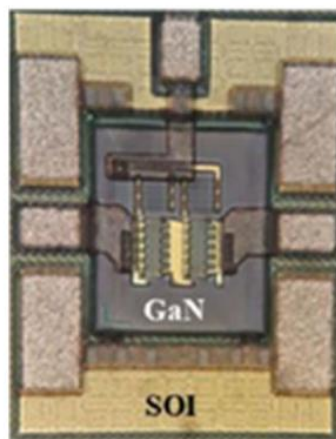
什么是先进封装技术？

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Silicon Photonics Co-Package Integration



硅光合封技术



GaN HEMT集成

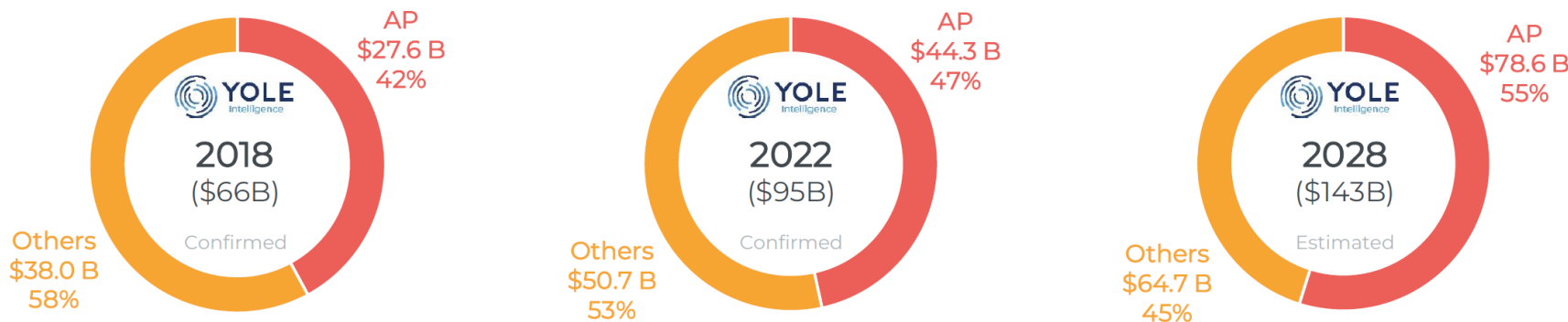
来源: Heterogeneous Integration Roadmap

先进封装产业发展趋势



■ 先进封装业发展迅猛

ADVANCED VS TRADITIONAL TECHNOLOGIES OVERVIEW 2018 to 2028
Revenue Split (\$B)

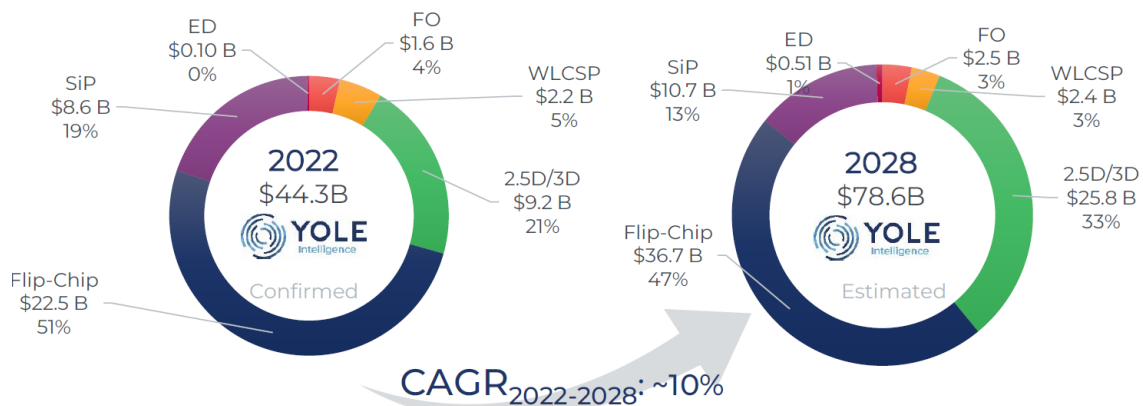


- 先进封装复合增长率10.6%，传统封装复合增长率4.1%。
- 2022年产值规模占比47%，预计2028年产值规模占比55%

先进封装产业发展趋势

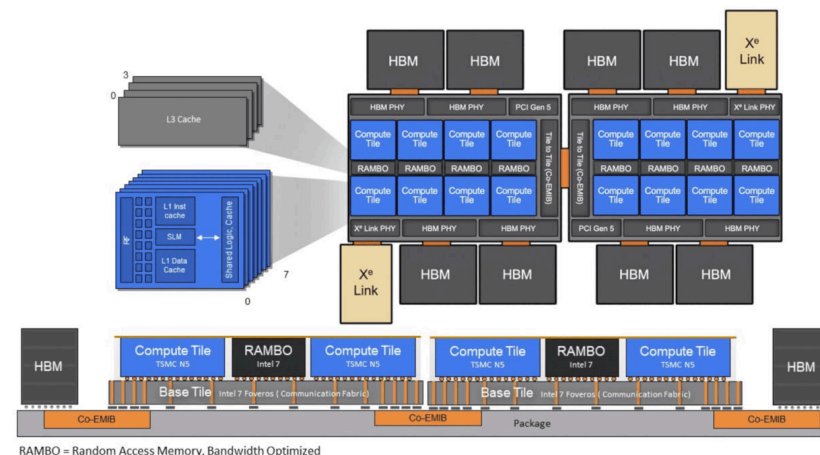
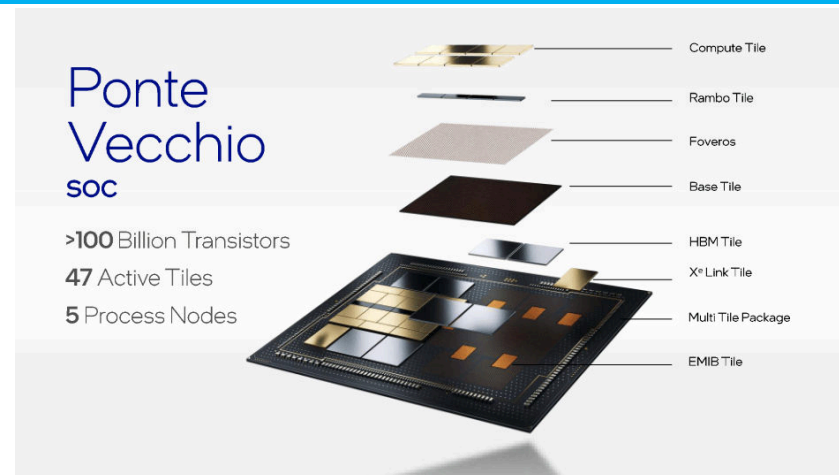


■ 高性能芯片需求牵引先进封装发展



Flip-chip	CAGR ₂₀₂₂₋₂₀₂₈ ~ 8.5%
SiP	CAGR ₂₀₂₂₋₂₀₂₈ ~ 3.6%
Fan-out	CAGR ₂₀₂₂₋₂₀₂₈ ~ 7.6%
Fan-in WLP	CAGR ₂₀₂₂₋₂₀₂₈ ~ 1.6%
2.5D/3D	CAGR ₂₀₂₂₋₂₀₂₈ ~ 18.7%
Embedded Die	CAGR ₂₀₂₂₋₂₀₂₈ ~ 30.4%

* Flip-Chip includes: FCCSP and FCBGA. It includes bumping, assembly and materials revenues.

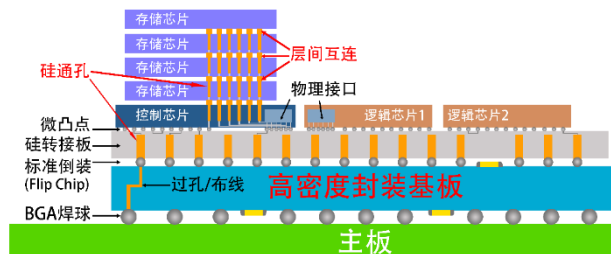


Intel Ponte Vecchio包含了47个Chiplets
采用2.5D封装技术

先进封装产业发展趋势

■ 产业链深度融合

以2.5D先进封装产业链为例：



提纲



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- 什么是先进封装技术?

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- 硅通孔互连 (TSV) 三维封装
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- 玻璃封装基板 (Glass Substrate)

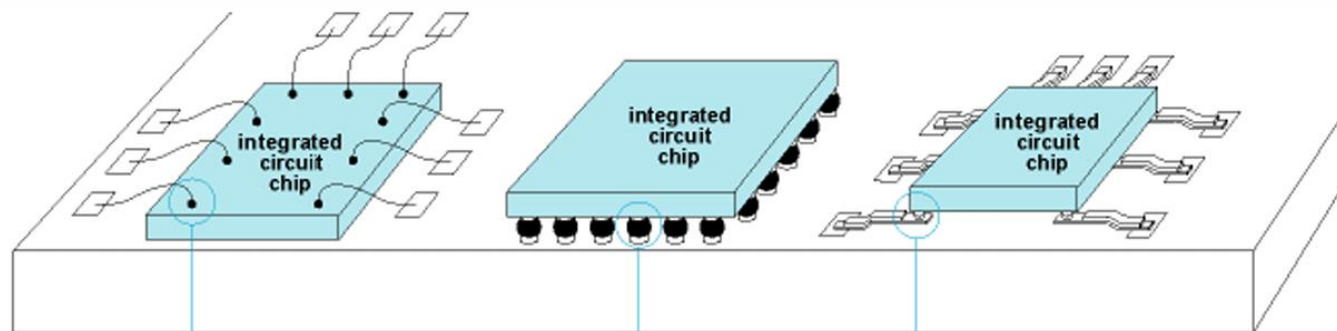
■ 先进封装案例

■ 先进封装技术发展的讨论



倒装芯片 (Flip-Chip)

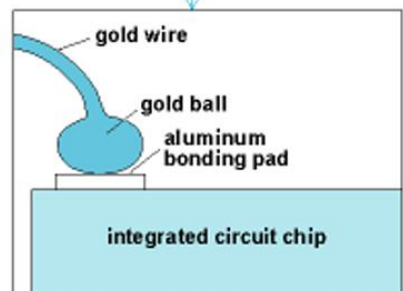
■ 芯片封装的常见3种互连形式：WB、FC、TAB



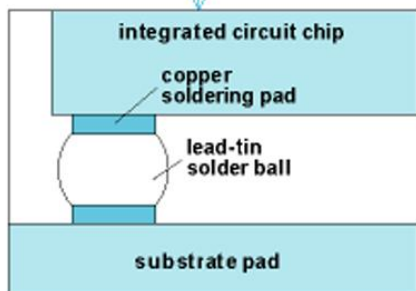
(a) 引线键合 (WB)

倒装芯片 (FC)

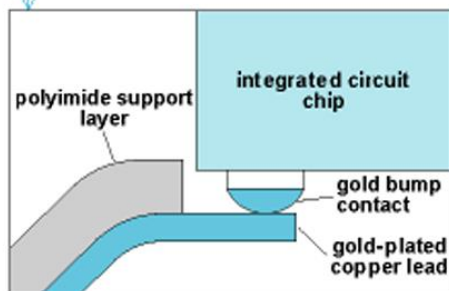
载带自动焊 (TAB)



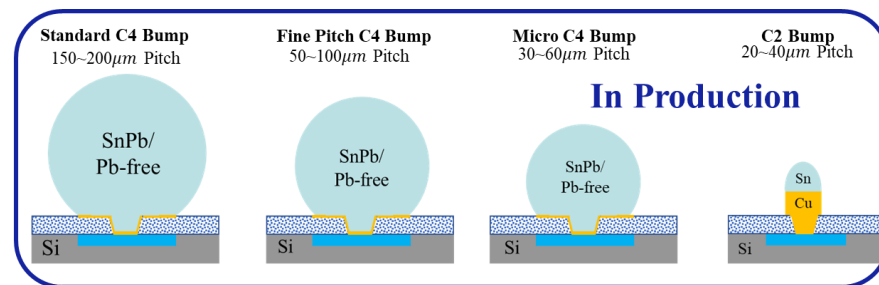
(b)



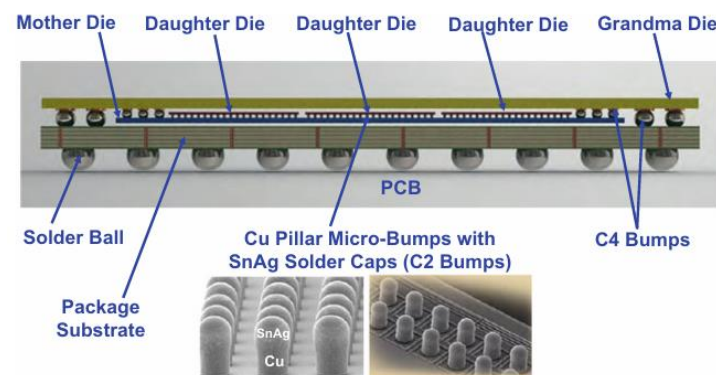
(c)



(d)



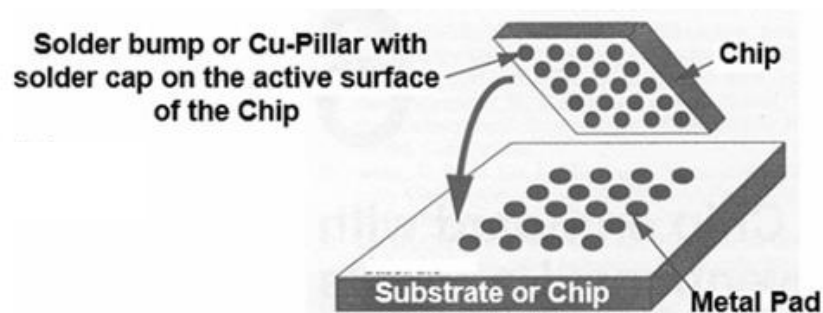
◆ 随着封装互连密度提升，倒装凸点的形态也在发生改变



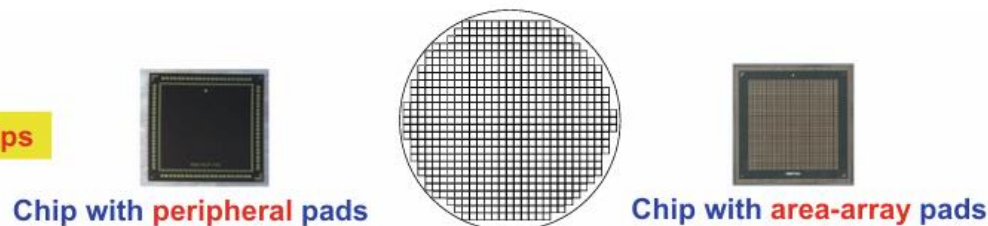
◆ 倒装芯片技术已经成为先进封装互连的关键组成部分

■ 先进封装中FC技术的要素：Wafer-Bumping、凸点键合、底部填充

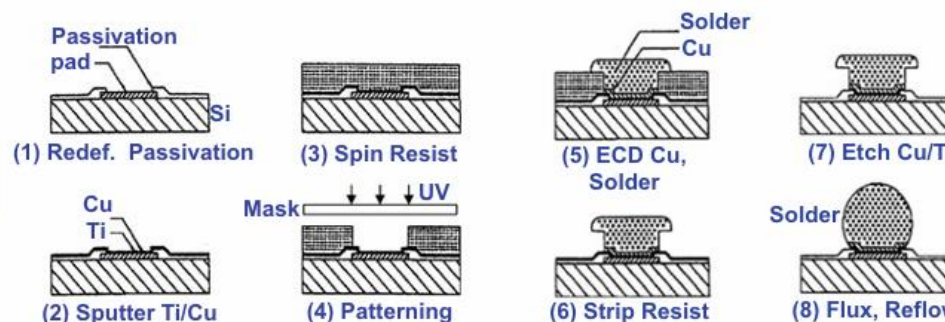
◆ FC技术定义为：芯片（Die）通过倒装凸点互连到基板（Substrate）或另一颗芯片（Die）上。



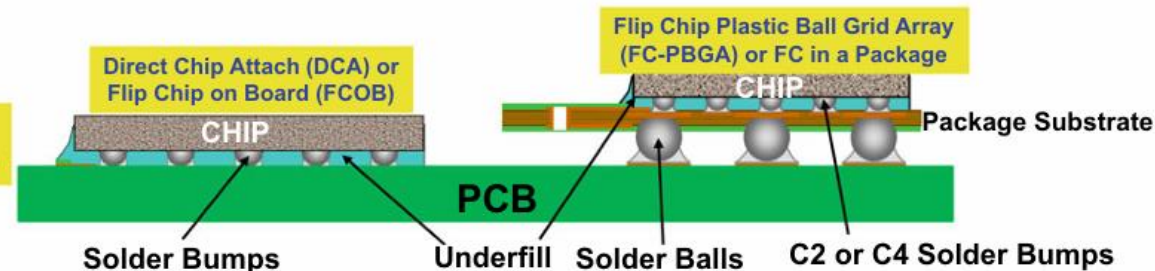
Wafer with Chips



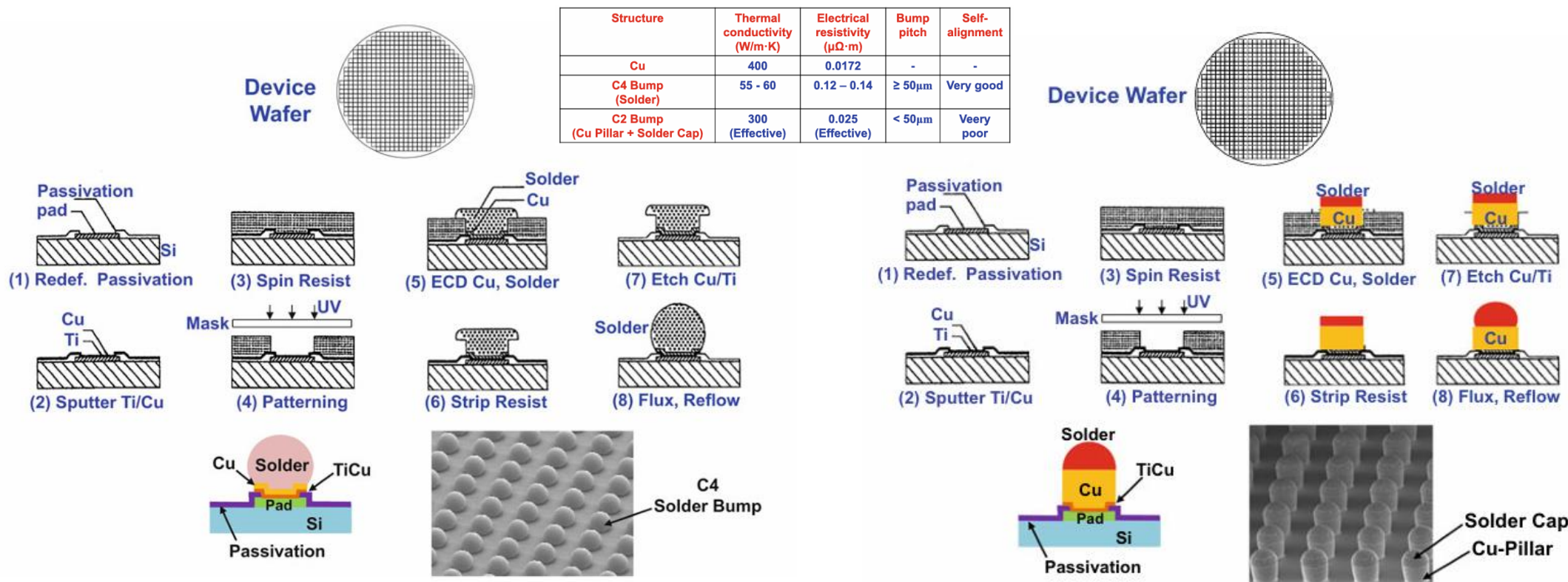
Wafer Bumping (e.g., C4 bump)



After dicing, Flip Chip Assembly



■ 晶圆级凸点（凸块）制作：Wafer-Bumping



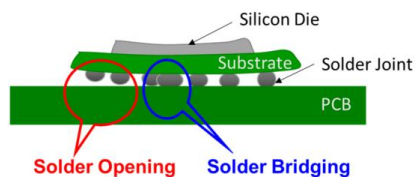
◆ C4焊料凸点（芯片-基板互连）制作工艺流程

注：C4凸点的全称是可控塌陷芯片连接（Controlled Collapse Chip Connect, C4）

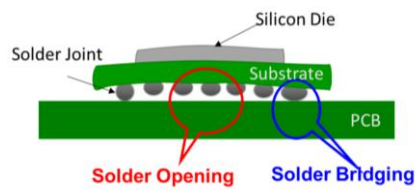
◆ C2凸点（芯片-芯片/基板互连）制作工艺流程

注：C2凸点相比C4凸点更适用于小节距互连，因铜柱可一定程度限制焊料外溢

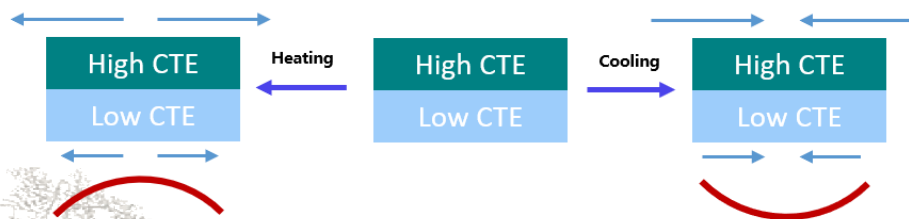
Flip-Chip倒装键合挑战：翘曲和Sn挤出



I. Concave Warpage of BGA Package

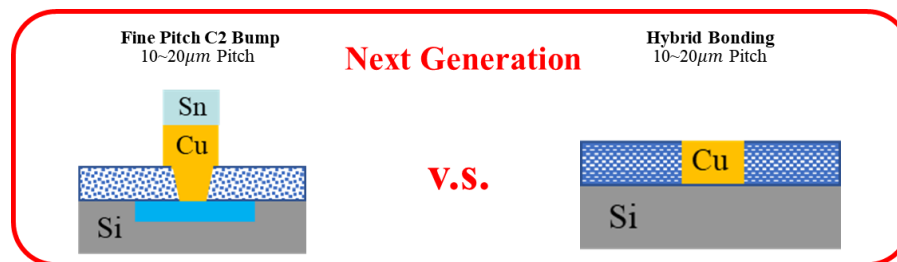
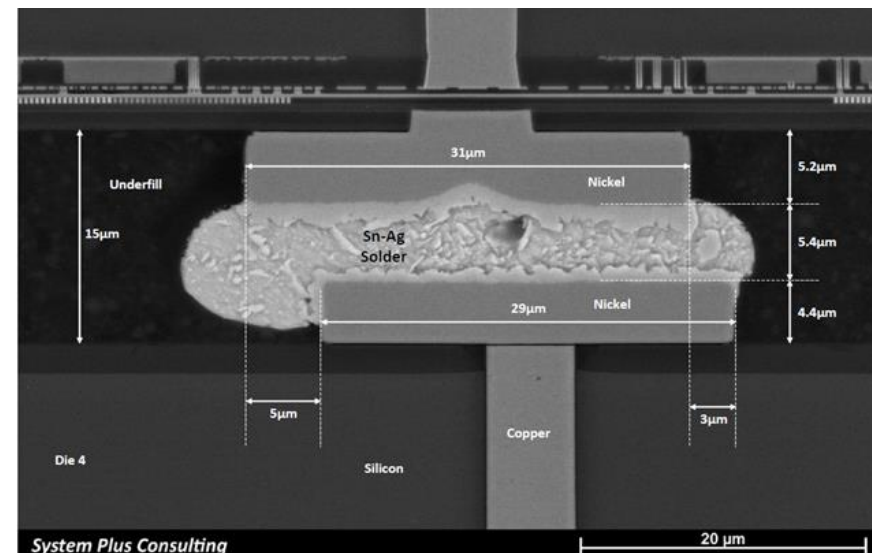
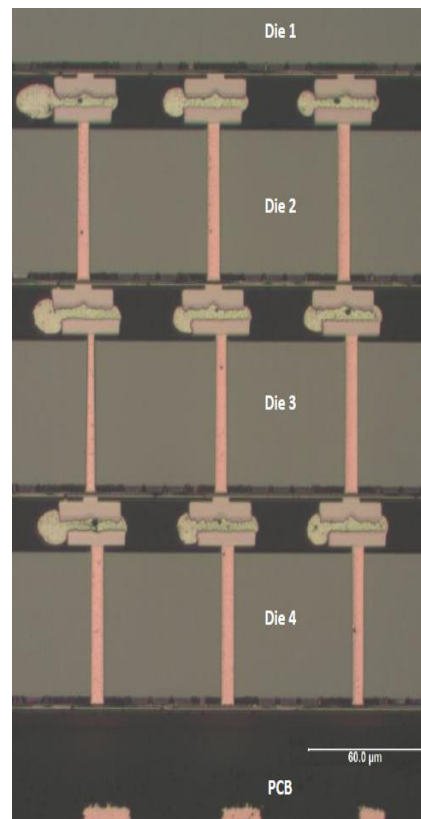


II. Convex Warpage of BGA Package



影响翘曲的因素：

CTE不匹配、温度变化、湿度影响、机械应力等



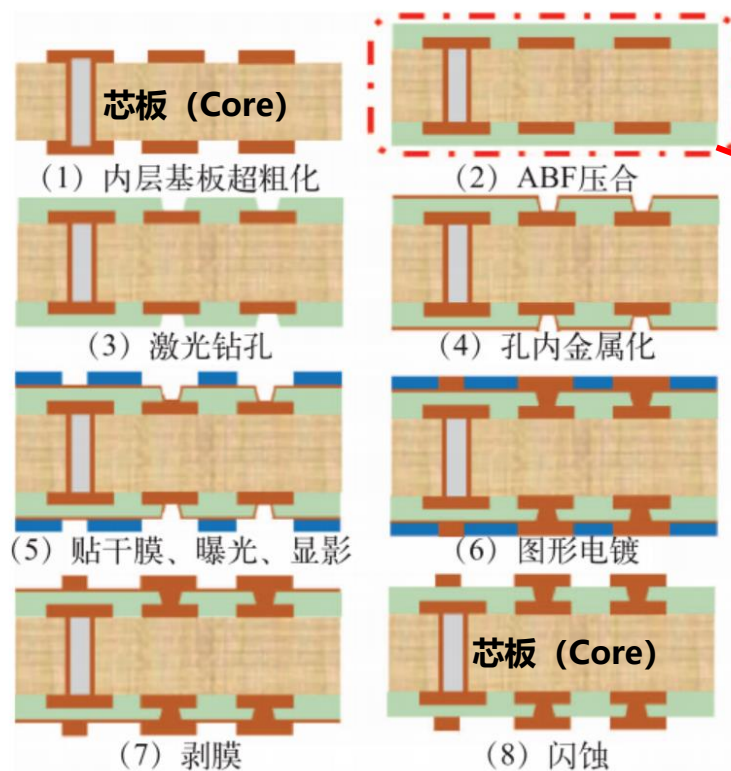
◆ 能否继续缩小微凸点节距？

■ 积层封装基板 (Build-up Substrate)

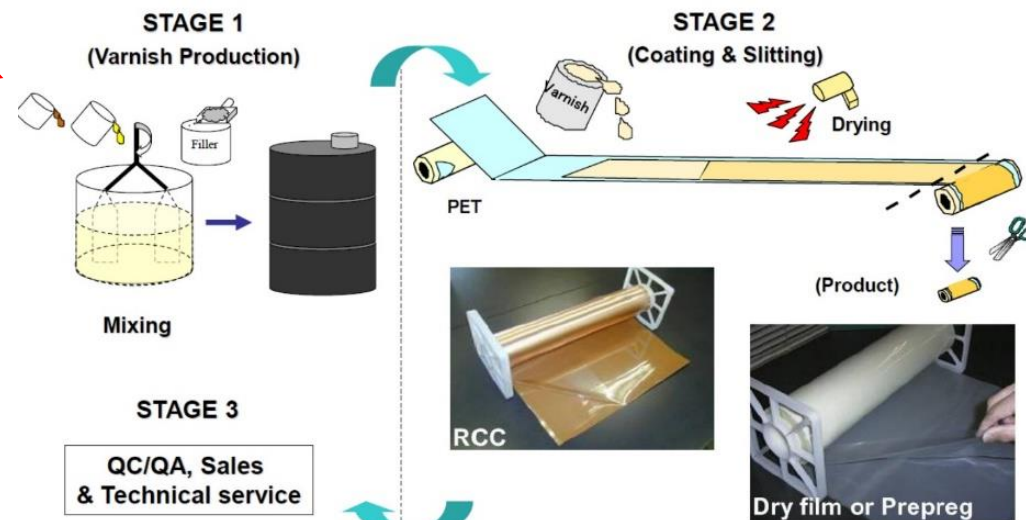
◆ FCBGA基板具有**超大尺寸**、**高叠层**和**精细线路**3个方面的特点

◆ 基板Core层一般为**玻璃纤维布**，起到强度和机械支撑作用

◆ 精细线路通常利用半加成法 (SAP) 工艺加工，FCBGA基板的线宽/线间距将减小到**5 μ m/5 μ m**以下



◆ 加成法 (SAP) 工艺流程

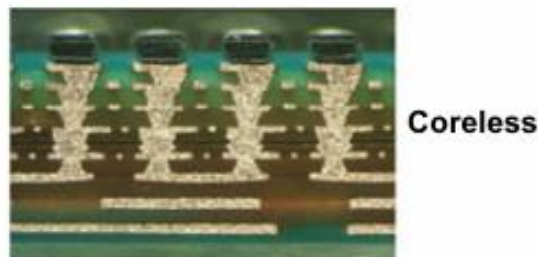
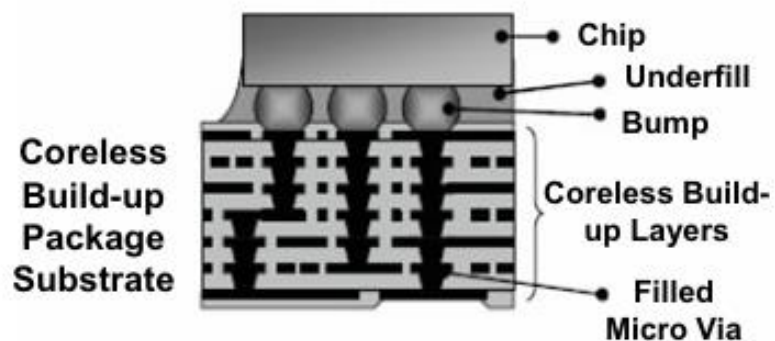
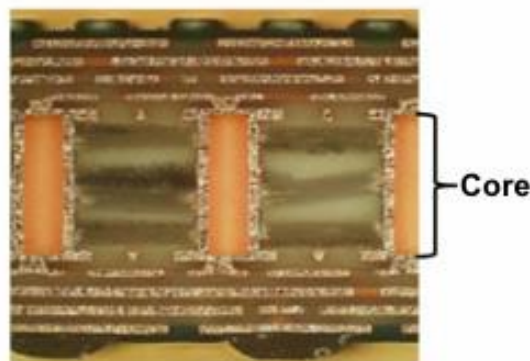
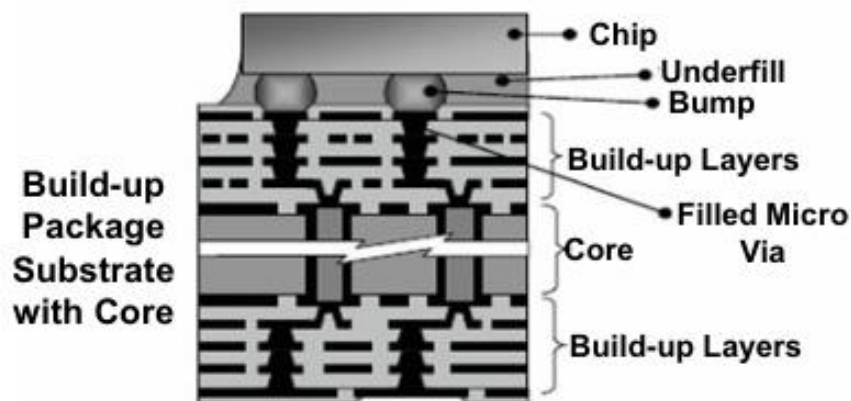


Source: Ajinomoto Fine-Techno Co. Inc.

◆ 目前FCBGA基板生产中的ABF薄膜主要由日本味之素公司垄断。

■ 积层封装基板 (Build-up Substrate)

◆ 随着产品迭代需要，FCBGA基板还发展出无Core层结构



◆ Coreless结构优点:

1. 制造成本更低
2. 基板布线能力更高
3. 基板电学性能更好
4. 基板外形尺寸更小

◆ Coreless结构缺点:

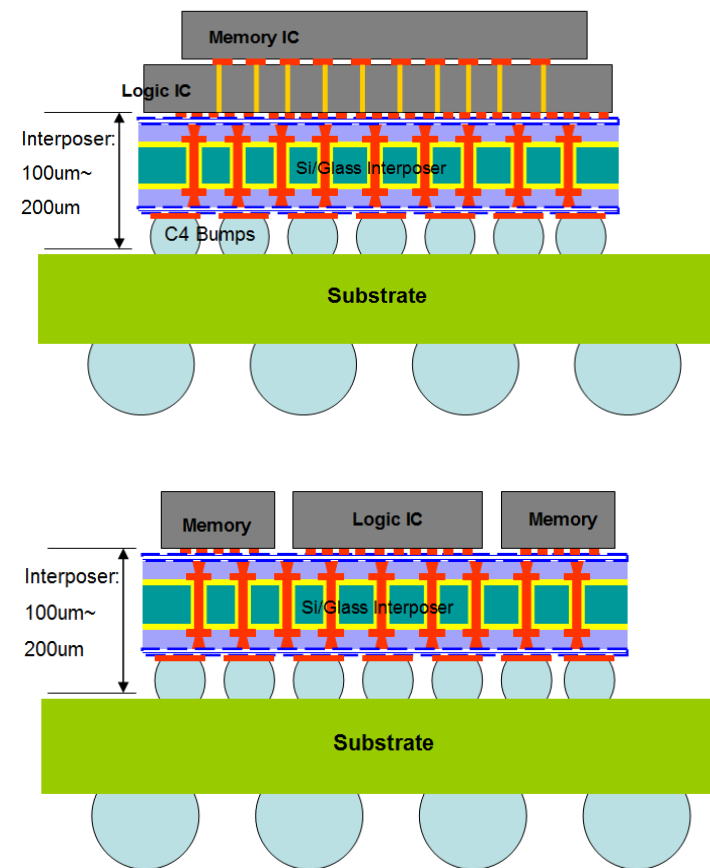
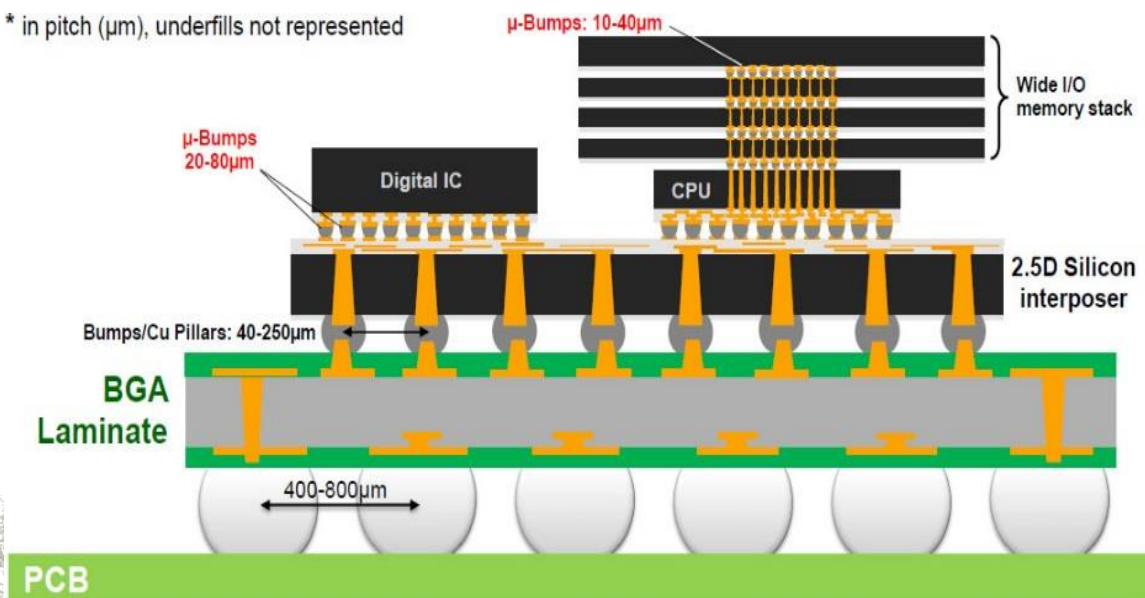
1. 翘曲问题更严重
2. 基板更容易分层
3. 基板刚度低，可焊性差
4. 新工艺需要新设备

基于TSV的三维封装

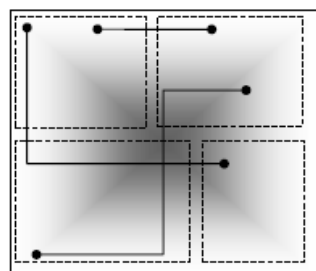
■ TSV —— Through Silicon Via (硅通孔技术)

● 2.5D & 3D

* in pitch (μm), underfills not represented



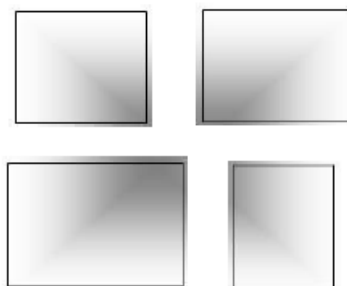
Large-scale logic circuits (LSI chip)



<Illustration of SoC>

Divided to ...

Sub-logic circuits on LSI chips

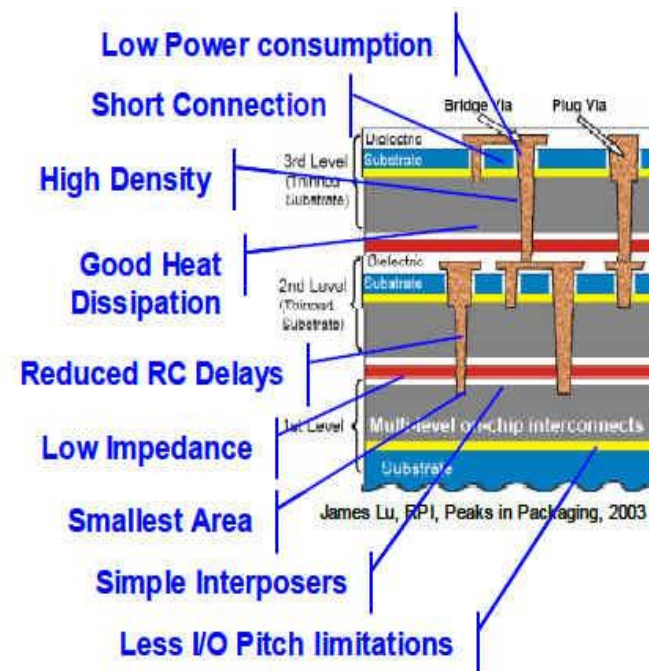
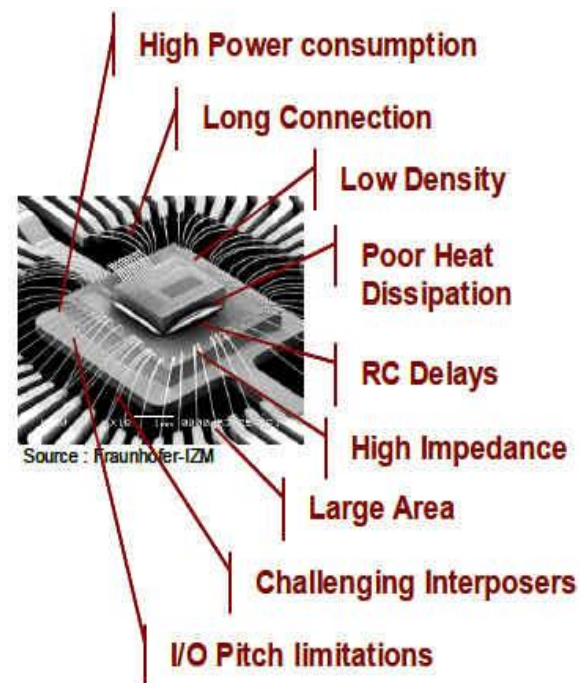
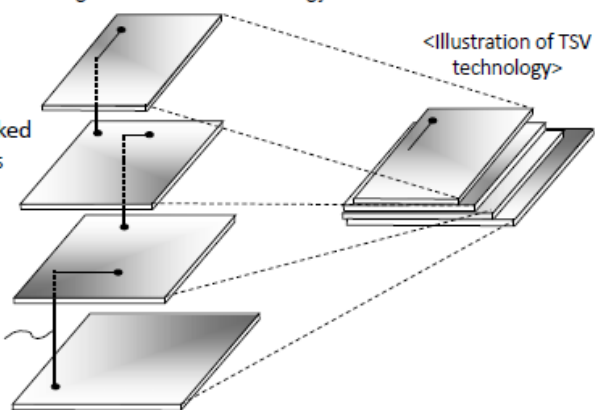


3D stacked
Chips

TSV

<Enlarged view of TSV technology>

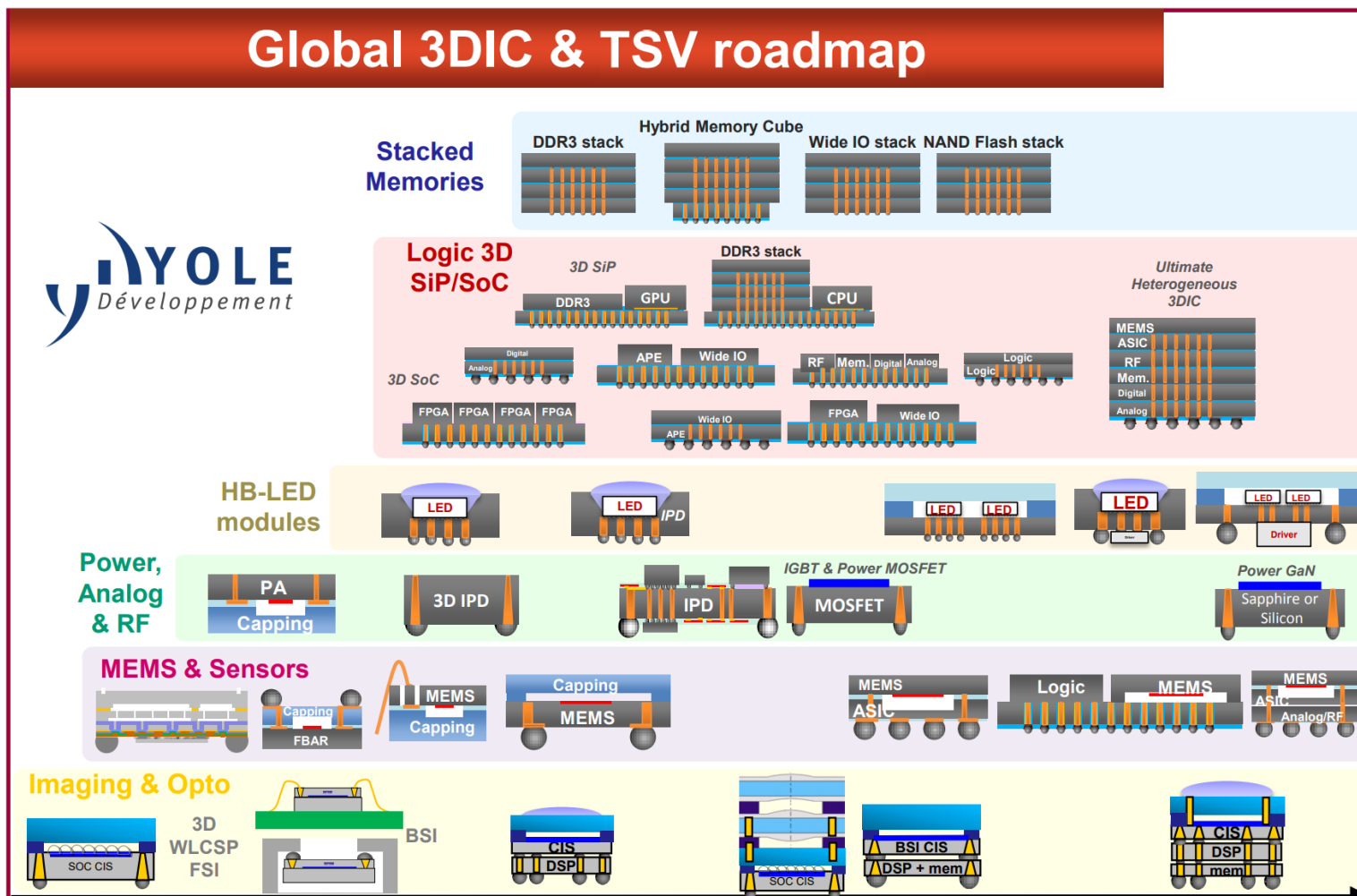
<Illustration of TSV technology>



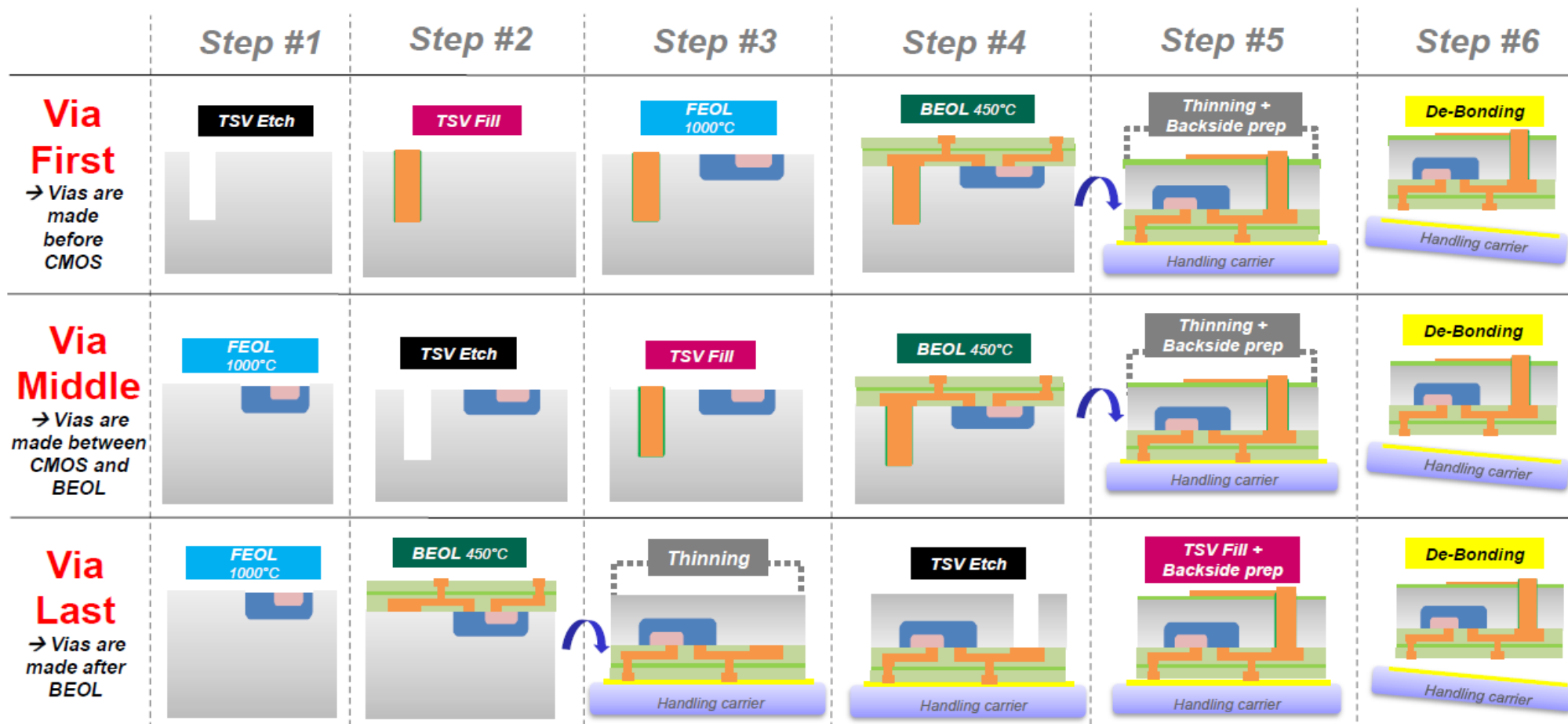
来源: Status and plans in 3D Technologies. 2007

■ TSV三维封装的应用

- 堆叠存储器
- 高性能2.5D/3D封装
- LED模组
- 电源模拟芯片
- MEMS传感器
- 图像光学器件
-



■ TSV技术路线



未见量产

主流

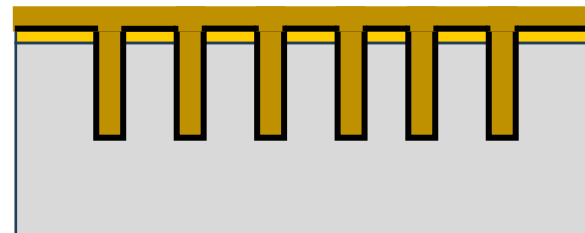
传感器应用

■ “中道” 工艺 (Via-Middle Process)

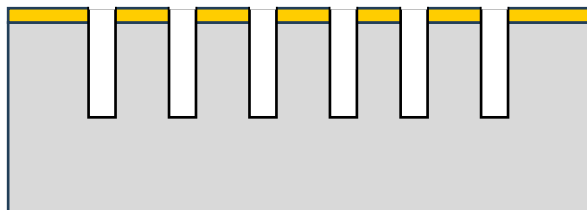
·前端器件有源层



·TSV 阻挡层/种子层沉积
·TSV电镀填充



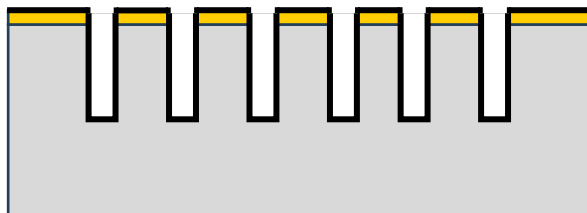
·TSV 深硅刻蚀



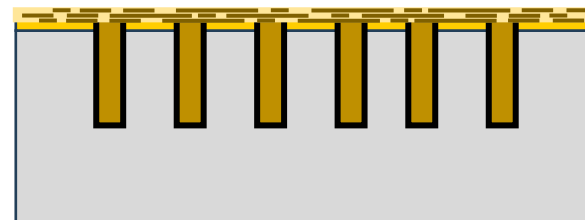
·化学机械抛光



·氧化层沉积



·后端互连线



FOEL

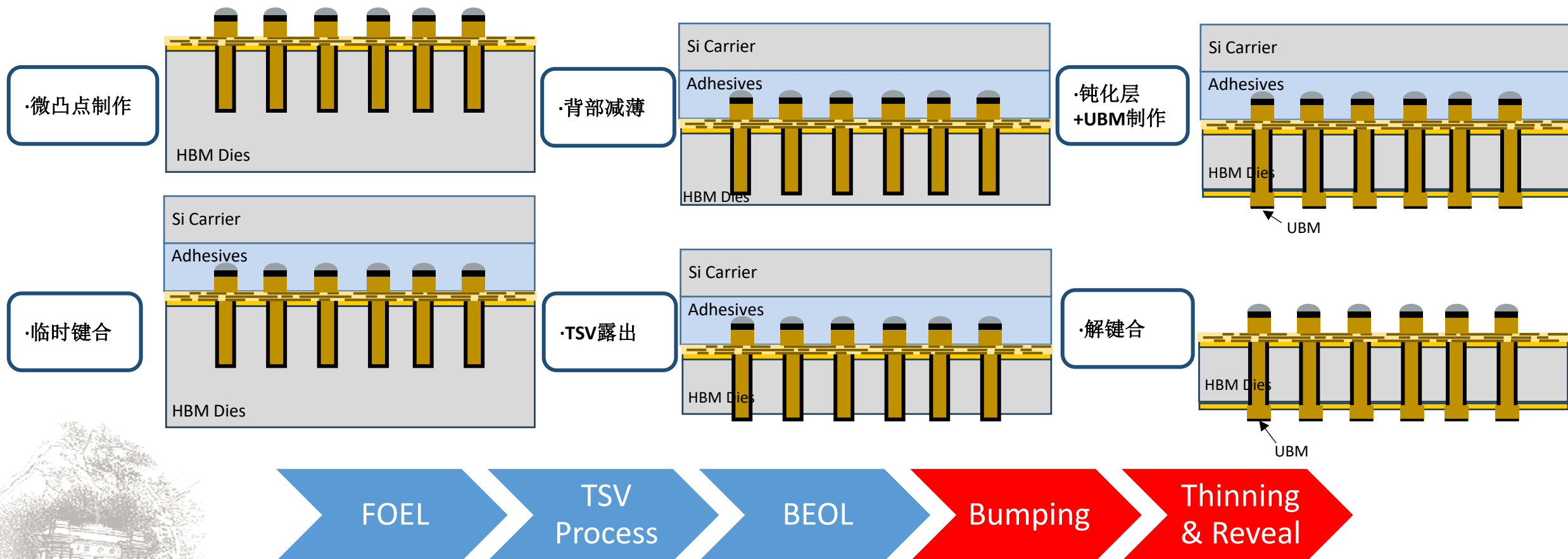
TSV
Process

BEOL

Bumping

Thinning &
Bonding

“中道” 工艺 (Via-Middle Process)



■ 高深宽比TSV的制作（刻蚀、填充）是三维集成中的重要技术

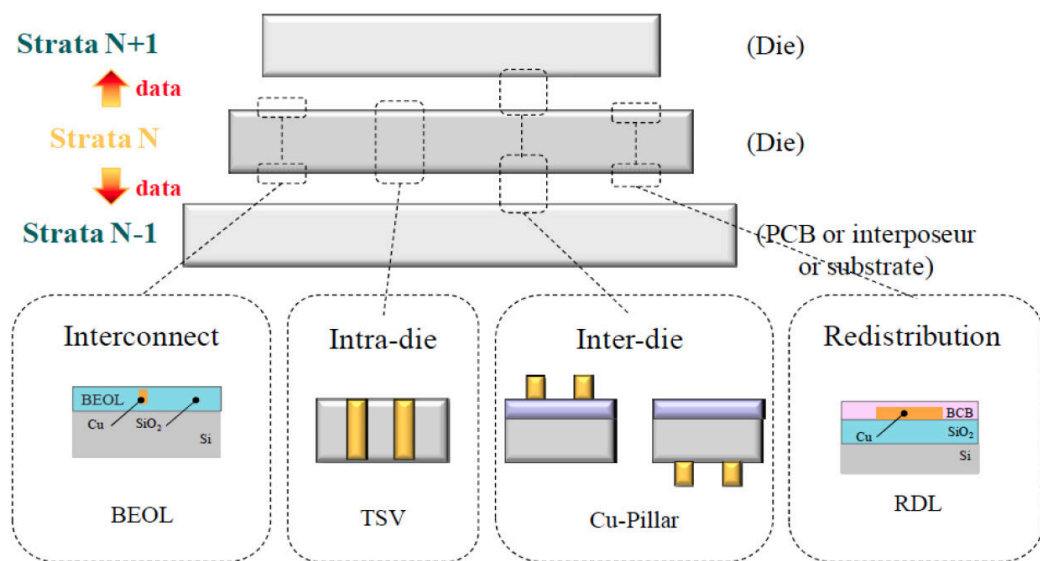
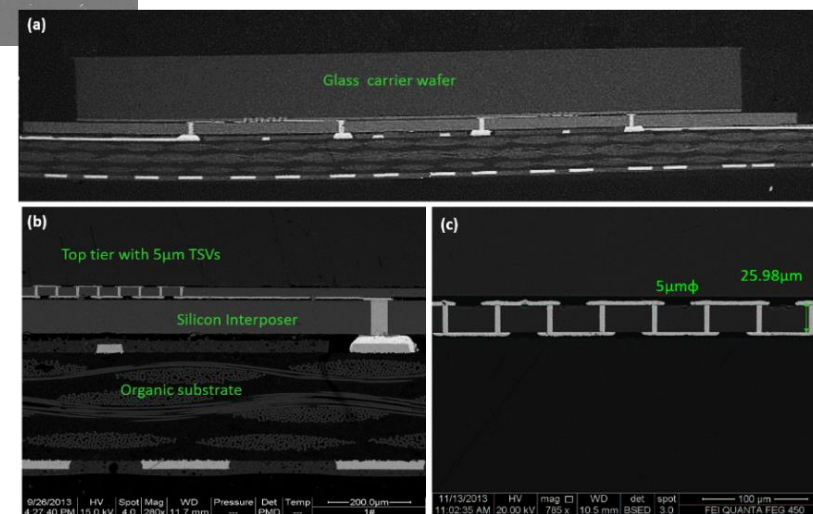
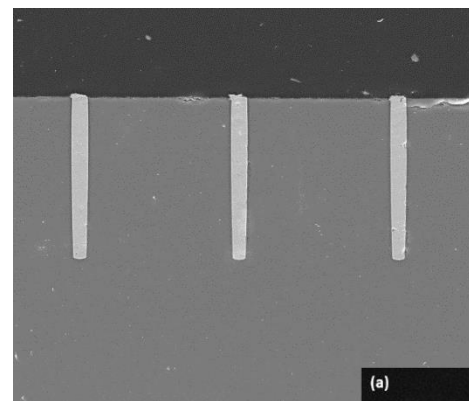


Figure 12: A schematic of 3D integration. Source: IEEE ECTC 2015

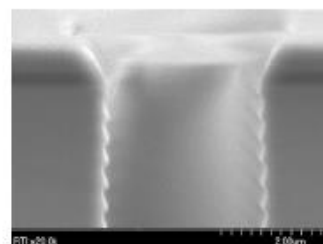
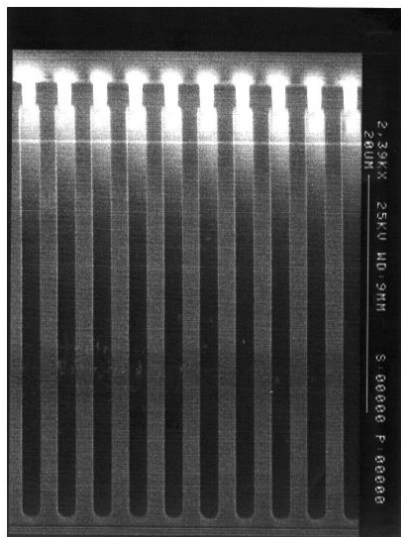


TSV直径: 5μm/40μm

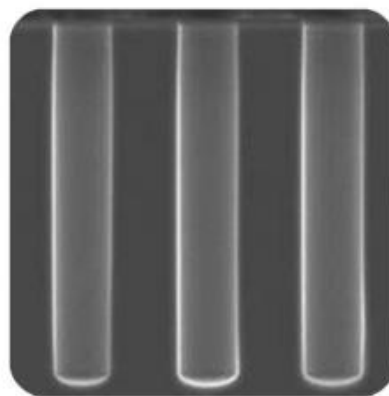
DRIE



“Bosch” Process

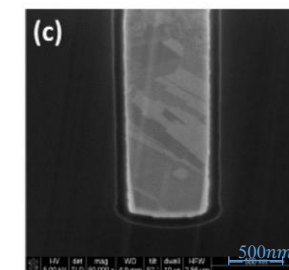
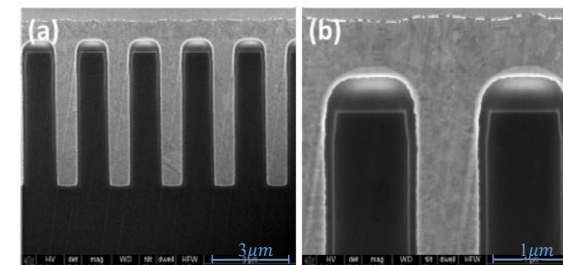
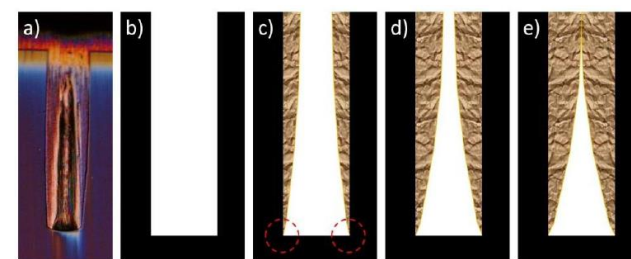
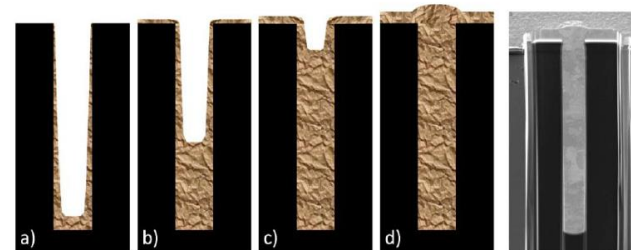
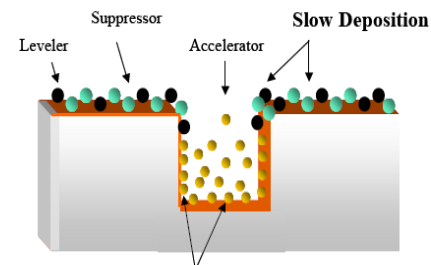


DRIE scallops (typical)

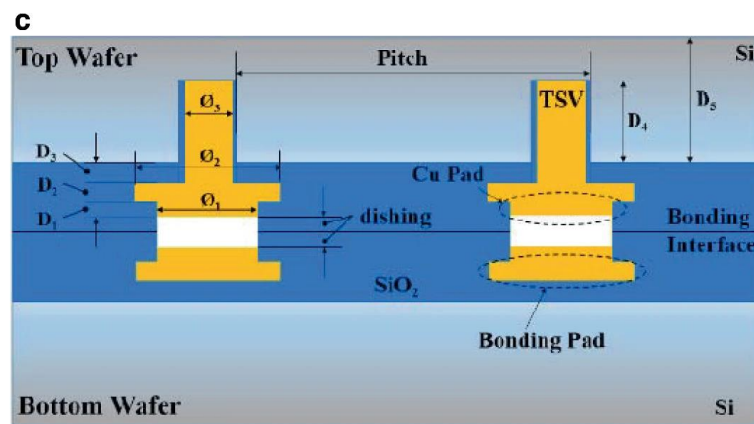
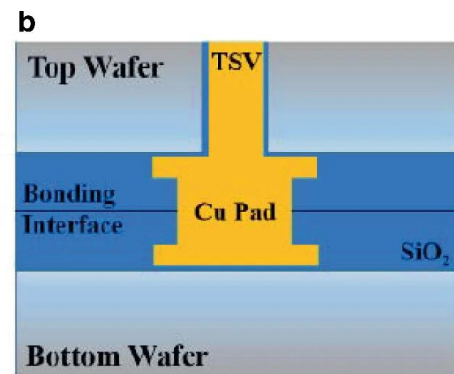
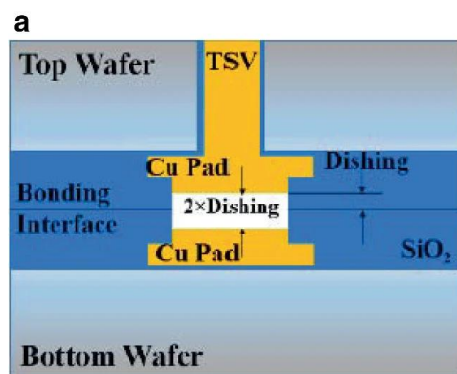
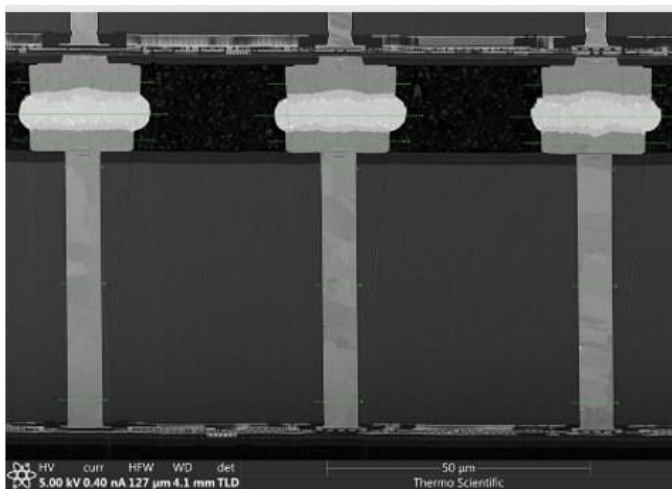


“Scallop-free” TSV etch (SPTS)

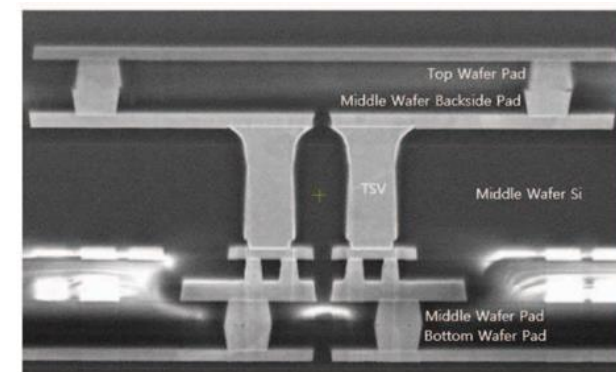
孔内填充



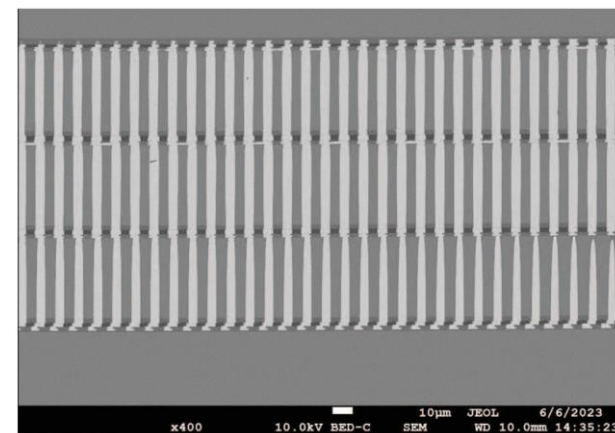
■ 片间键合技术：从微凸点走向混合键合



2021 John. Lau Semiconductor Advanced Packaging



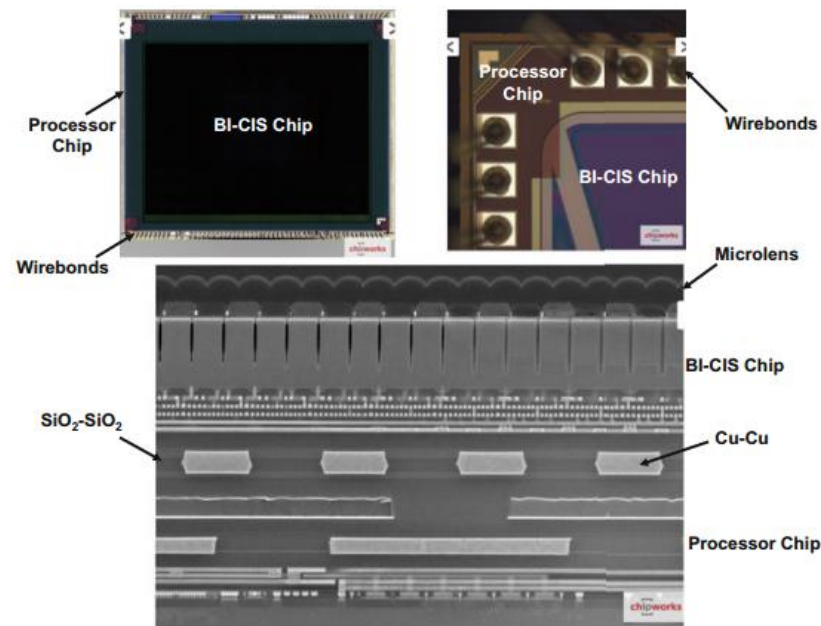
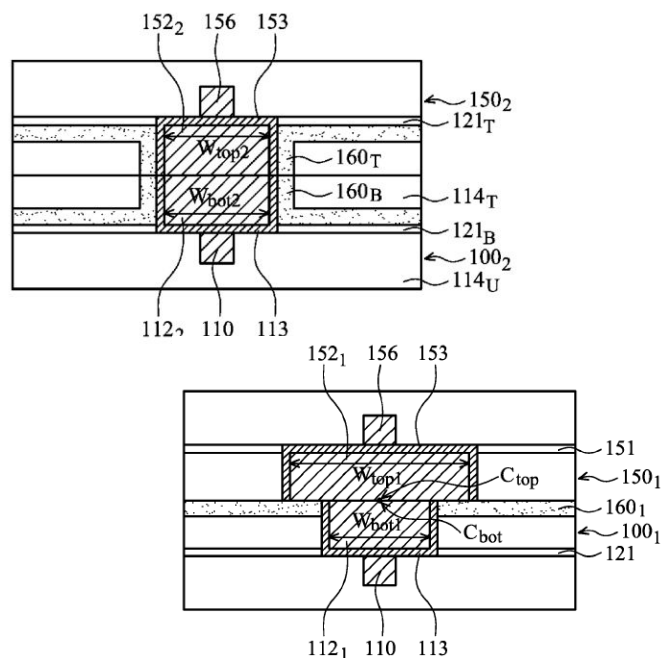
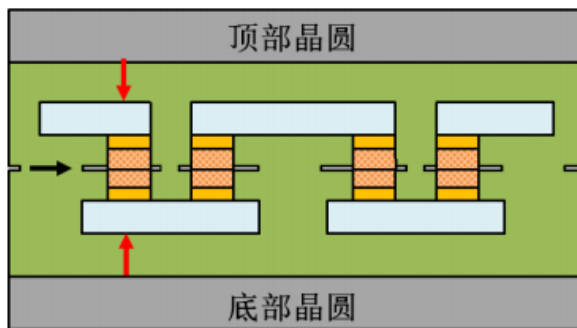
2021 ECTC. Sumsang



2024 ECTC. IME

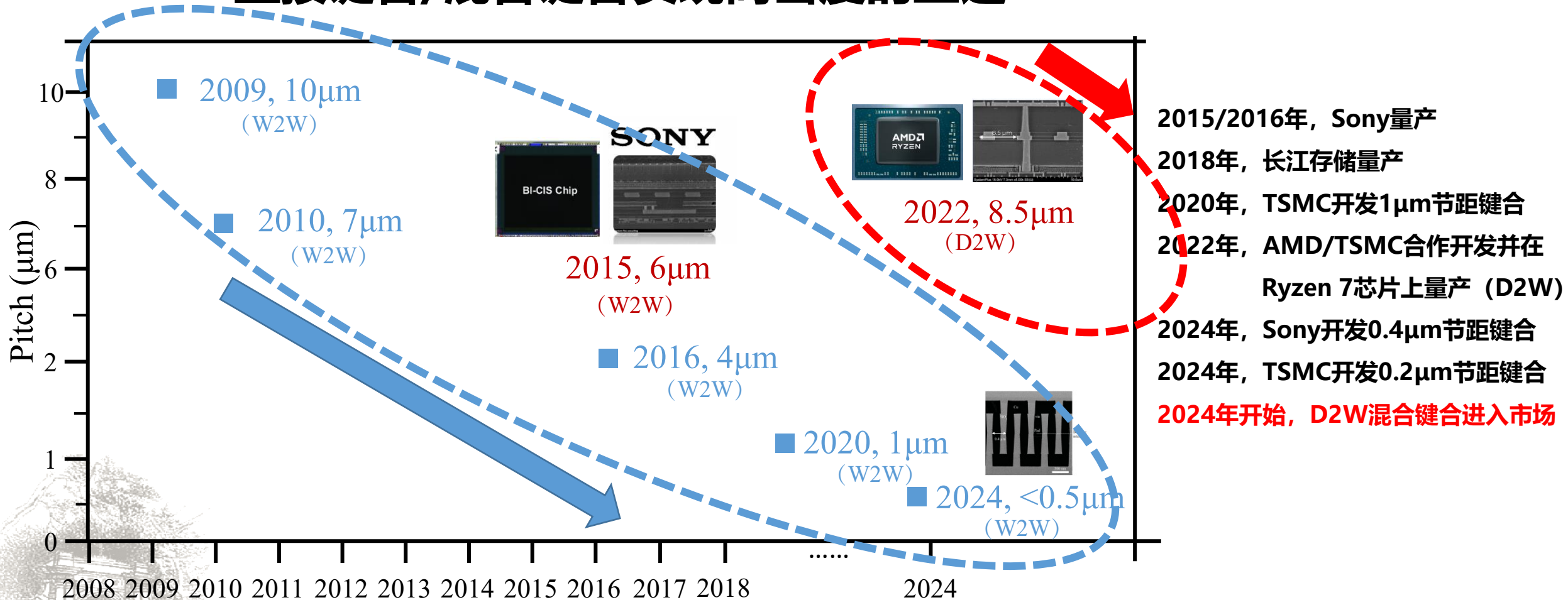
■ Cu-Cu直接键合/混合键合实现高密度的互连

- 2005年, Ziptronix公司提出直接键合技术 (Direct Bonding Interconnect, DBI)
- 2015年, TSMC获得了名为Hybrid bonding mechanisms for semiconductor wafers的专利, 该专利主要描述了Hybrid bonding中应对扩散的结构以及机制



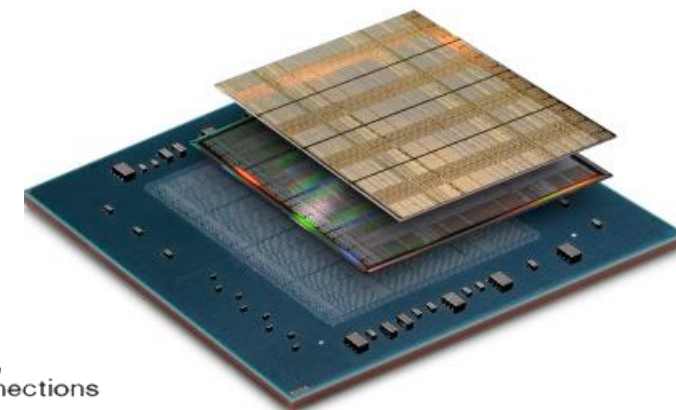
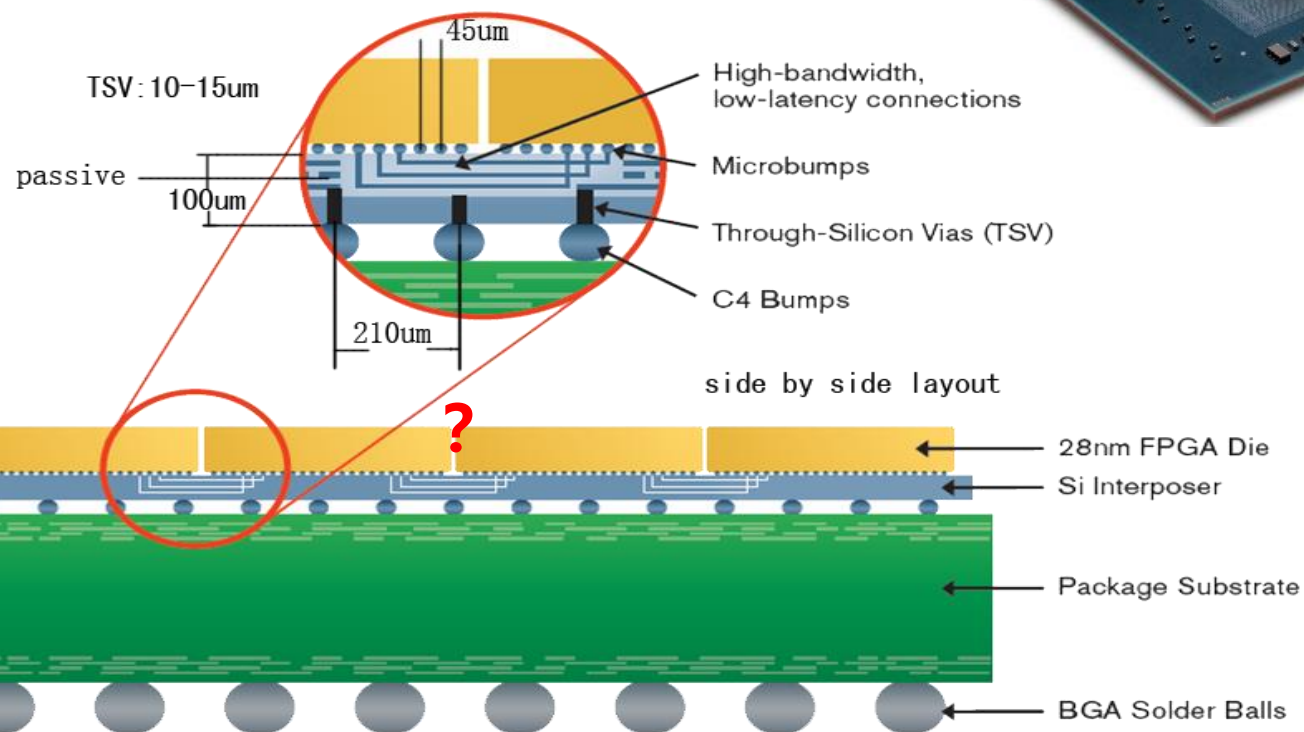
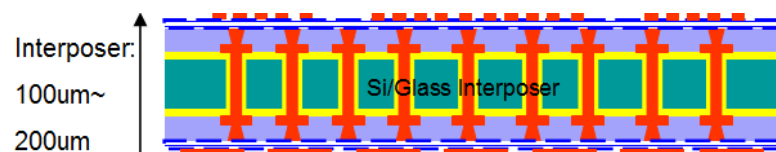
Y. Kagawa et al, Novel stacked CMOS image sensor with advanced Cu₂Cu hybrid bonding, *IEDM*, 2016

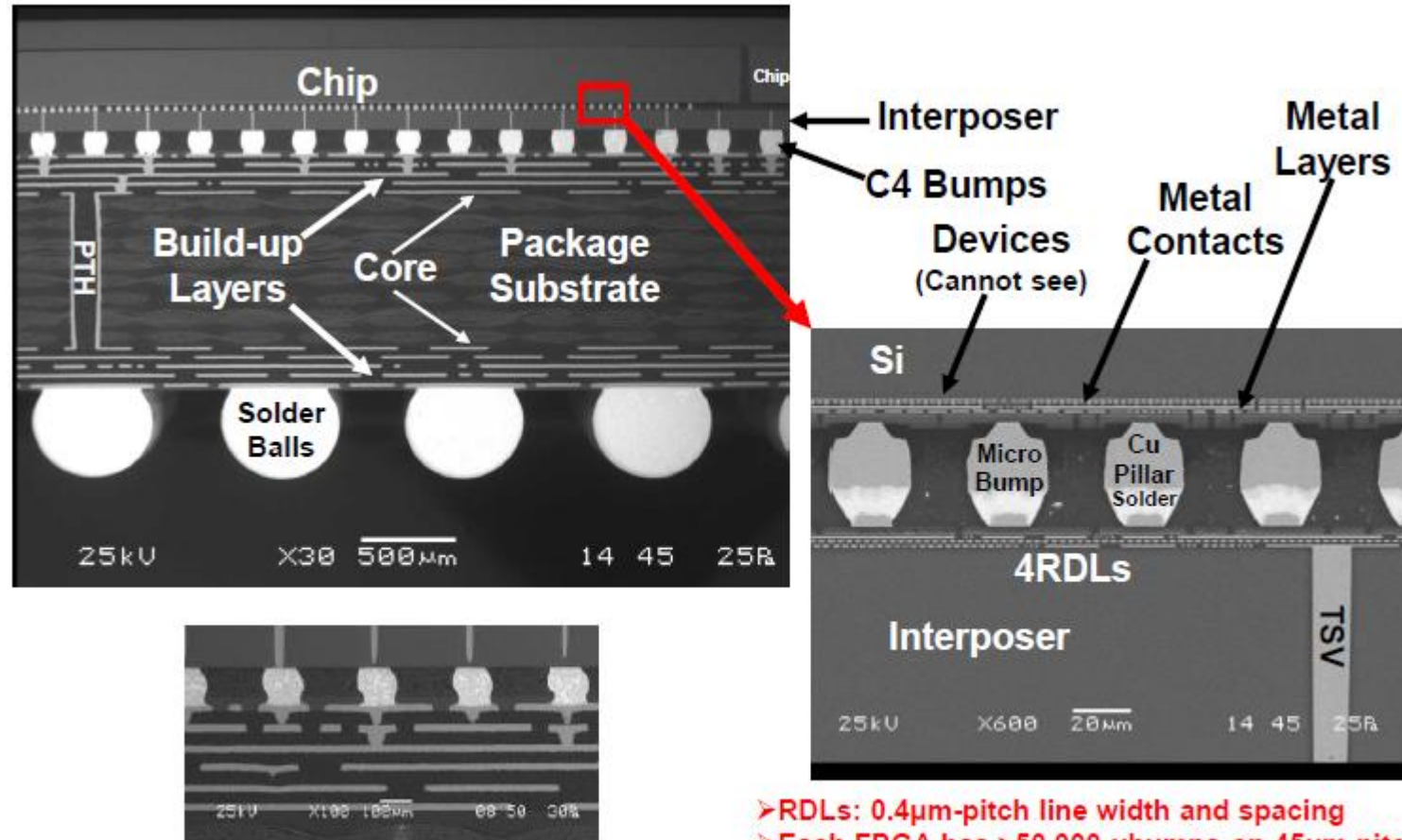
■ Cu-Cu直接键合/混合键合实现高密度的互连



■ Xilinx Virtex7 FPGA中的2.5D集成 (2015年)

● Stacked Silicon Interconnect





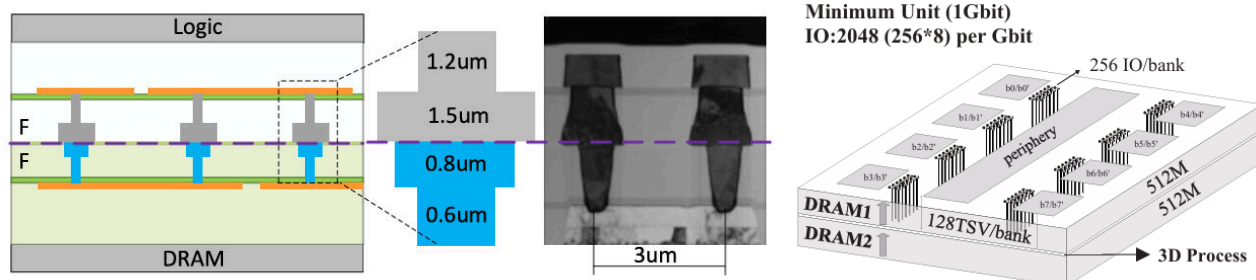
The package substrate is at least (5-2-5)

- RDLs: 0.4µm-pitch line width and spacing
- Each FPGA has >50,000 µbumps on 45µm pitch
- Interposer is supporting >200,000 µbumps

■ 武汉新芯SeDRAM (2022&2023)

实现了Logic和DRAM以及双层DRAM的HB堆叠

DRAM+SoC: $3\mu\text{m}$ pitch $1.1 \times 10^5/\text{mm}^2$



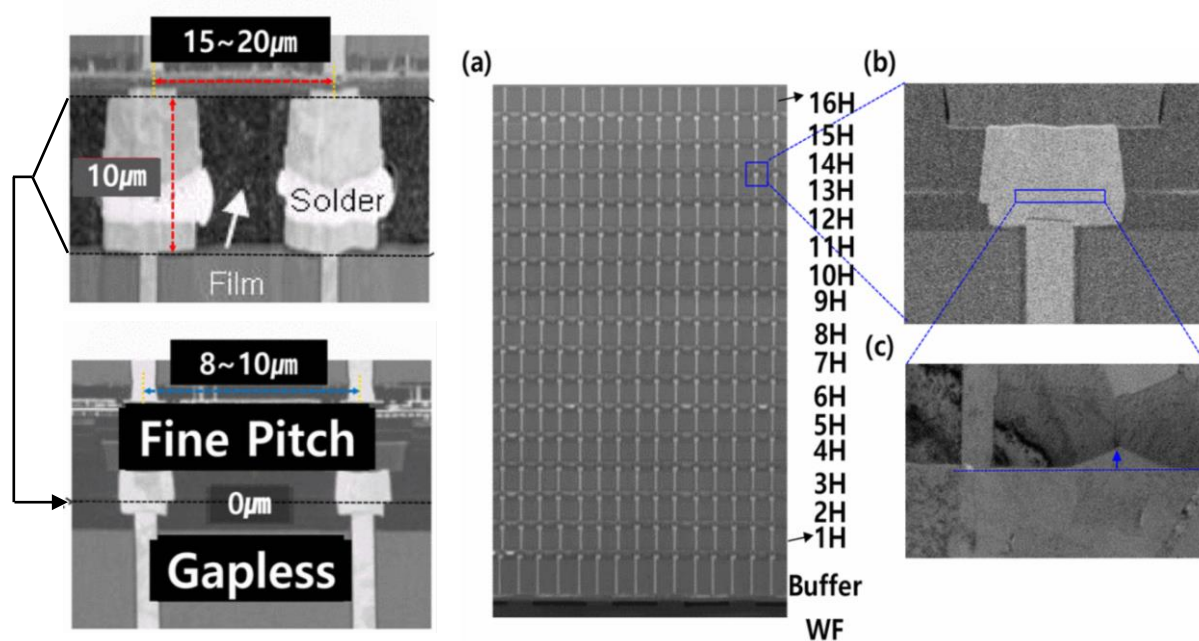
Typical 3D scheme performance benchmark

Technology	3D-IC F2F	SoIC F2F	HB F2F
Bump density	1.0x	16.0x	160.0x
Speed (1/wire delay)	1.0x	11.9x	59.5x
Bandwidth density (bump density $\times$ speed)	1.0x	19.1x	955.0x
Power efficiency (energy/bit)	1.0x	0.05x	0.02x

■ Samsung HBM4 (ECTC2023)

采用D2W混合键合工艺, 实现16层DRAM堆叠

实现了更薄的封装堆叠和更高密度的垂直互连。

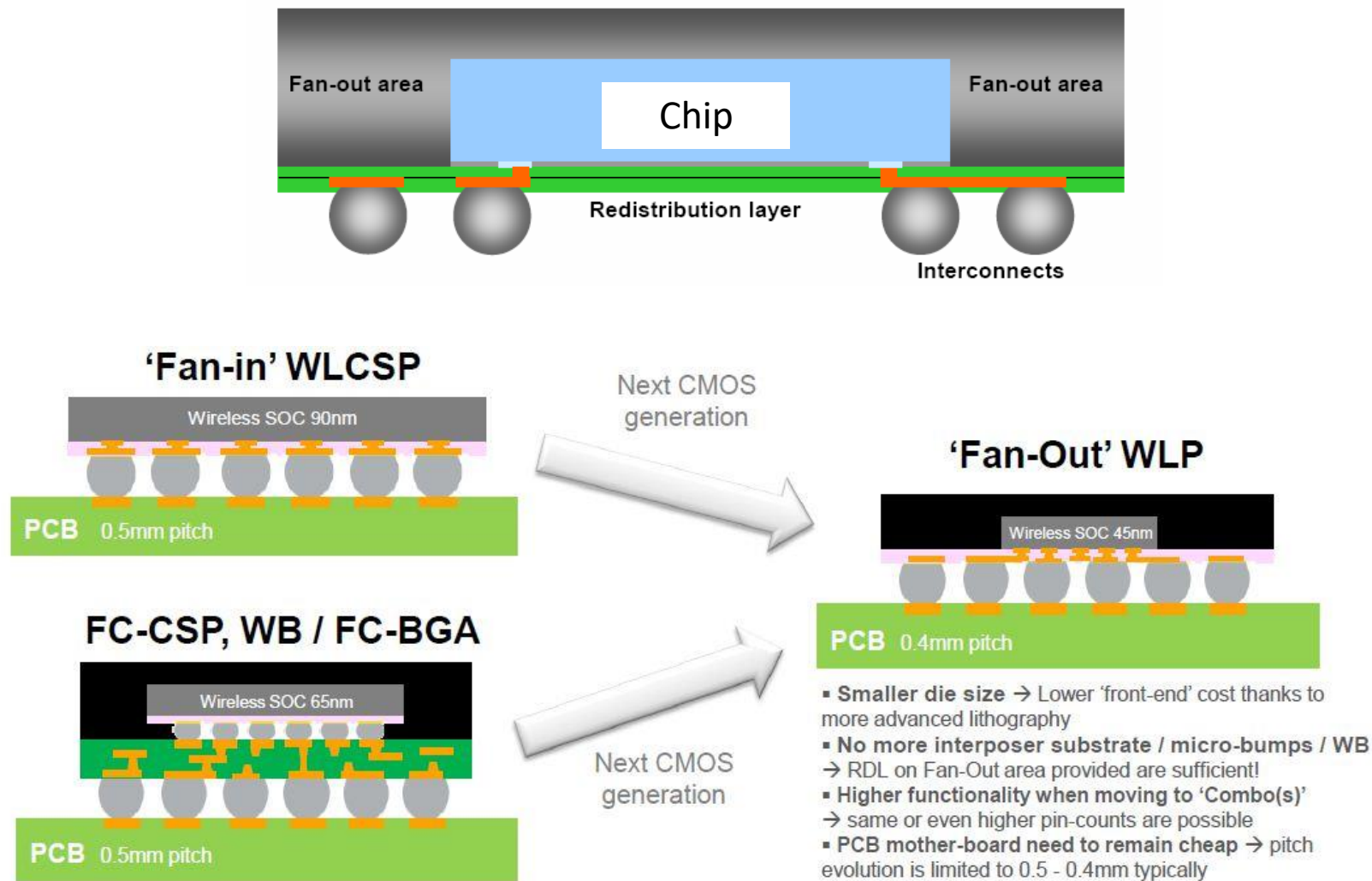


Fan-out WLP



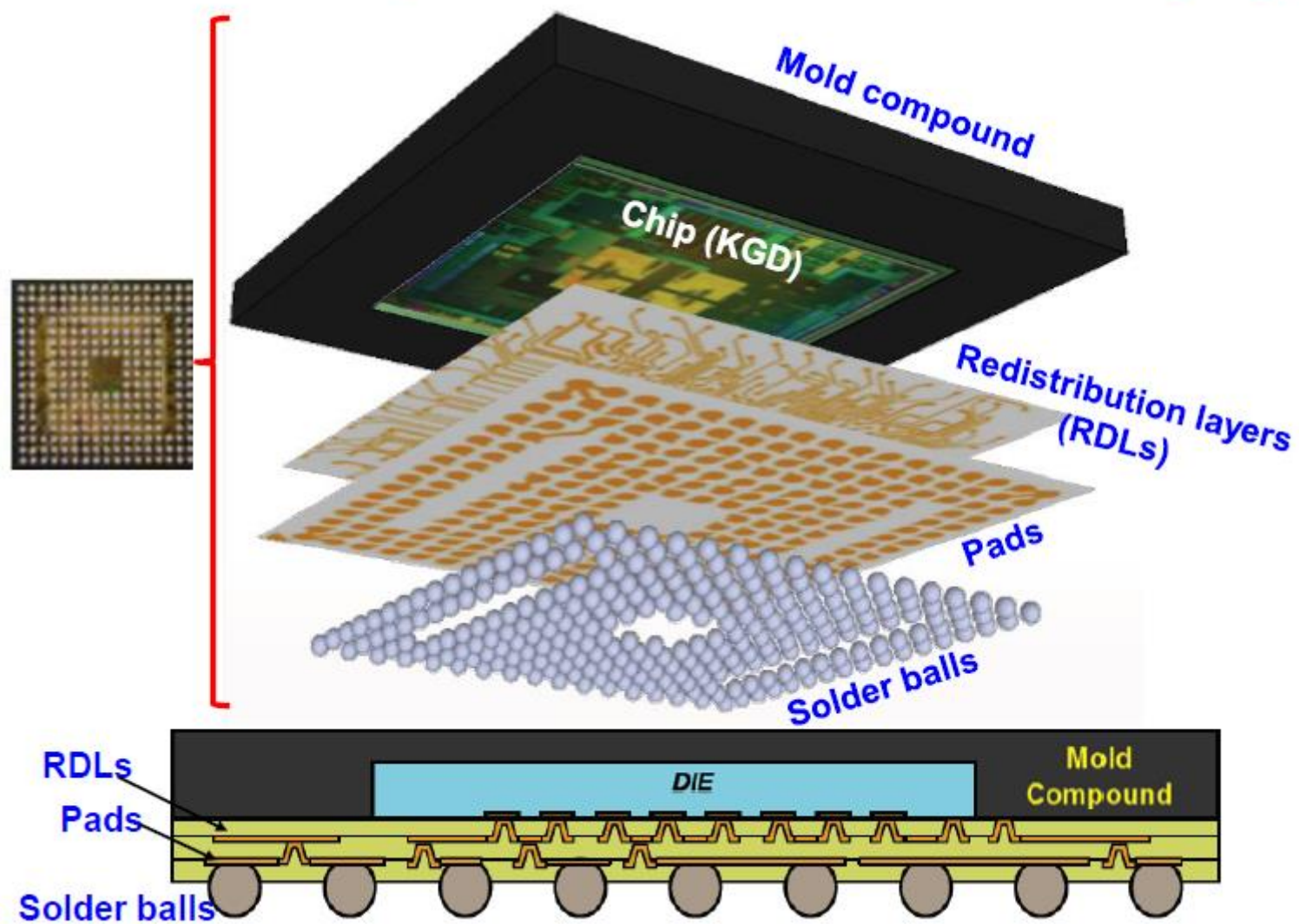
- 成本
- 外形
- 性能优化
- 功能拓展
- 兼容工艺
- 可靠性

■

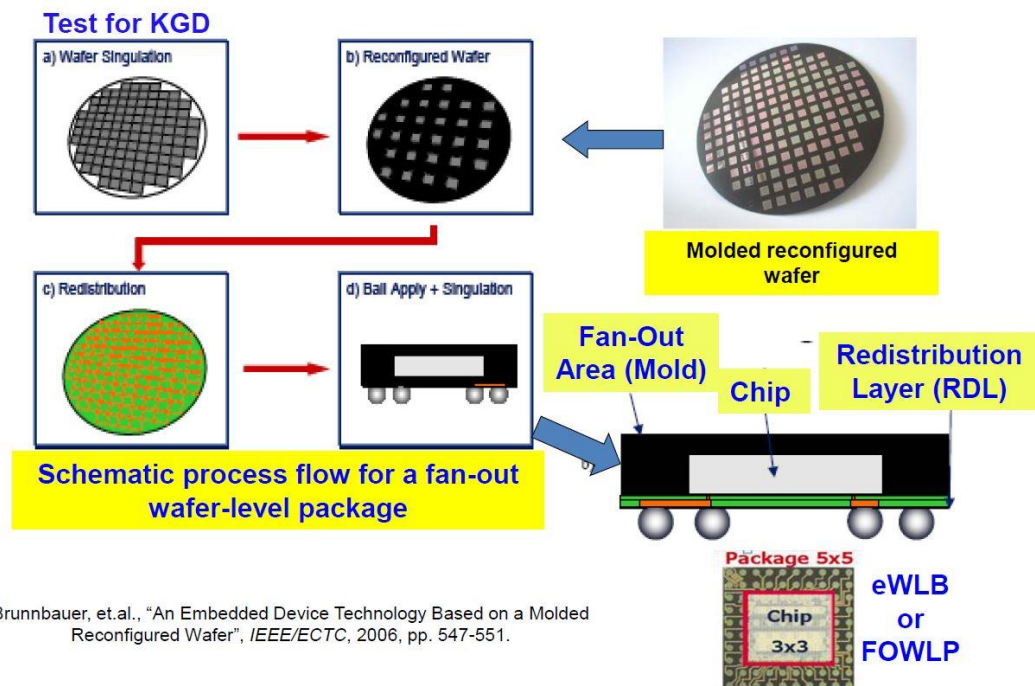


- Smaller die size → Lower 'front-end' cost thanks to more advanced lithography
- No more interposer substrate / micro-bumps / WB → RDL on Fan-Out area provided are sufficient!
- Higher functionality when moving to 'Combo(s)' → same or even higher pin-counts are possible
- PCB mother-board need to remain cheap → pitch evolution is limited to 0.5 - 0.4mm typically

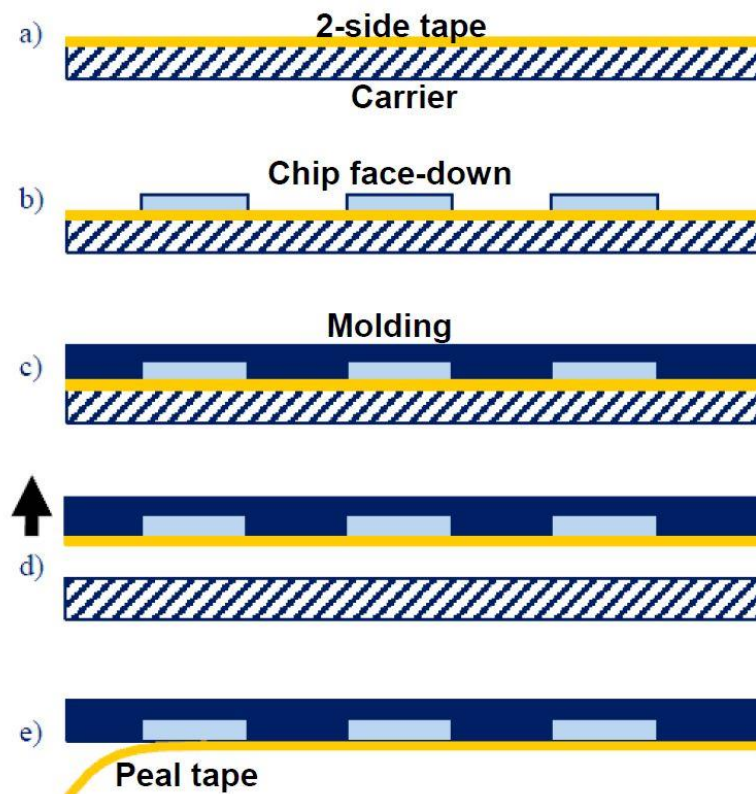
来源: Semiconductor Advanced Packaging. John Lau



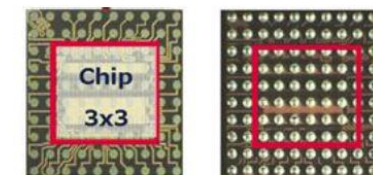
■ Infineon公司的eWLB (Embedded Wafer-Level Ball Grid Array)



Brunnbauer, et.al., "An Embedded Device Technology Based on a Molded Reconfigured Wafer", *IEEE/ECTC*, 2006, pp. 547-551.

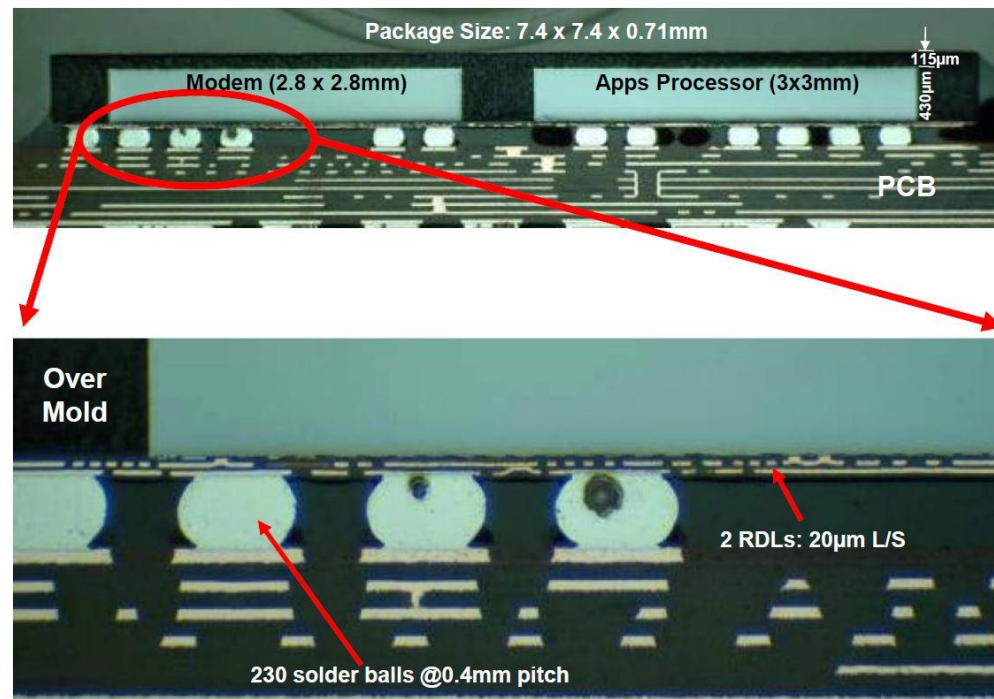


Infineon picked a modified, commercially available tape, which is equipped with a **thermo-release layer**. It is losing its adhesive properties once it is heated above a specific temperature, which is higher than any processing temperature before.



➤ eWLB的应用

- Baseband, RF switch/transceiver, PMIC, Audio codec, MCU, RF radar, Connectivity ICs



HTC Desire 606W (SPREADTRUM SC8502), 2013年

➤ TSMC为A10开发InFO(Integrated Fan-Out) WLP

● Chip-Frist (Die-Up)工艺

US 9,000,584 B2 (Publication Date: April 7, 2015)

PACKAGED SEMICONDUCTOR DEVICE WITH A MOLDING COMPOUND AND A METHOD OF FORMING THE SAME

Jing-Cheng Lin, Hsinchu County (TW); Jui-Pin Hung, Hsinchu (TW); Nai-Wei Liu, Fengshan (TW); Yi-Chao Mao, Zhongli (TW); Wan-Ting Shih, Touwu Township (TW); and Tsan-Hua Tung, Hsinchu (TW)

Assigned to Taiwan Semiconductor Manufacturing Company, Ltd., (TW)

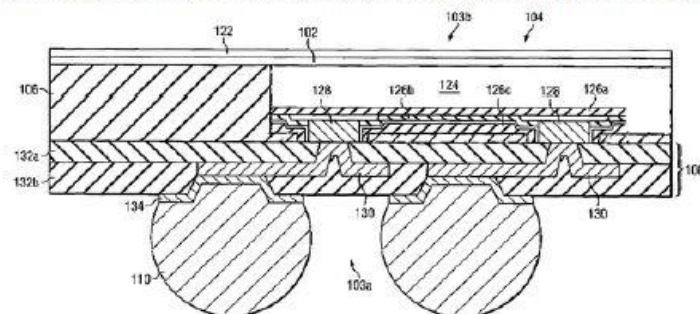
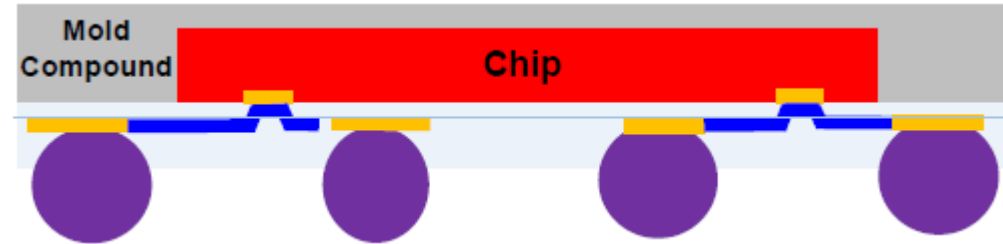


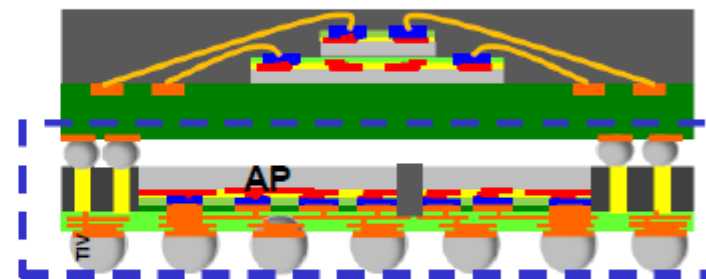
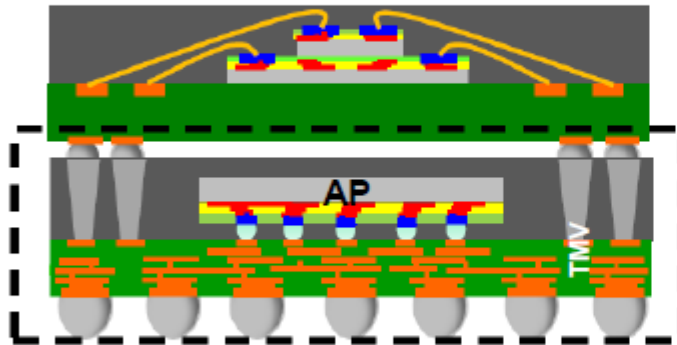
FIG. 1I also shows a more detailed view of the die 104 and the wiring layer 108, in accordance with some embodiments. The view of the die 104 and wiring layer 108 are exemplary; alternatively, the die 104 and wiring layer 108 may comprise other configurations, layouts and/or designs. In the embodiment shown, the die 104 includes a substrate 124 comprising silicon or other semiconductive materials. Insulating layers 126a and 126b may comprise passivation layers disposed on the substrate 124. Contact pads 128 of the die 104 may be formed over conductive features of the substrate such as metal pads 127, plugs, vias, or conductive lines to make electrical contact with electrical components of the substrate 124, which are not shown.



Conventional PoP for
Application Processor (AP)
chipset

TSMC's PoP for
Application Processor (AP)
chipset

A10

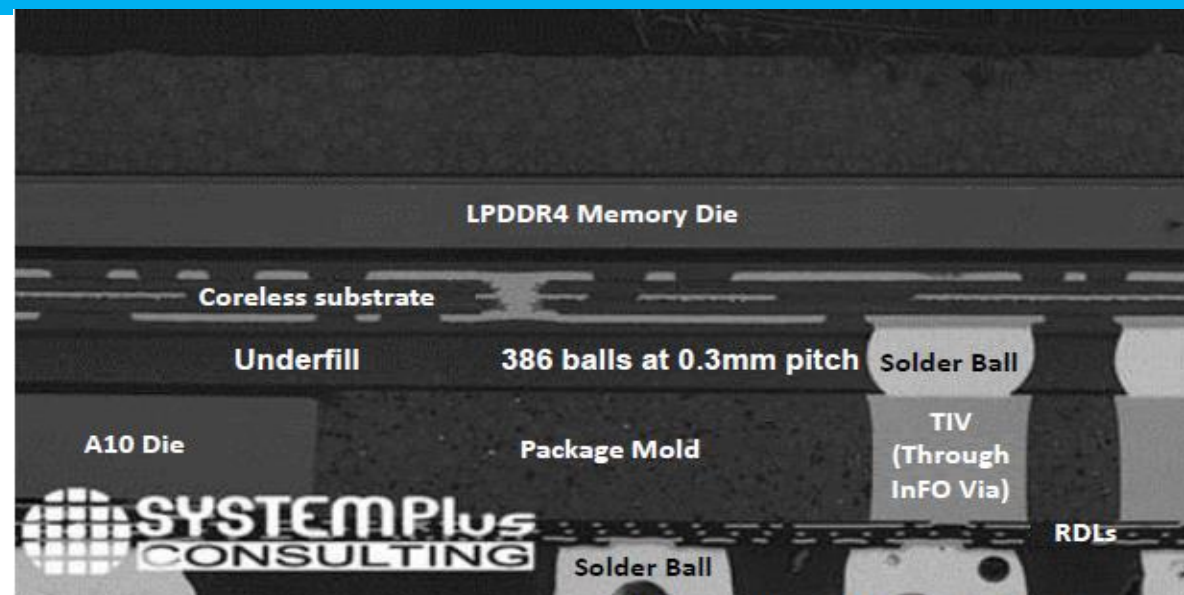


Eliminated wafer bumping, solder
reflow, flux cleaning, underfilling,
and package substrate.
Lower Profile!

Fan-out + Package-on-Package



PoP sizes: 15.5mm x 14.4mm x 825 μ m



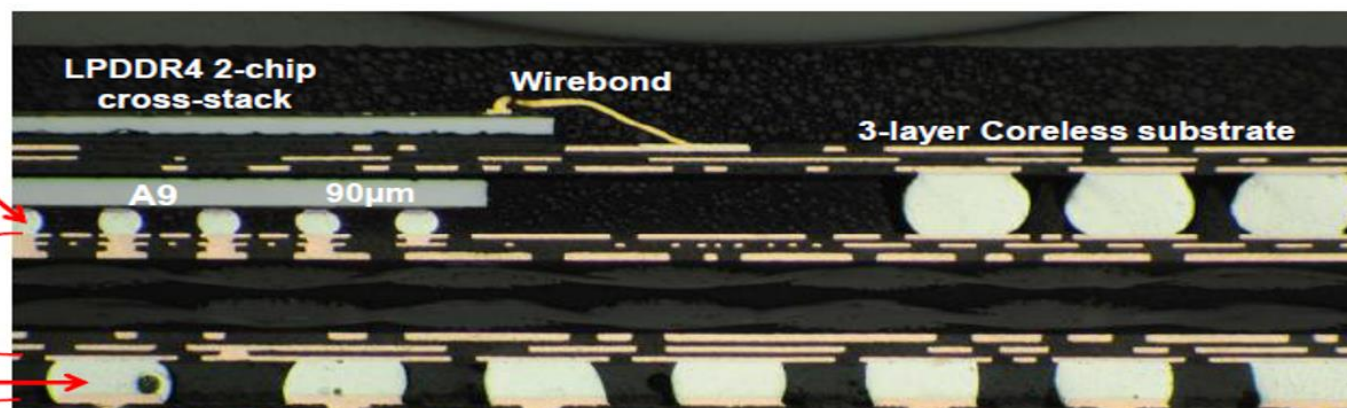
~1300 solder balls at 0.4mm pitch



150 μ m
pitch
staggered
C4 bumps

2-2-2 build-up
substrate

0.4mm pitch
solder balls

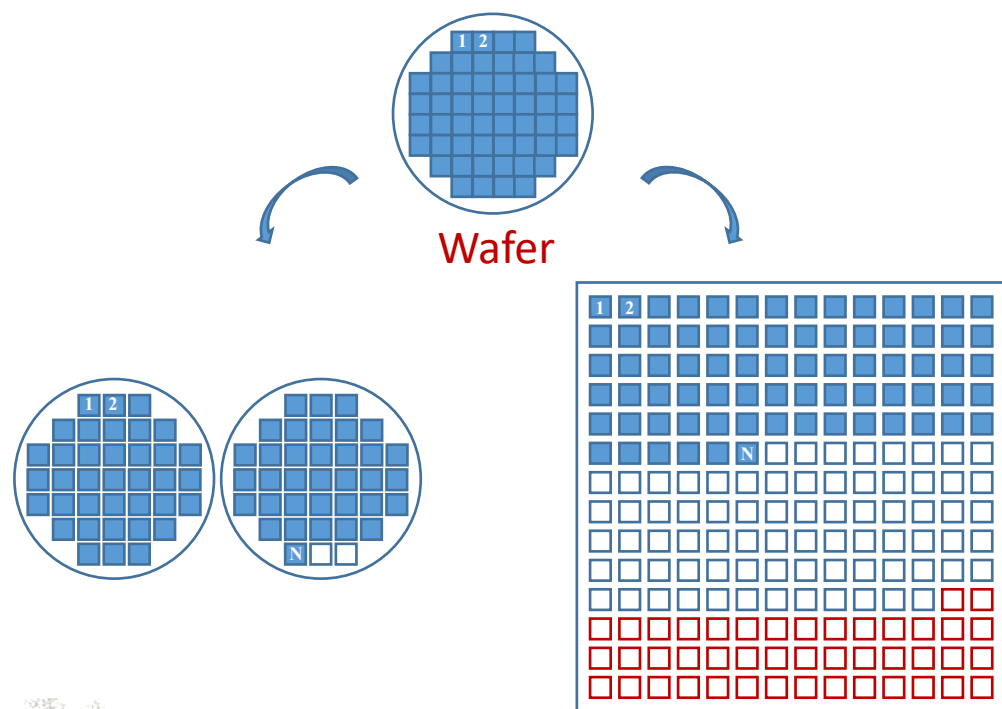


Fan-out PLP



■ 扇出面板级封装 (Fan-out Panel Level Package, FOPLP)

- ◆ 利用**面板代替晶圆**进行**芯片重排**，可以**放置更多的芯片**，成本也比FOWLP便宜。具体推算从200mm过渡到300mm大约能节省**25%**的成本，从300mm过渡到板级，则能节约**66%**的成本。



Fan Out Wafer Level Package (FOWLP)

300mm晶圆

芯片占用面积: 64%

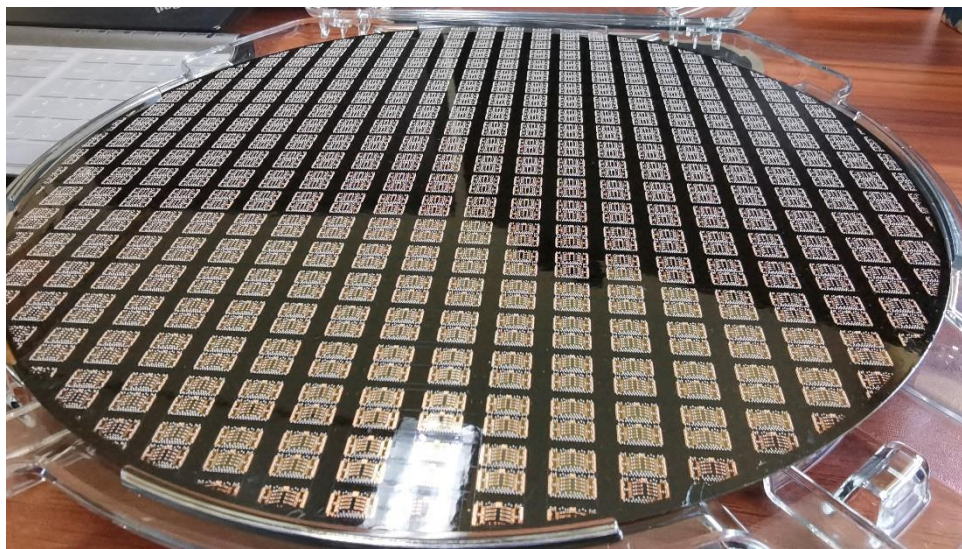
Fan Out Panel Level Package (FOPLP)

515x500mm面板

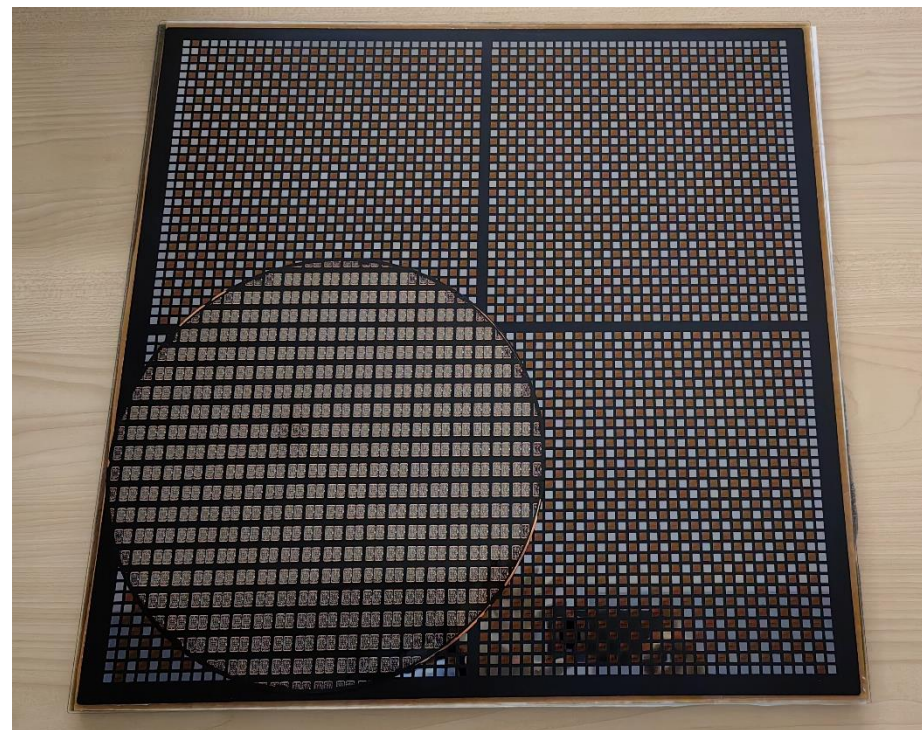
芯片占用面积: 93%



■ 多芯片集成的扇外型封装和板级扇外型封装



1CMOS+4GaAs的扇外型封装，可实现在相控阵
雷达、5G通信等的应用
工作频率可达40GHz，产品布线层数2层



玻璃封装基板



■ 2023年9月，Intel宣布未来投入研发玻璃基板以支持高性能处理器

Motivation for Glass Core Substrates

Organic Substrate

Organic substrates leverage traditional PCB-like cores with woven glass laminates

- Provides a low cost, easily manufacturable material set with off the shelf laminates available from leading suppliers

Glass Core Substrate

Glass core substrate enable significant improvement to both electrical and mechanical properties

- Tunable Modulus and CTE closer to silicon → Large form factor enabling
- Dimensional stability → Improved feature scaling
- High (~10x) through-hole density → improved routing and signaling
- Low Loss → High speed signaling
- Higher Temperature capability → Advanced Integrated Power Delivery

Glass Core has similar properties as Si → Dimensional stability and ability to scale

*Based on internal analysis of Intel products and projections of future Intel products.

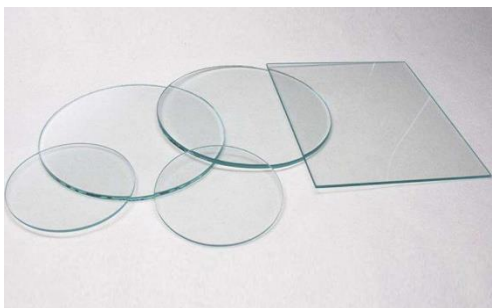
intel 6

Characteristic	Material				
	Ideal Properties	Silicon	Organic	Glass	
Electrical	• High resistivity				Excellent Fair Difficult
	• Low loss				
	• Electric Isolation				
	• Low RF dielectric constant				
Physical	• Smooth surface finish				
	• Hermeticity				
	• Optical transparency				
	• Thickness				
	• Embedded Components				
Thermal	• Large area availability				
	• Ultra thin				
Mechanical	• High strength				
	• High modulus				
Chemical	• Resistance to process chemicals				
Processability	• Ease of Via formation and metallization				
	• Via pitch				
	• Via filling materials				
	• L/S				
Reliability	• CTE matched to Si and PWB				
Cost	• Cost per I/O at 25µm pitch				

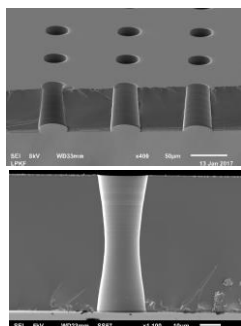
◆ 玻璃基板相比传统有机基板具有**散热能力强、信号完整性好和理论互连密度高**等优势，还可以**集成光学器件**。

■ 玻璃（基材）在先进封装中的应用

玻璃

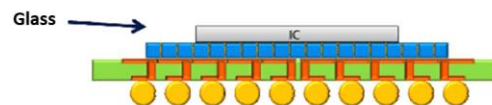


玻璃通孔
TGV

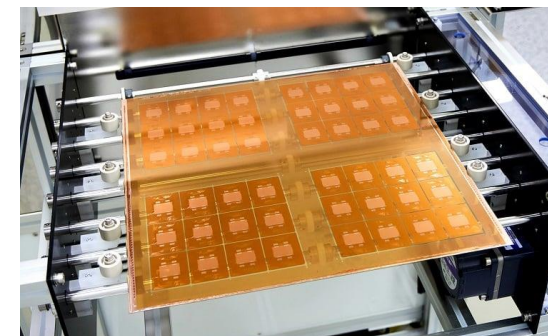
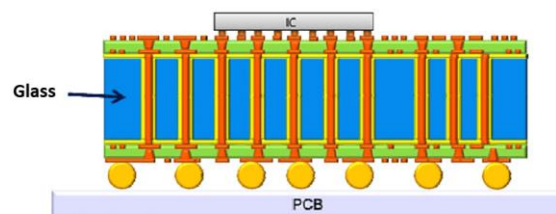


Glass-Two Approaches

Option A: Glass as an Interposer

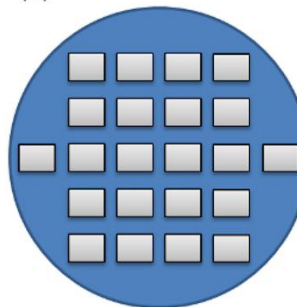


Option B: Glass as a Package Substrate

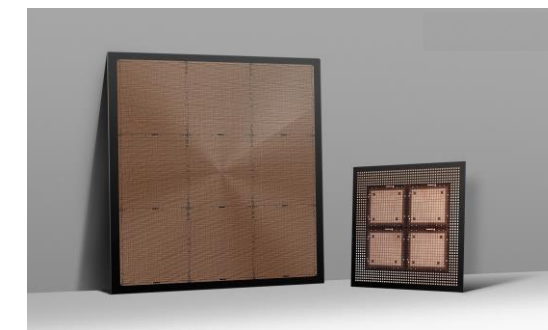
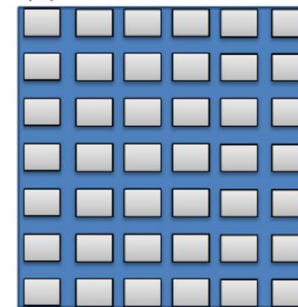


玻璃转接板或玻璃基板

(a)



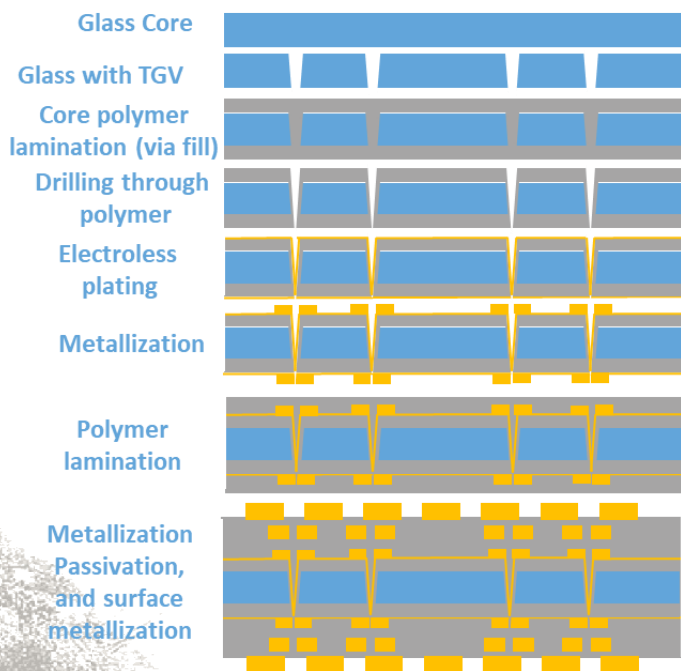
(b)



扇外型封装的临时承载玻璃面板

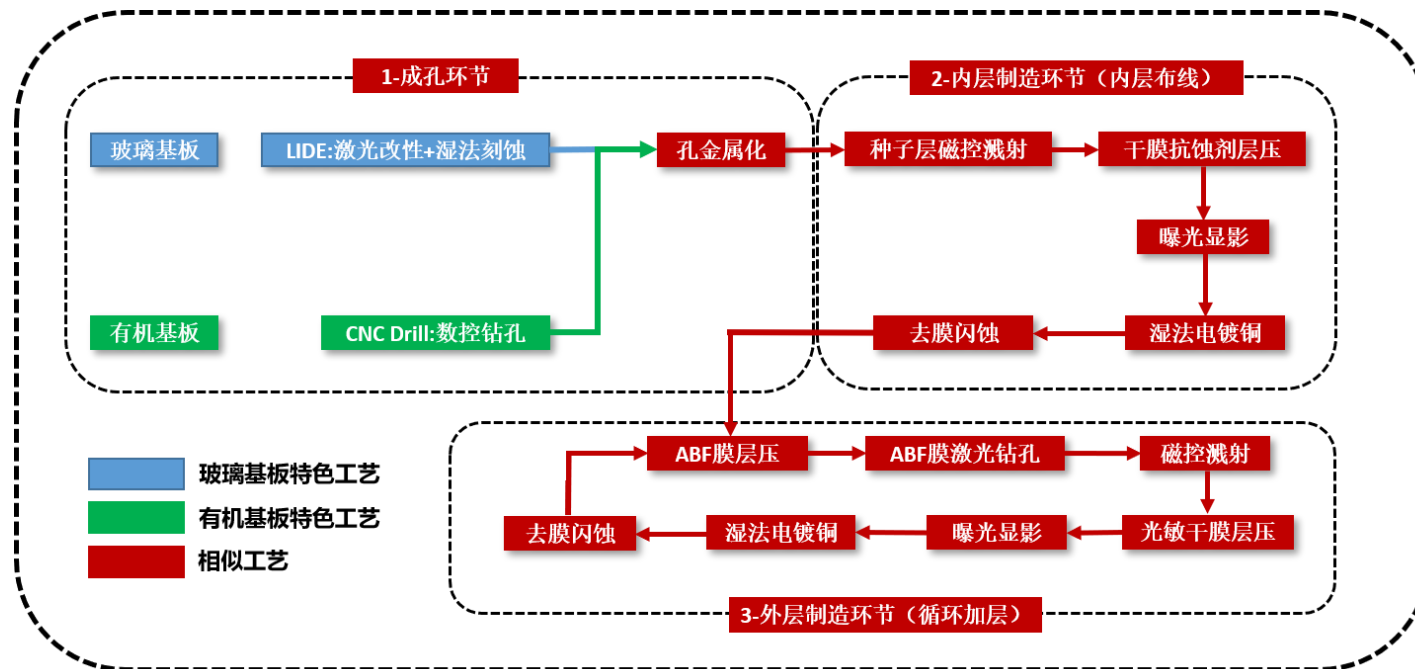
■ 玻璃基板制造：使用与有机基板相同的SAP工艺

- ◆ 实际制造流程：预先对玻璃进行玻璃通孔，通过真空层压技术将干膜层压到玻璃基板上，随后通过UV或准分子激光进行图案化；此外还需对玻璃基板进行种子层及导线的电镀、PVD、CMP等工艺。



玻璃基板工艺流程概览

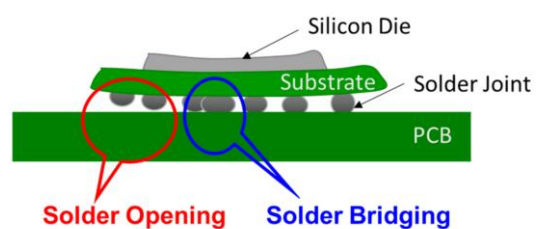
Sources: Yole Development



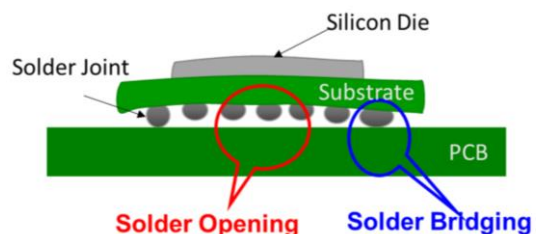
玻璃基板与FC-BGA基板工艺流程对比

Sources: 半导体材料与工艺, 中金公司研究部

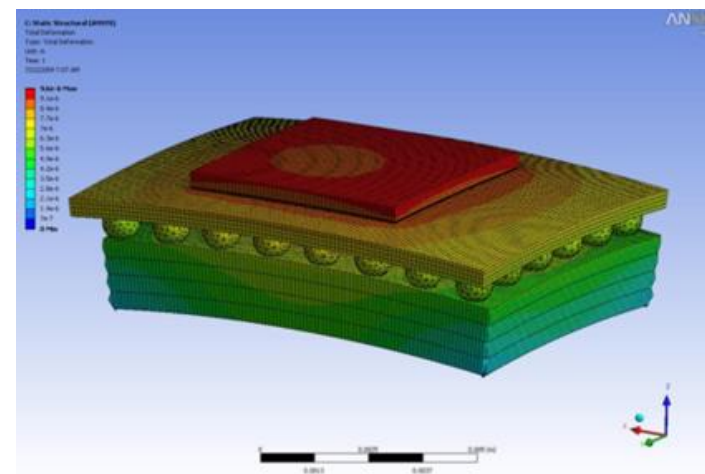
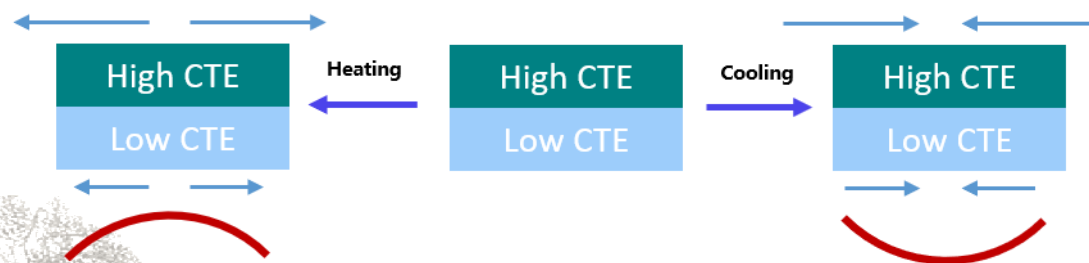
■ 热膨胀系数(CTE)匹配



I. Concave Warpage of BGA Package



II. Convex Warpage of BGA Package

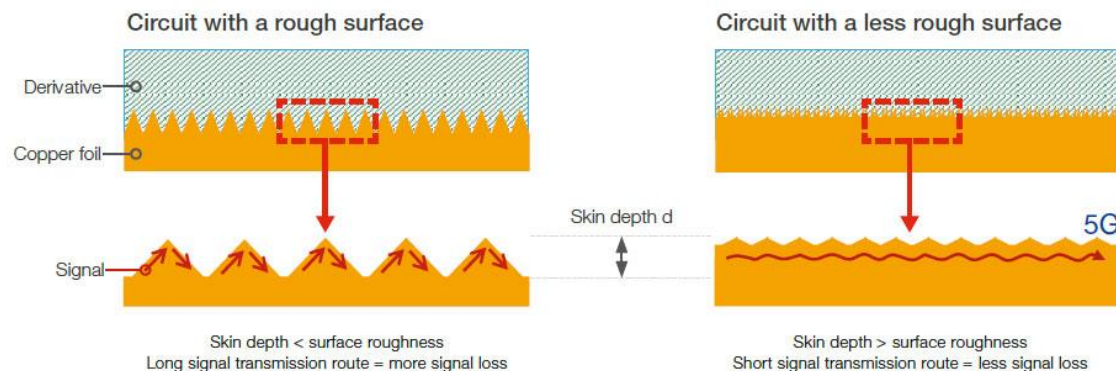
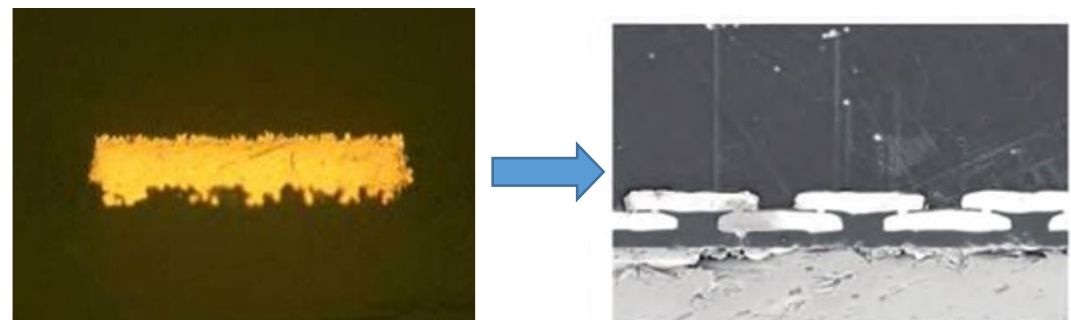
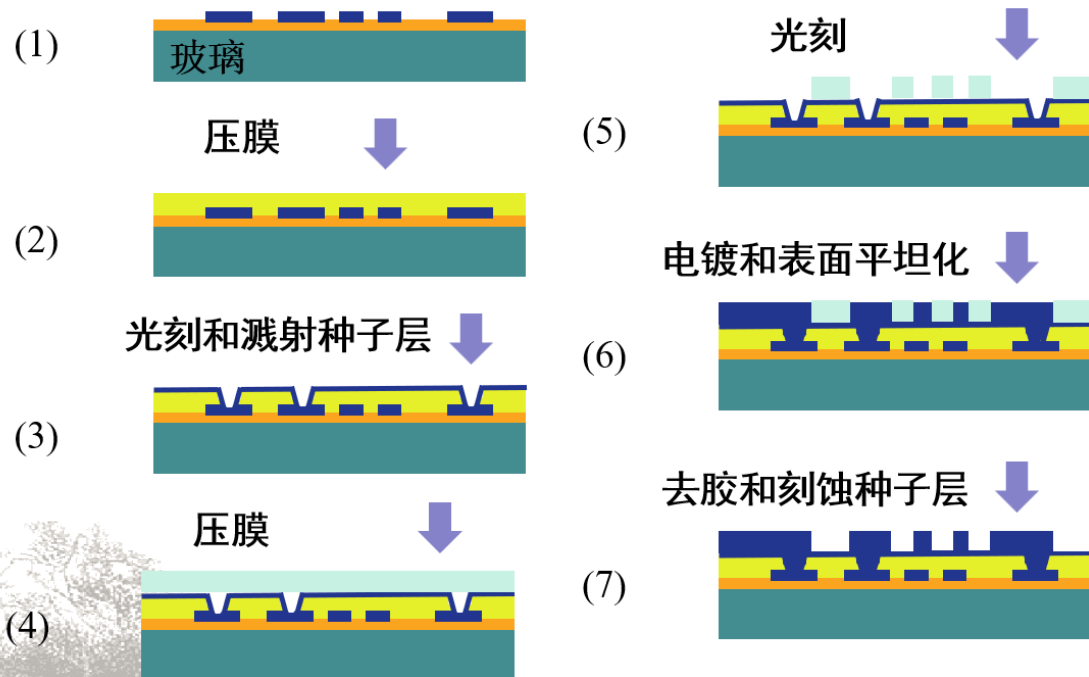


材料	FR-4	玻璃	硅
热膨胀系数 (ppm/K)	18	5~10	2.5
热导率 (W/m*K)	0.3	1.4	130

◆ 玻璃与硅的CTE更加接近且可调控，散热能力更佳；可以有效抑制翘曲，适合大尺寸芯片或多颗芯粒集成。

■ 高密度布线是玻璃基板的优势

◆ 相对于有机材料衬底而言，**玻璃表面的粗糙度小**，所以在玻璃上可以进行**高密度布线**。



◆ 更小的边缘粗糙度可**减少导线趋肤效应**。

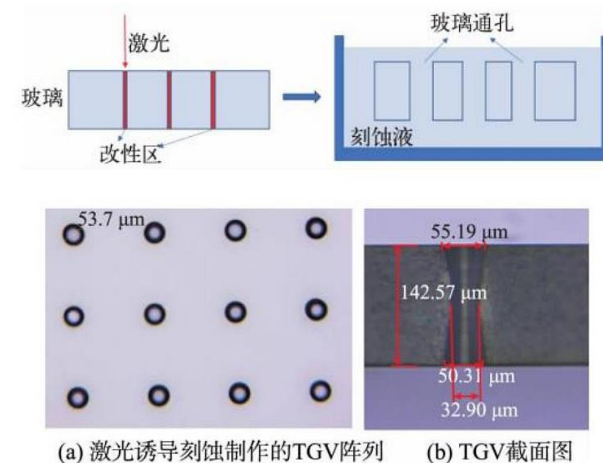
■ TGV是目前玻璃基板的技术瓶颈之一

难点1：刻蚀技术

- ◆ 玻璃材料质地较脆，在通孔刻蚀过程中容易产生裂纹。
- ◆ 难以制作高深宽比的通孔

- ◆ 通常采用激光诱导刻蚀法（LIDE）

玻璃通孔制备方式	加工品质	生产效率	生产成本
喷砂法	孔径大、节距大、表面质量较差	中：双面喷砂	高：需用到掩膜板、研磨
聚焦放电法	密度高、深宽比高、垂直度差	低：单孔制作	中：设备维护成本
等离子刻蚀法	侧壁粗糙度低、损伤小	中：工艺复杂，刻蚀速率低	高：需用到掩膜板、光刻胶
激光烧蚀法	粗糙度大、存在侧裂纹	低：单孔作业	中：激光设备维护成本
电化学放电法	孔径大、节距大、垂直度差	中：单孔成型快	低：设备简单
光敏玻璃法	密度高、深宽比高、侧壁粗糙度好、垂直度好	高：工艺简单，整片作业	中：需用到掩膜板
激光诱导刻蚀法	密度高、深宽比高、无损伤、垂直度差	中：单孔成型快	高：设备维护成本



数据来源：《玻璃通孔技术研究进展》（陈力等），广发证券发展研究中心

数据来源：《三维集成封装中光敏玻璃通孔制备工艺研究》（王刚），广发证券发展研究中心

难点2：填孔技术

- ◆ 玻璃通孔的侧壁光滑，与常用金属（如Cu）的粘附性较差，容易造成玻璃衬底与金属层之间的分层现象。
- ◆ 目前研究方向主要包括金属实孔填充和孔内电镀薄层

提纲



■ 问题

- 什么是集成电路封装?
- 什么是先进封装技术?

■ 业界关注的封装技术

- 倒装芯片技术 (Flip Chip)
- 硅通孔互连 (TSV) 三维封装
- 扇出型圆片级封装 (Fan-out WLP)
- 玻璃封装基板 (Glass Substrate)

■ 先进封装案例

■ 先进封装技术发展的讨论

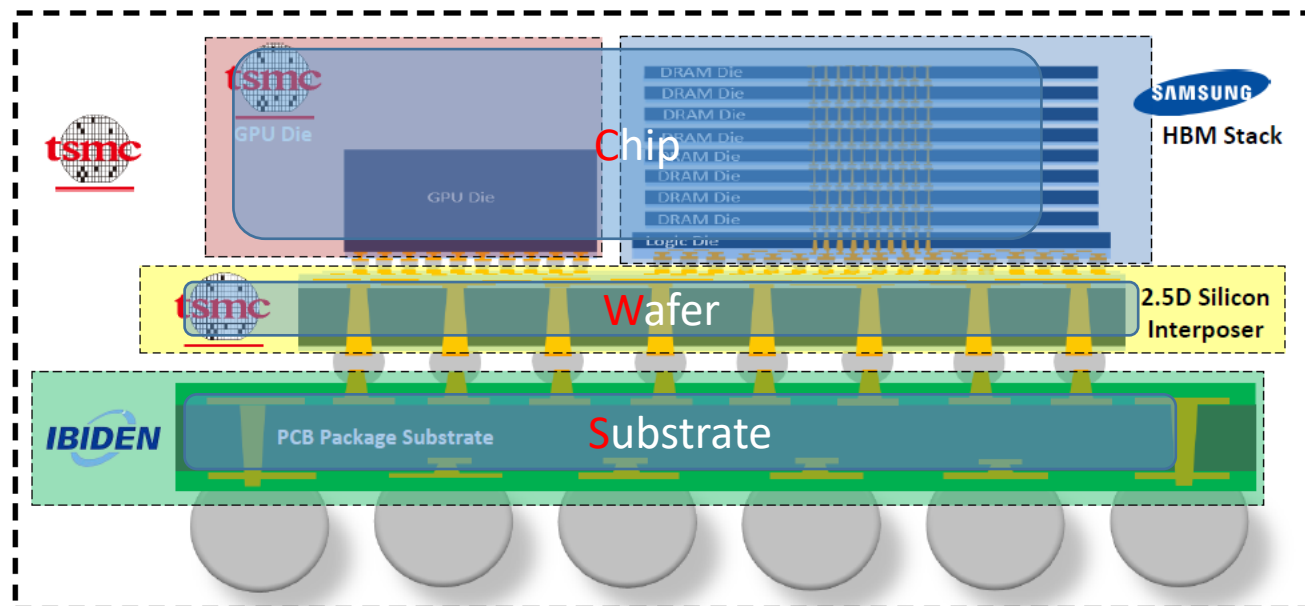
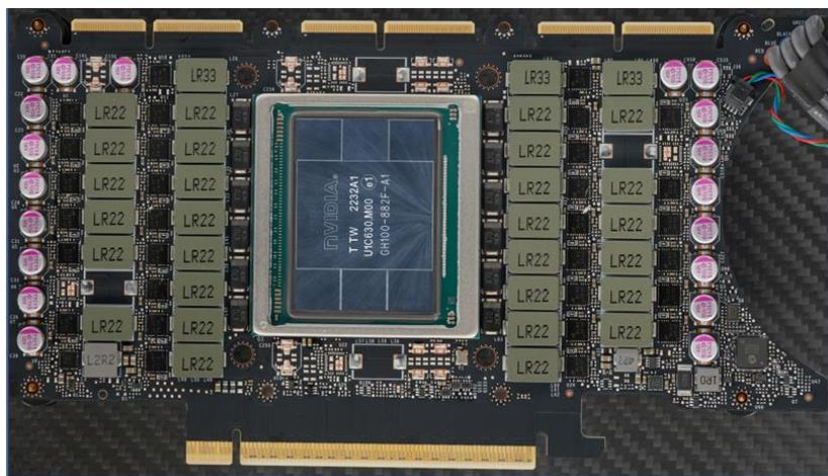


Nvidia H100加速器



■ 服务器GPU，2022年3月上市

- 采用CoWoS实现提升带宽、降低功耗和减小尺寸的目标

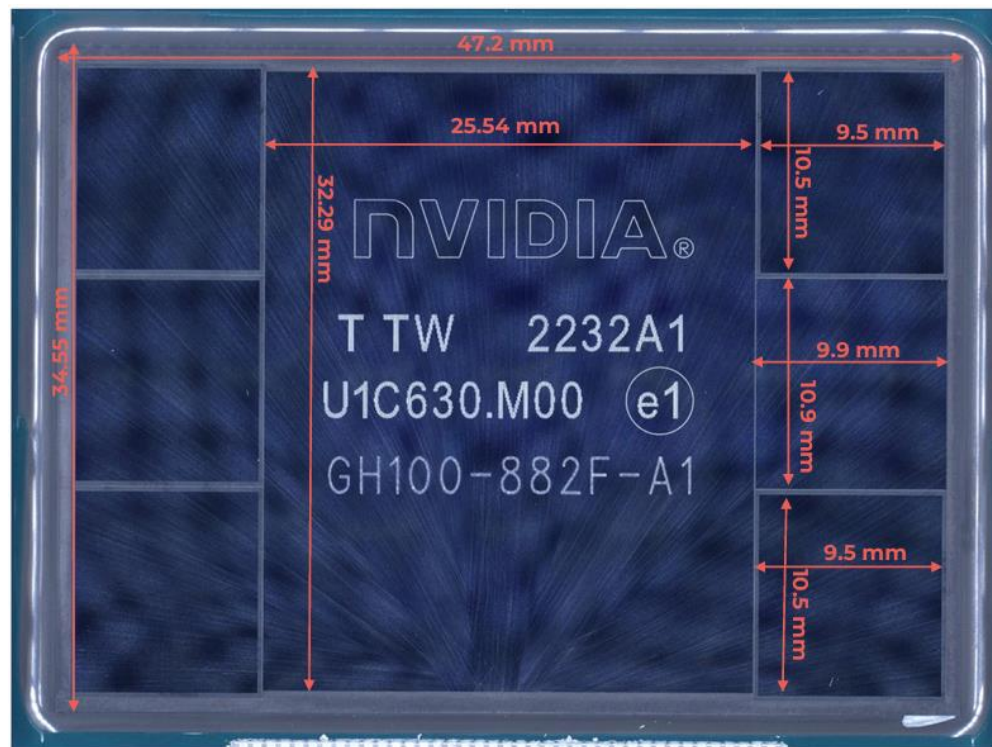
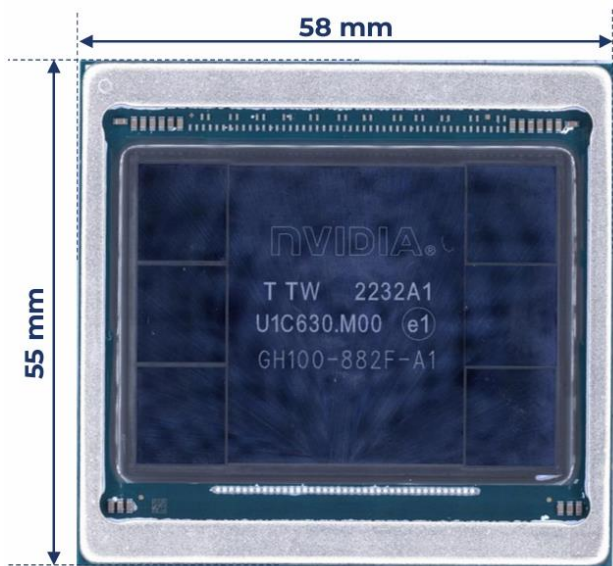


Nvidia H100 Tensor Core GPU, Yole Development

Nvidia H100加速器



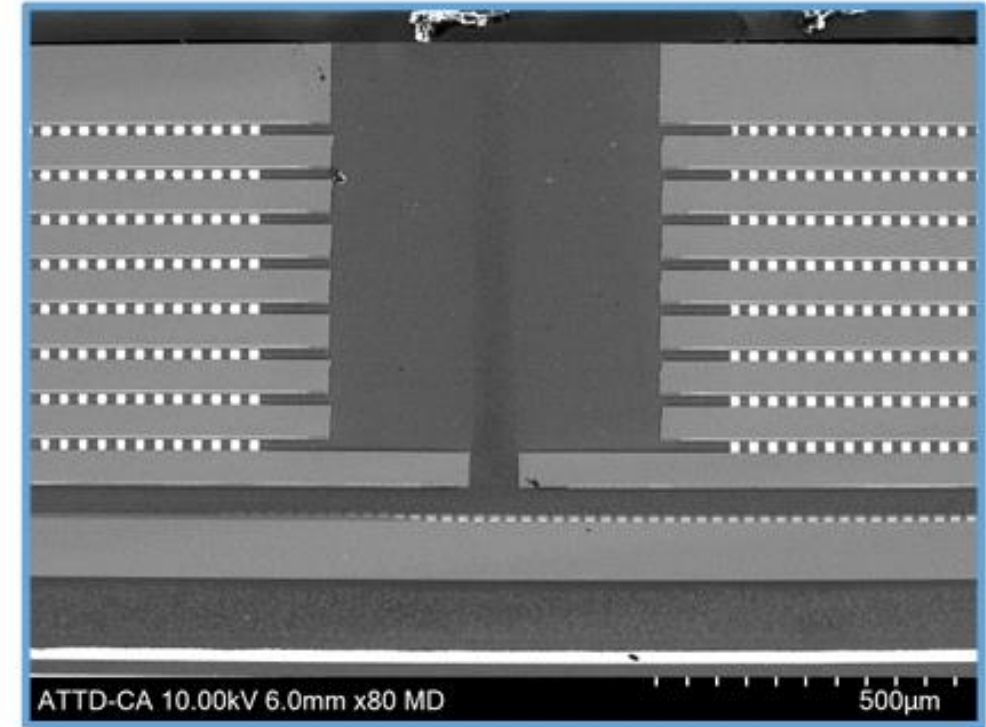
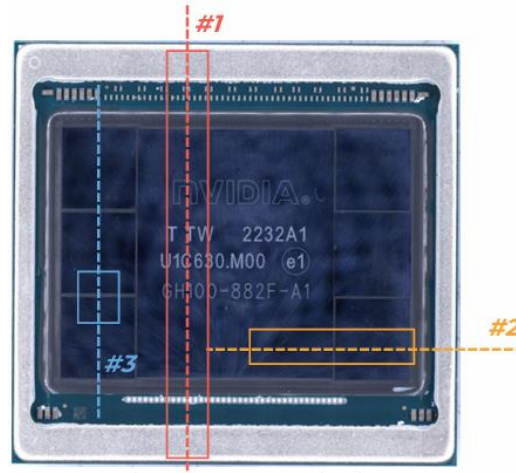
■ GPU芯片尺寸825mm² (25.54mm*32.29mm)



- GPU芯片
 - 825mm² (25.54mm*32.29mm)
- Interposer芯片
 - 1631mm² (47.2mm*34.6mm)
- HBM Logic芯片
 - 109.34mm² (11mm*9.94mm)
- HBM DRAM芯片
 - 99.75mm² (10.5mm*9.5mm)
- Filler芯片
 - 107.91mm² (10.9mm*9.9mm)

Nvidia H100 Tensor Core GPU, Yole Development

Nvidia H100加速器

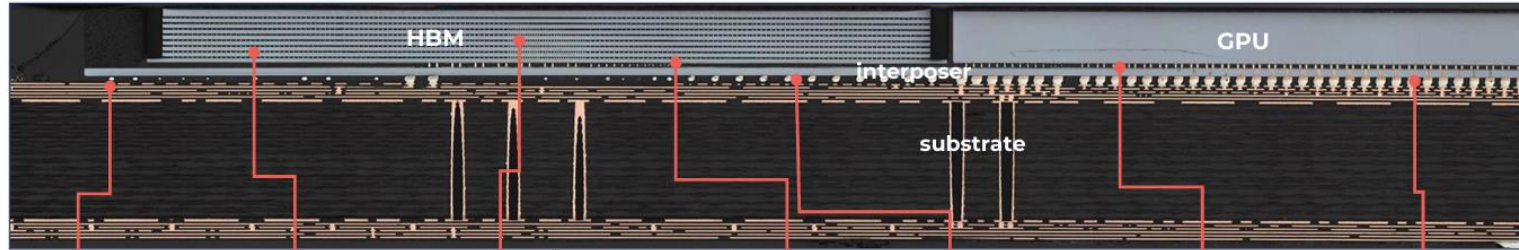


Nvidia H100 Tensor Core GPU, Yole Development

Nvidia H100加速器



■ HBM2e Stack



Package Cross Section 2

Micro-bump
between
**interposer &
substrate edge**
✓ Diameter 150 μm
✓ Pitch 300 μm

Micro-bump
between **DRAM
dies at the edge**
✓ Diameter 14.6 μm
✓ Pitch 60 μm

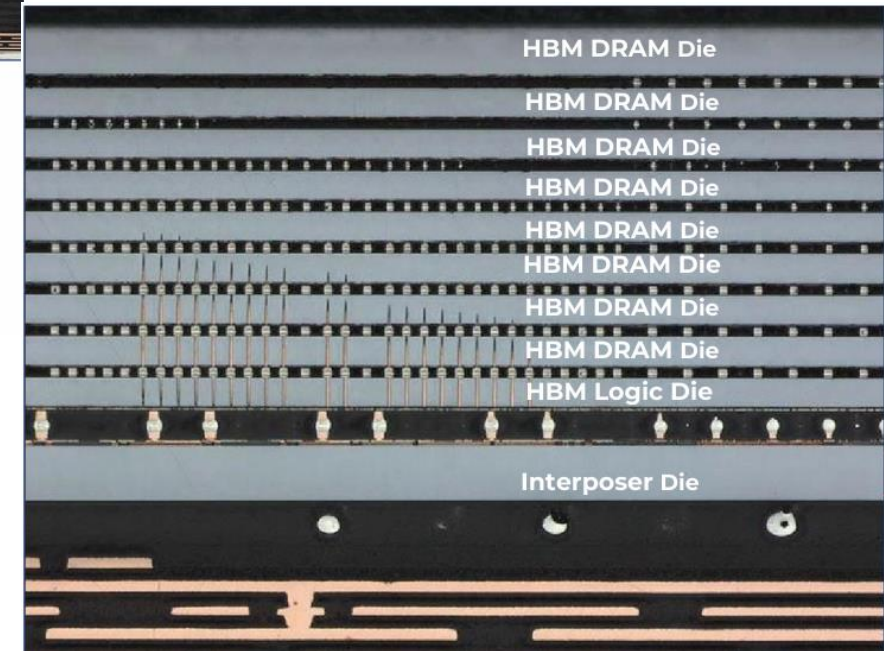
Micro-bump
under **HBM
DRAM TSV**
✓ Diameter 14.6 μm
✓ Pitch 29 μm

Micro-bump
between **logic
die &
interposer**
✓ Diameter 25 μm
✓ Pitch 96 μm

Micro-bump
between
**interposer &
substrate under
HBM**
✓ Diameter 146 μm
✓ Pitch 192 μm

Micro-bump
between **GPU &
interposer**
✓ Diameter 23 μm
✓ Pitch 76 μm

Micro-bump
between
**interposer &
substrate under
GPU**
✓ Diameter 115 μm
✓ Pitch 195 μm

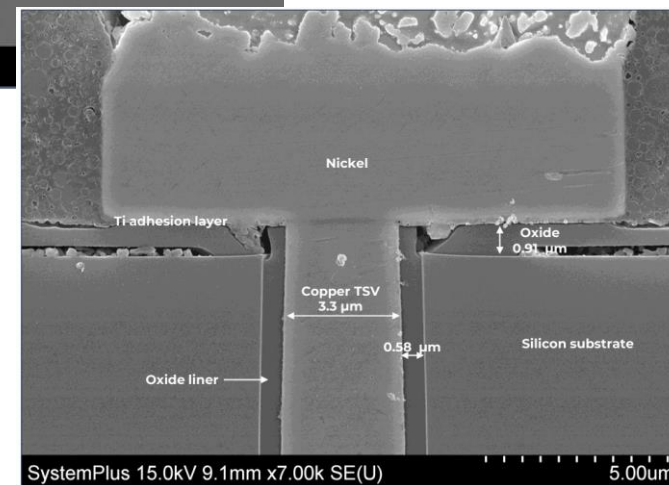
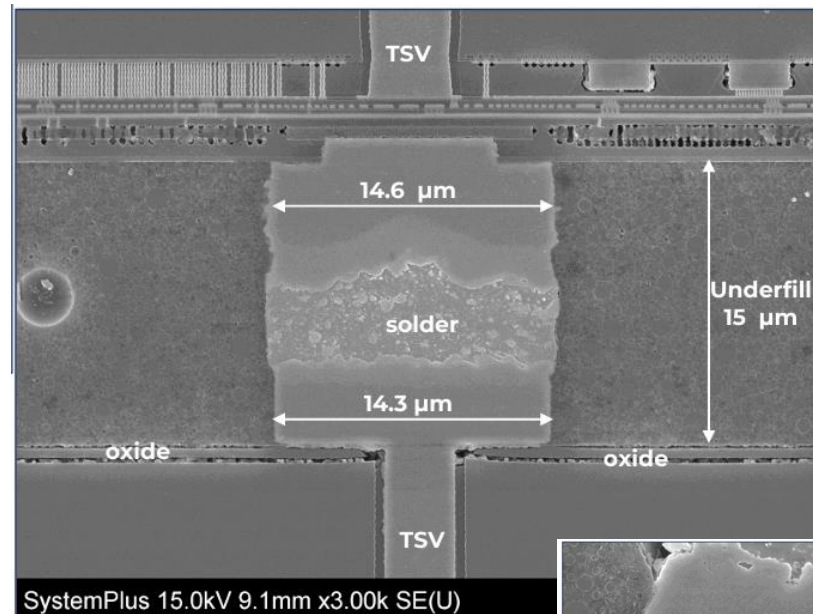
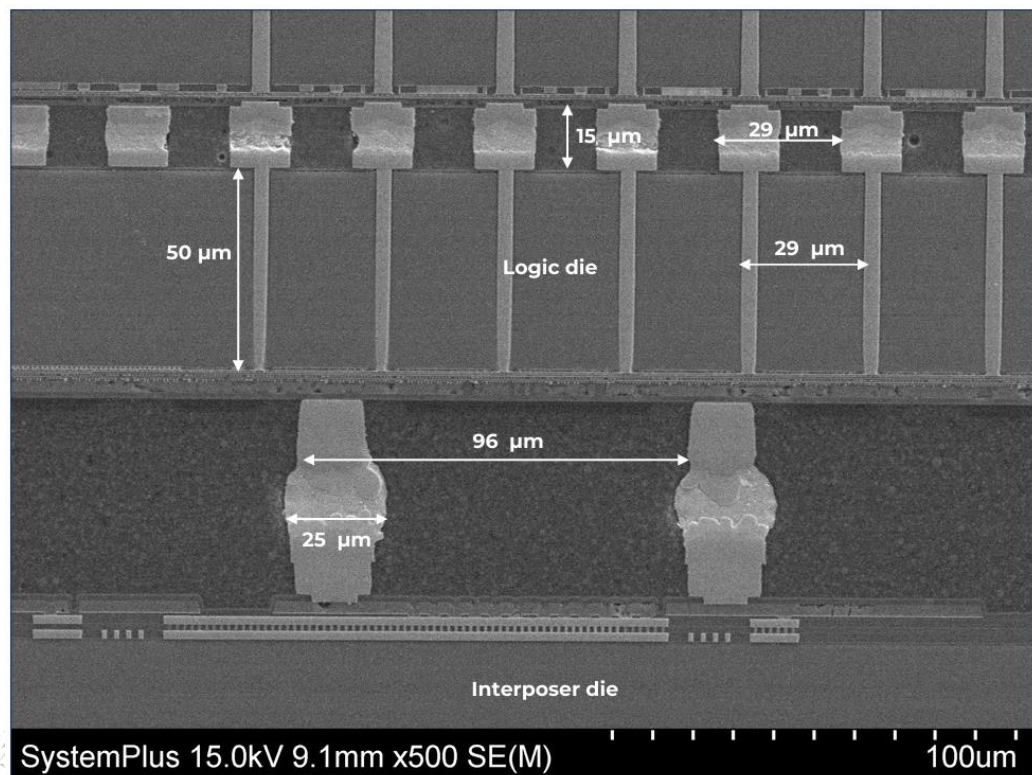


Nvidia H100 Tensor Core GPU, Yole Development

Nvidia H100加速器



■ HBM2e Stack

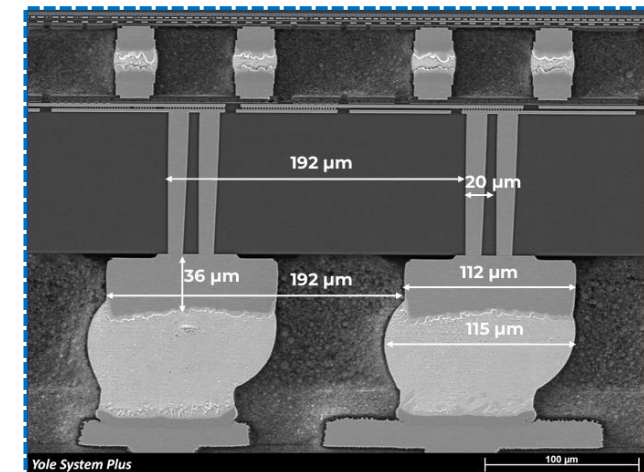
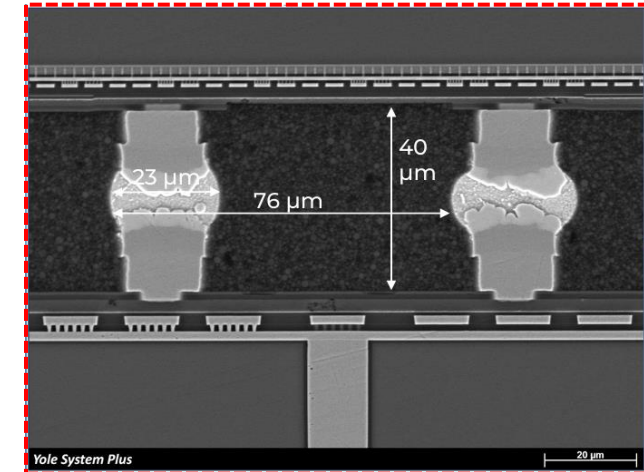
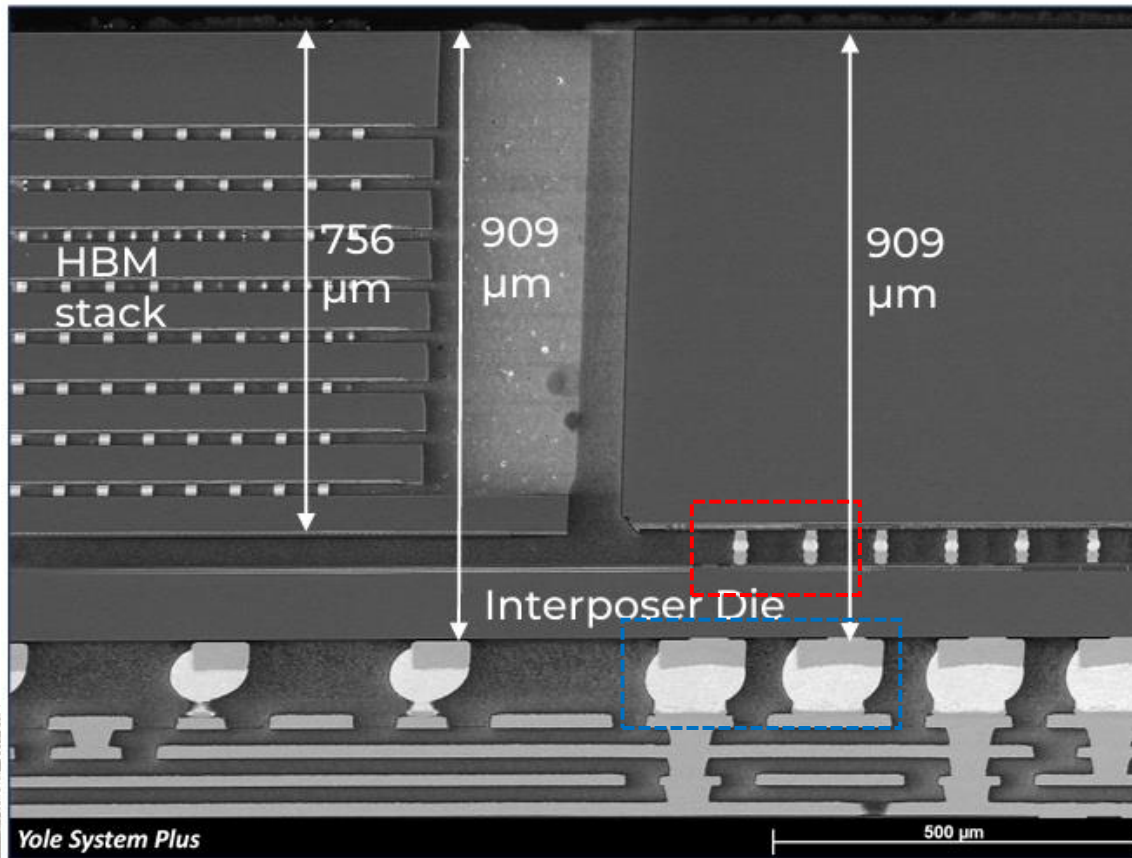


Nvidia H100 Tensor Core GPU, Yole Development

Nvidia H100加速器



■ Interposer

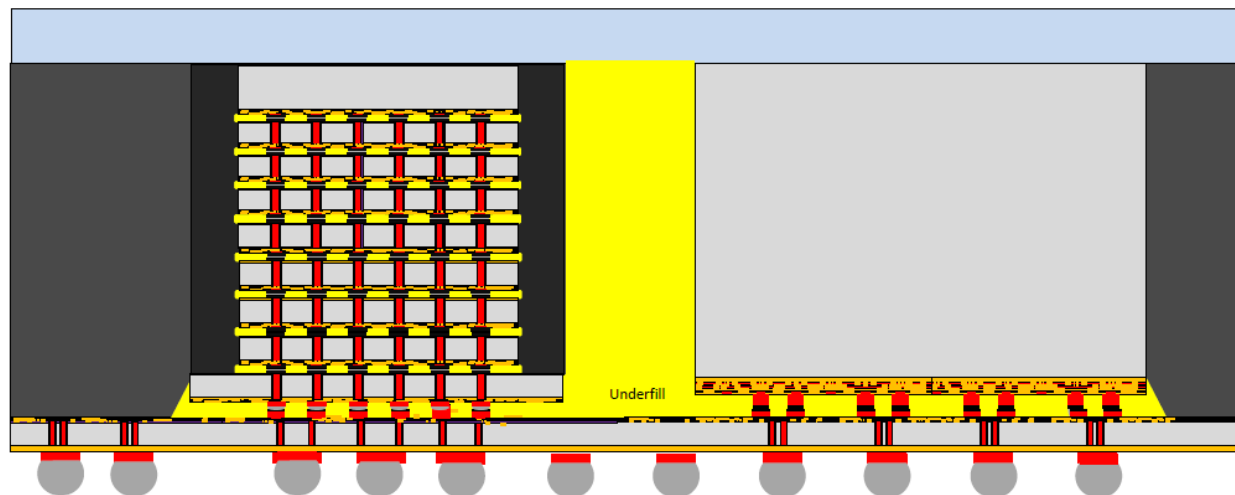
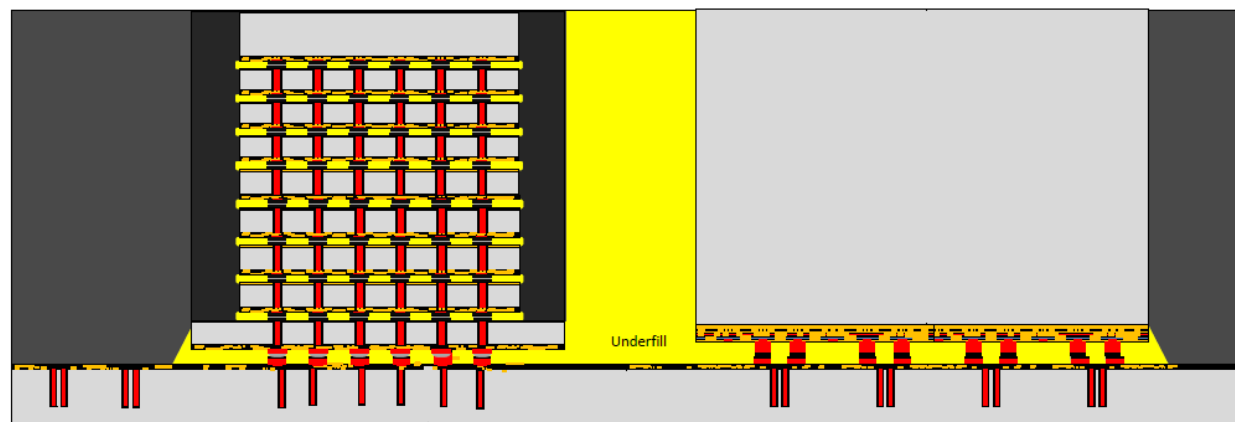
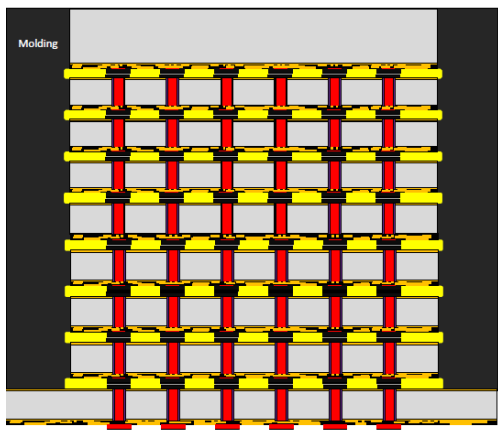


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Nvidia H100加速器



■ CoW



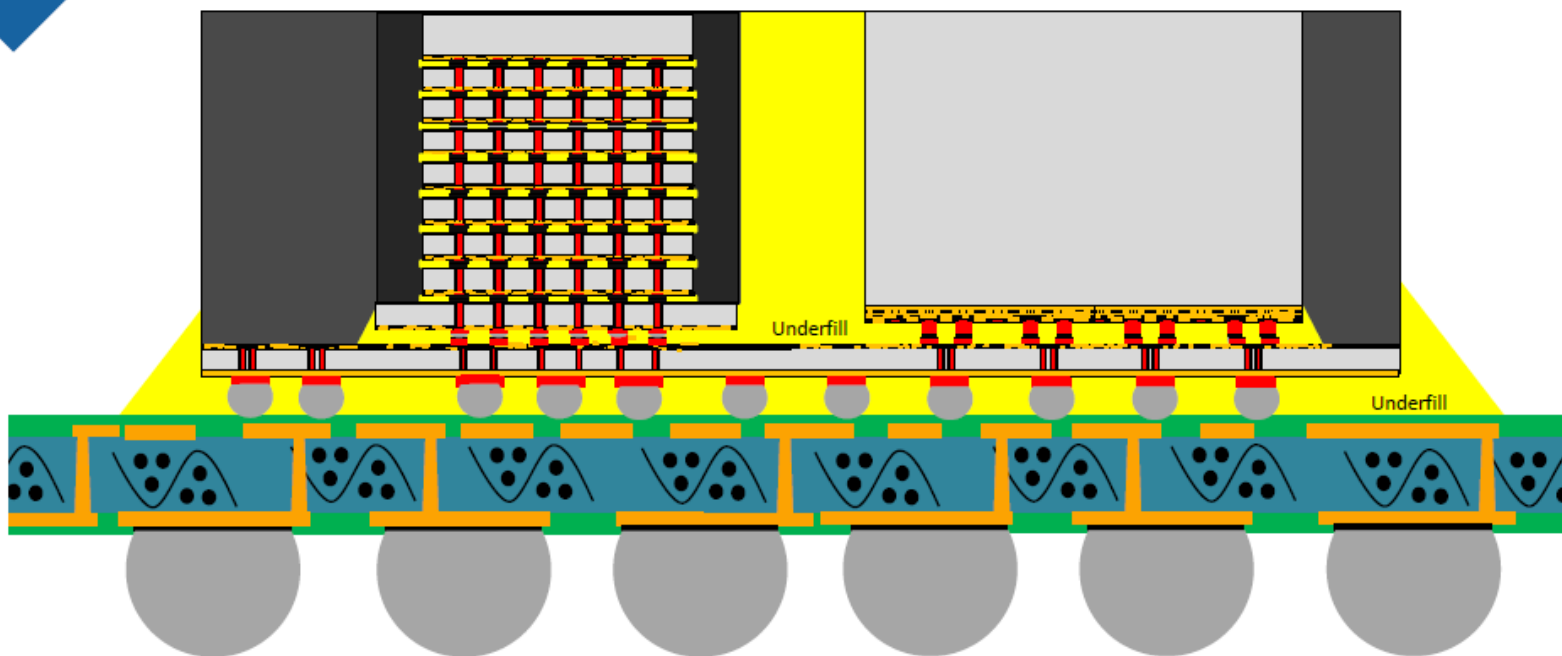
Nvidia H100 Tensor Core GPU, Yole Development



Nvidia H100加速器



■ CoWoS



Nvidia H100 Tensor Core GPU, Yole Development



Nvidia H100加速器



Bumps after center before edge

✓ Diam. $23\mu\text{m}$

✓ Pitch. $76/94\mu\text{m}$

Bumps at die center

✓ Diam. $23\mu\text{m}$

✓ Pitch. $76/94\mu\text{m}$

Bumps at die edge

✓ Diam. $23\mu\text{m}$

✓ Pitch. $76/94\mu\text{m}$

BGA Bump Under GPU

✓ Pitch. $192\mu\text{m}$

BGA Bump Under HBM(center)

✓ Pitch. $192\mu\text{m}$

μ Bumps under TSV

✓ Diam. $15\mu\text{m}$

✓ Pitch. $29\mu\text{m}$

μ Bumps edge of DRAM

✓ Diam. $15\mu\text{m}$

✓ Pitch. $60\mu\text{m}$

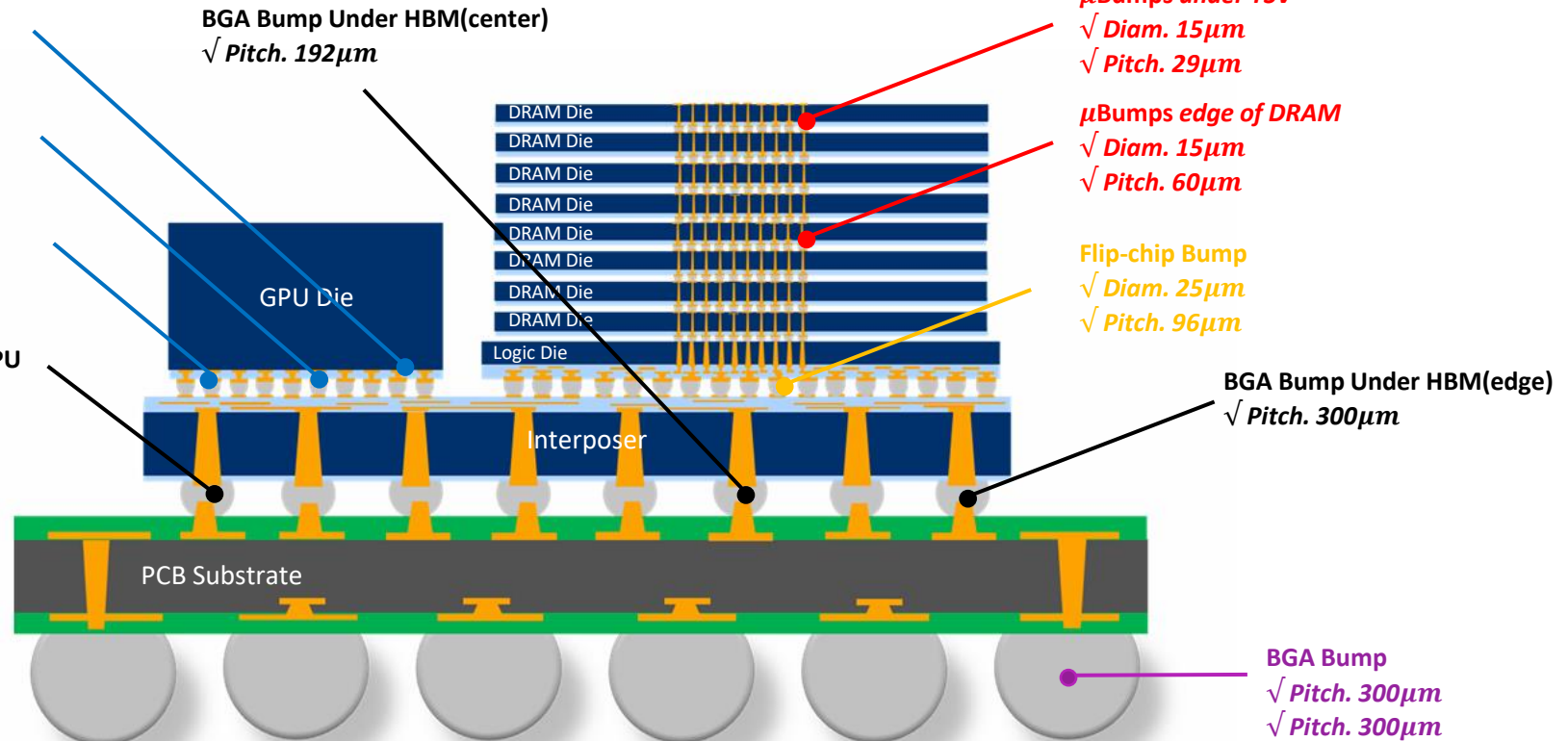
Flip-chip Bump

✓ Diam. $25\mu\text{m}$

✓ Pitch. $96\mu\text{m}$

BGA Bump Under HBM(edge)

✓ Pitch. $300\mu\text{m}$



Package Structure:

- 3D Packaging: 5 stacked dies with TSV & μ Bumps (HBM stack).
- 2.5D Packaging: HBM stack and GPU stacked with μ Bumps and a silicon interposer holding TSV.
- Flip-chip BGA: silicon interposer flip-chipped to a 12-layers PCB substrate

提纲



■ 问题

- 什么是集成电路封装?
- 什么是先进封装技术?

■ 业界关注的封装技术

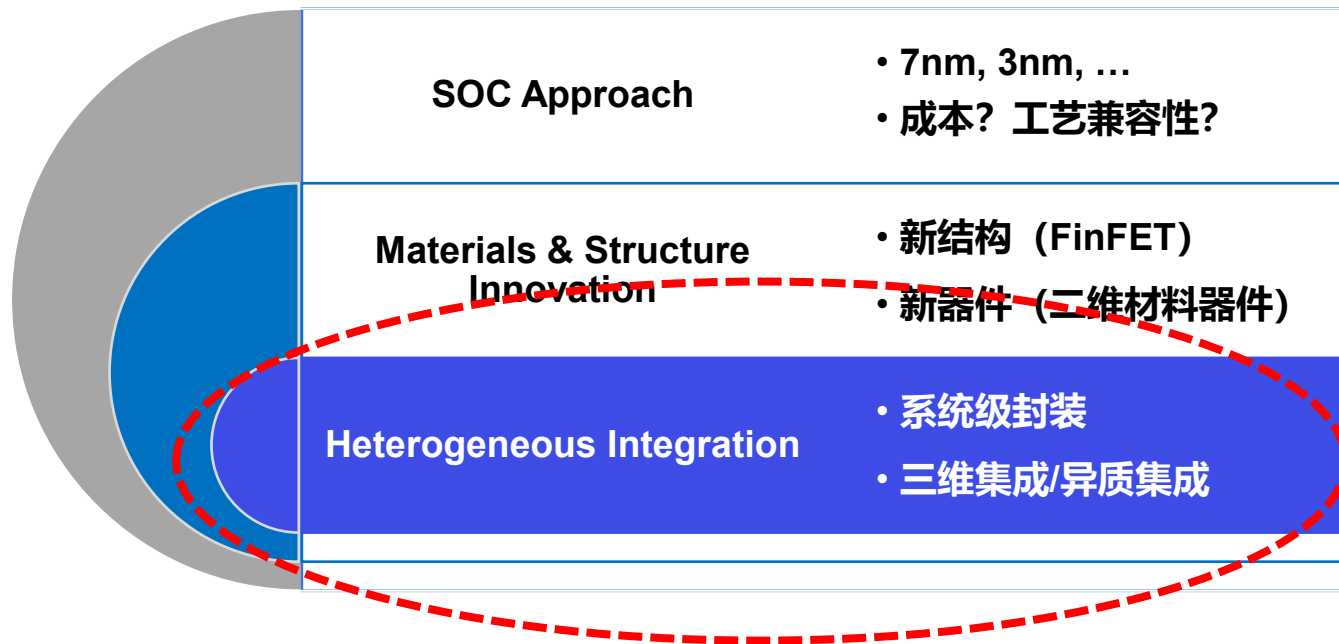
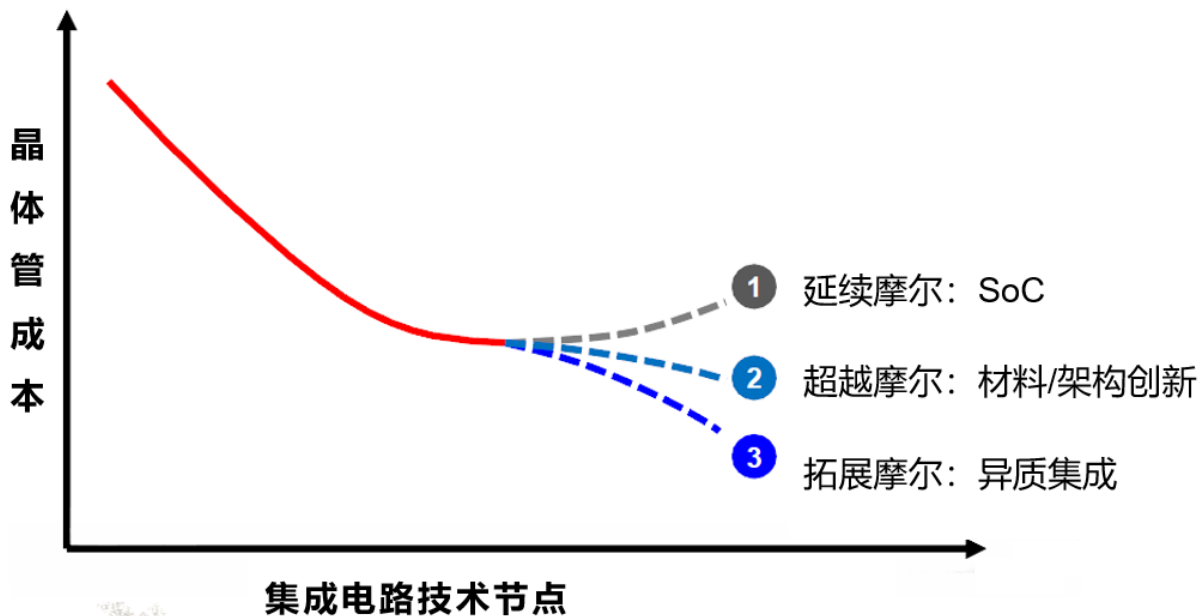
- 倒装芯片技术 (Flip Chip)
- 硅通孔互连 (TSV) 三维封装
- 扇出型圆片级封装 (Fan-out WLP)
- 玻璃封装基板 (Glass Substrate)

■ 先进封装案例

■ 先进封装技术的讨论



关于异质集成



功能提升

成本降低

工艺兼容

ECTC2020 Keynote Innovative Heterogeneous Integration Technologies — The New Path Going Forward, Douglas Yu, TSMC

异质集成与Chiplet



■ Chiplet的定义?

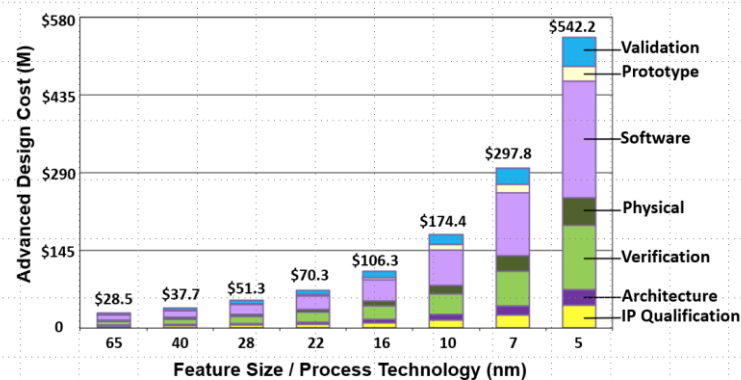
The naming of new technologies can be difficult and often inaccurate. Over time, standards are developed and language becomes aligned.

The cost of leading-edge nodes, combined with the lack of scaling of significant design blocks (ex: analog) and die sizes reaching reticle size, is driving disaggregation (splitting up) of chip functions into their best price/performance nodes requiring new technologies to interconnect these functions.

We are voicing our support in naming these small IP blocks “chiplets”. It is not a perfect name but fits into our vernacular well and is gaining acceptance in our industry. Our definition of “chiplet”, and what it is not, follows.

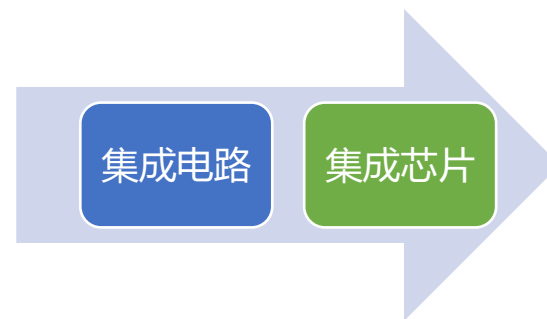
<https://eps.ieee.org/technology/definitions.html>

Design Cost for Advanced Nodes in Semiconductors



It will take another \$1 billion for 5nm process development.

Sources: International Business Strategies



Some envision, that in time, **the greater use of chiplets will drive the package to become the new SoC**, and chiplets become the new "IP". However, for this to be viable across packaging companies, there must be standard/common communication interfaces between chiplets. There are several solutions today, but standards must be chosen.

异质集成与Chiplet



- **Chiplet（小芯片、芯粒）技术已经用于关键的高性能计算芯片，包括AI芯片、服务器CPU芯片、GPU芯片等，是近年刚刚兴起的关键新技术。**
 - 芯粒技术通过复用常用的芯片模块，并进行快速组装和重新配置，实现一个更大的集成芯片技术，是高算力芯片的关键技术路径。
 - **标准化**：2022年3月，英特尔、AMD、Arm、台积电等十大巨头成立UCIe联盟，并发布UCIe接口标准1.0，推动Chiplet技术标准化、形成生态。**2024年8月，发布UCIe2.0。**
 - **平台化**：2022年8月，美国总统签署《芯片与科学法案》。总统科技顾问委员会建议：打造一个包含完整软件栈的Chiplet平台，以大幅减少完成新产品验证所需要的成本和时间。
2022年9月，欧盟委员会建议开发基于芯粒的系统芯片基础架构，推出RISC V+Chiplet平台。



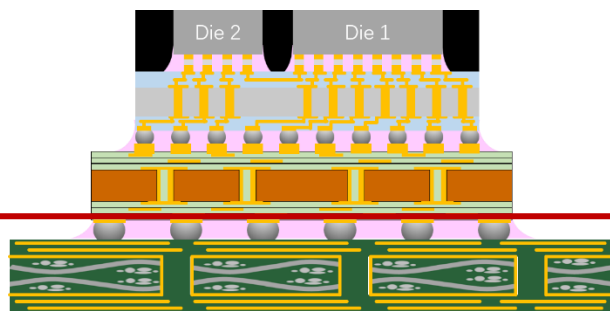
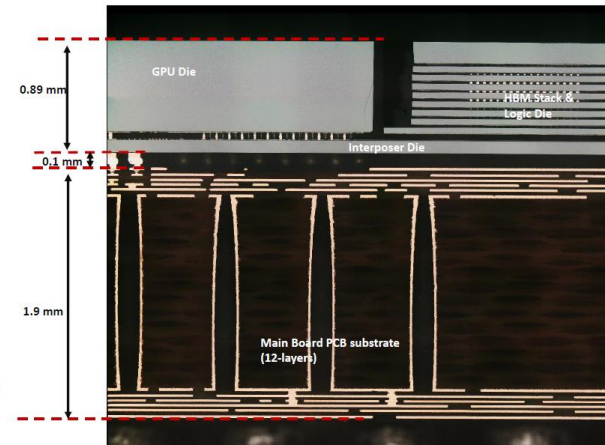
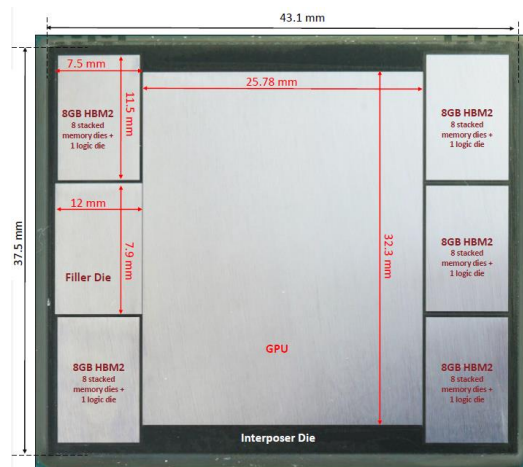


异质集成与Chiplet

- Chiplet（小芯片、芯粒）技术已经用于关键的高性能计算芯片，包括AI芯片、服务器CPU芯片、GPU芯片等，是近年刚刚兴起的关键新技术。
 - 2023年10月17日，美国商务部将壁仞科技等中国科技企业列入实体清单。
 - 2023年10月22日，中国科协发布2023重大科学问题（10个）、工程技术难题（9个）和产业技术问题（10个），人工智能、新能源、高性能材料、生命科学等领域重大问题受到关注。
 - “如何发挥我国信息通信产业优势，快速实现芯粒(Chiplet)技术和产业突破？”列为产业技术问题之一
 - 2023集成芯片与芯粒技术白皮书（2023年10月，中国计算机学会）

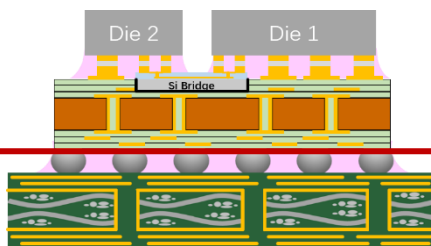
异质集成与Chiplet

- 异质集成是芯片封装方法
- Chiplet是芯片设计方法
- Chiplet依赖异质集成实现



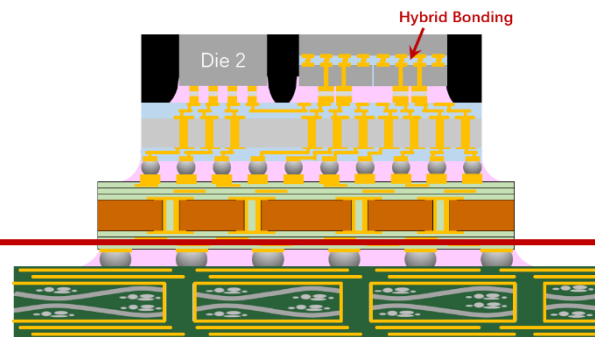
2.5D/3D TSV

Xilinx (2011)、Samsung (2014)、
AMD (2015)、Intel (2016)、...



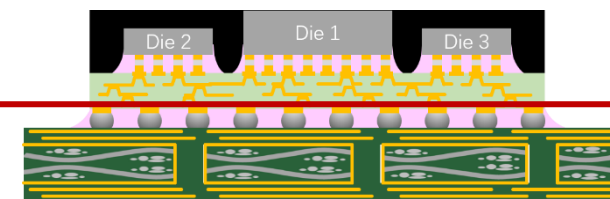
EMIB

Intel (2017)、AMD (2022)、...



Hybrid Bonding

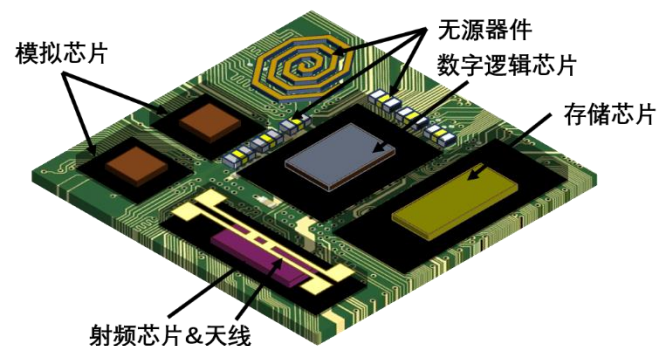
YMTC (2019)、AMD (2022)、
Tesla (2022)、...



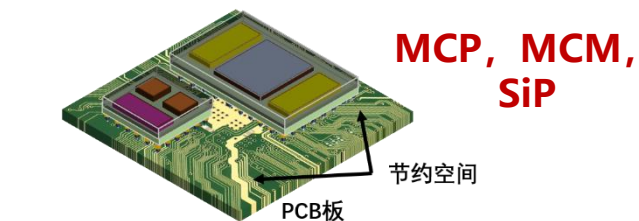
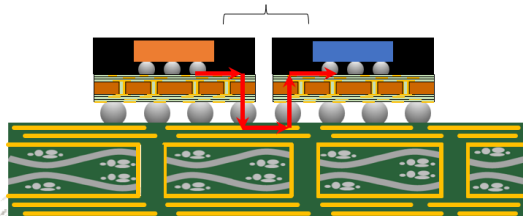
UHD FO

GUC (2021)、Xilinx (2021)、
Tesla (2022)、...

电子产品的制造

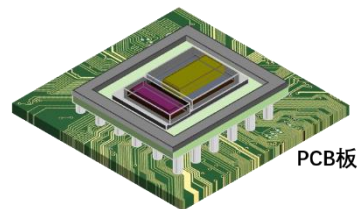
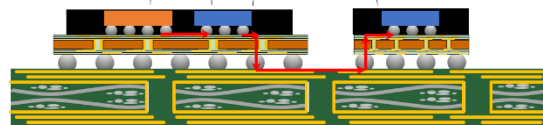


通常跨度为厘米级



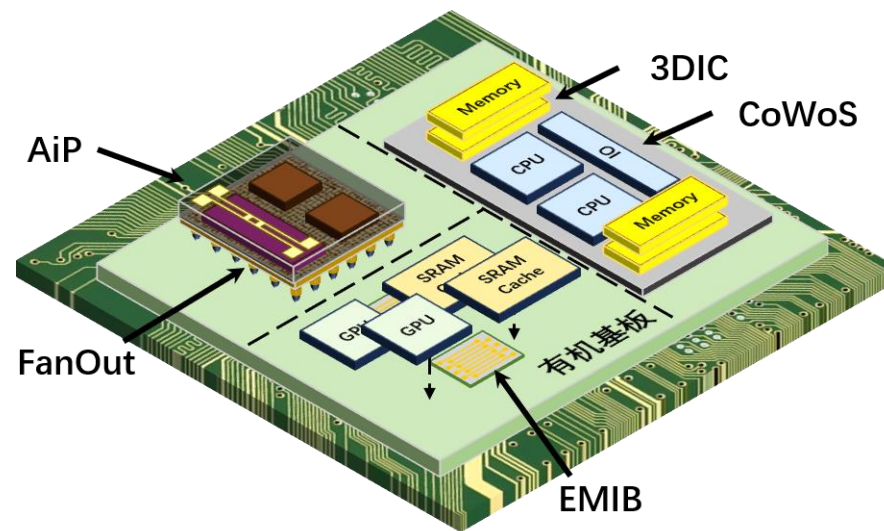
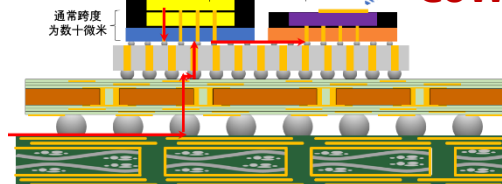
通常跨度为毫米级

通常跨度为厘米级



通常跨度为百微米级

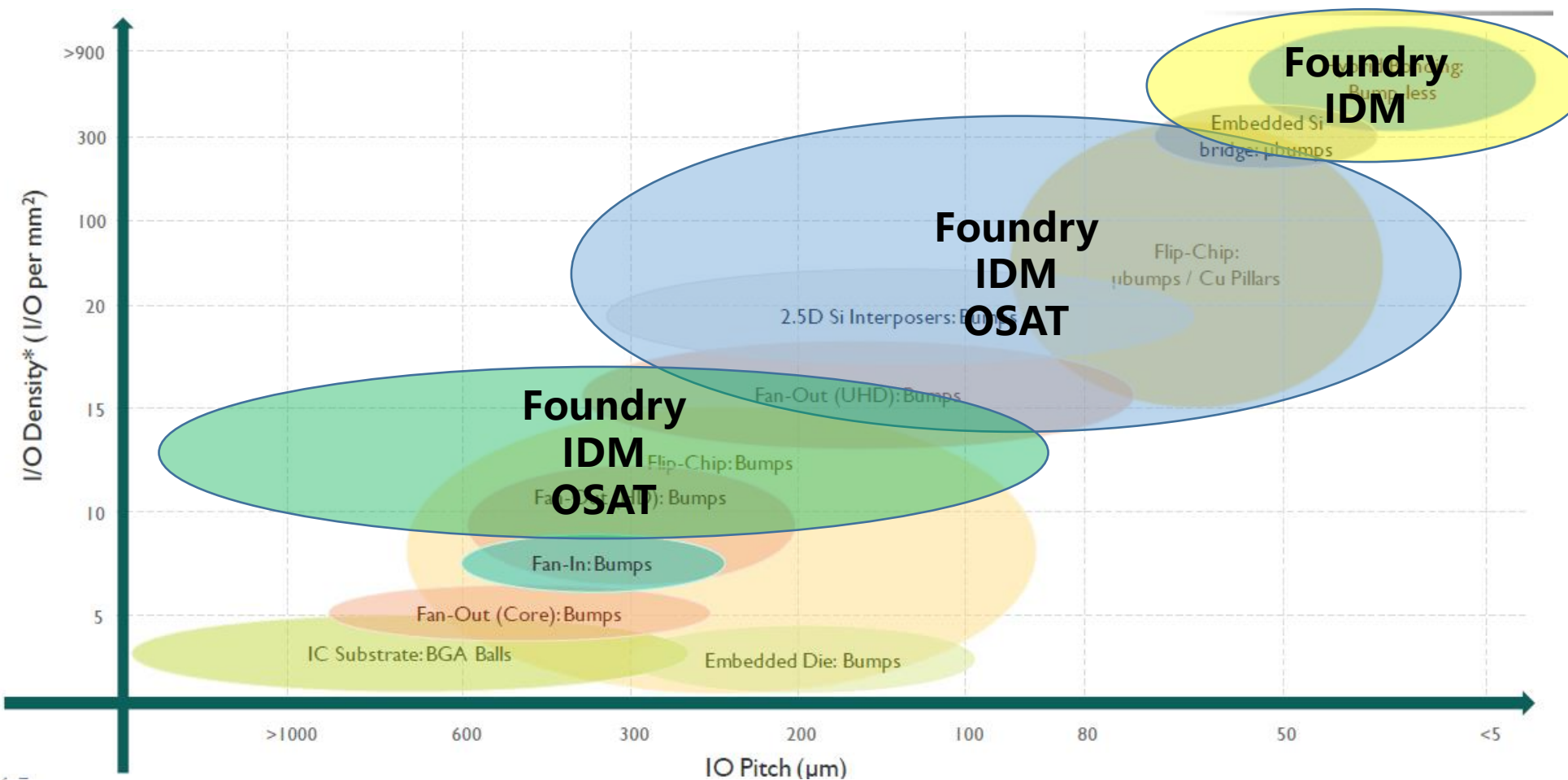
CoWoS



Foundry对先进封装的布局



■ 在高端应用芯片的封装上，OSAT面临挑战

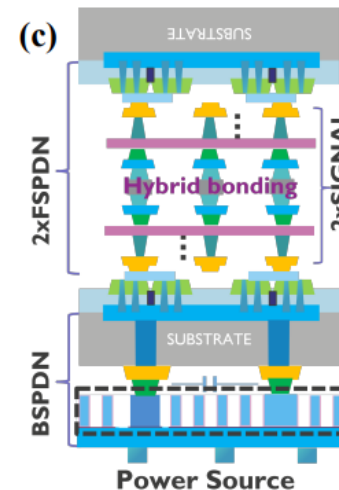
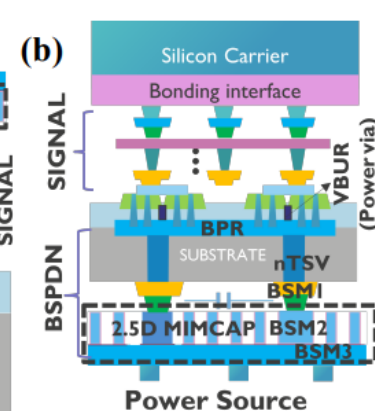
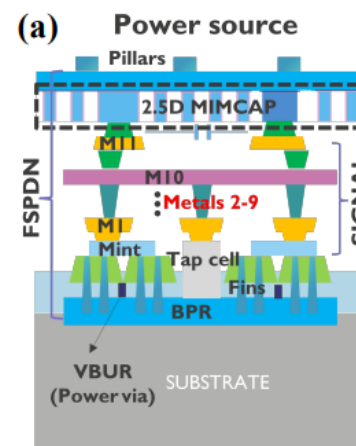
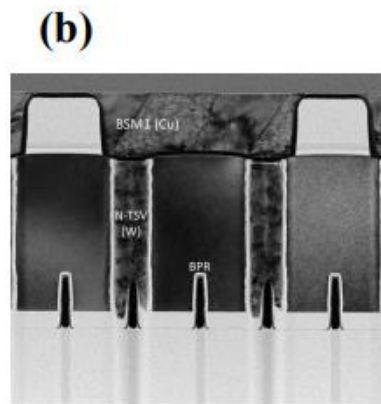
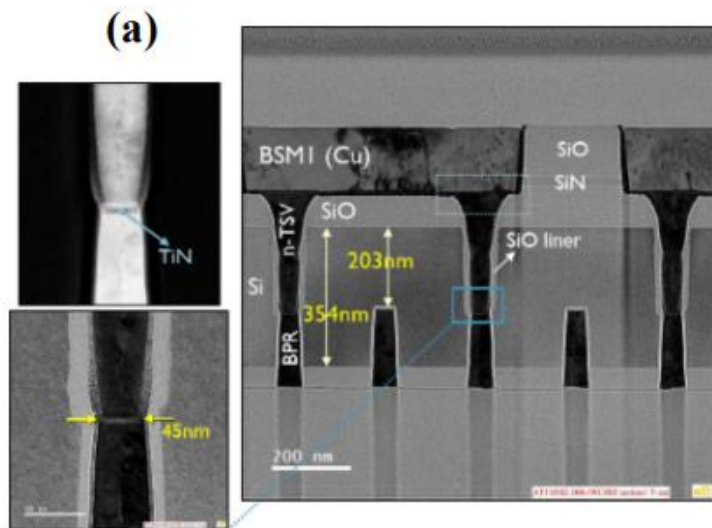


来源: Yole

Foundry对先进封装的布局



■ 技术趋势：融合先进制程与先进封装

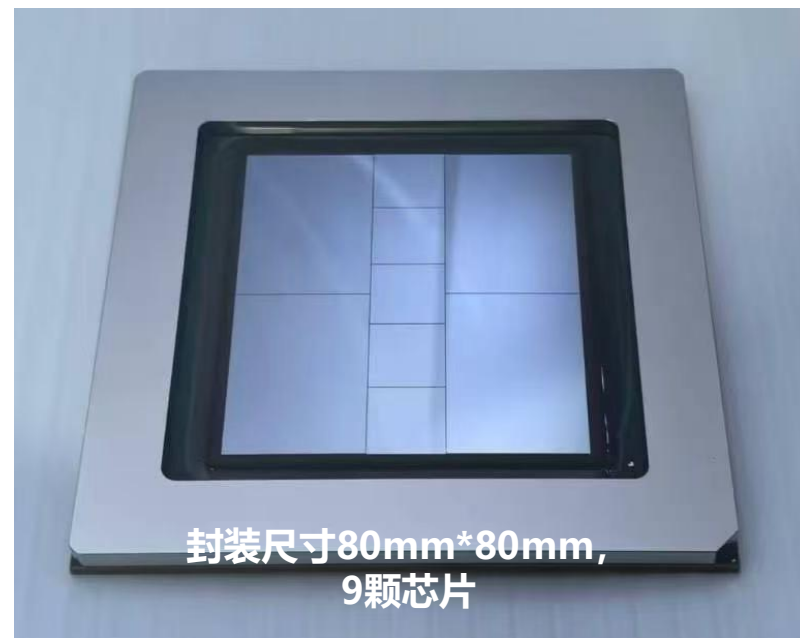
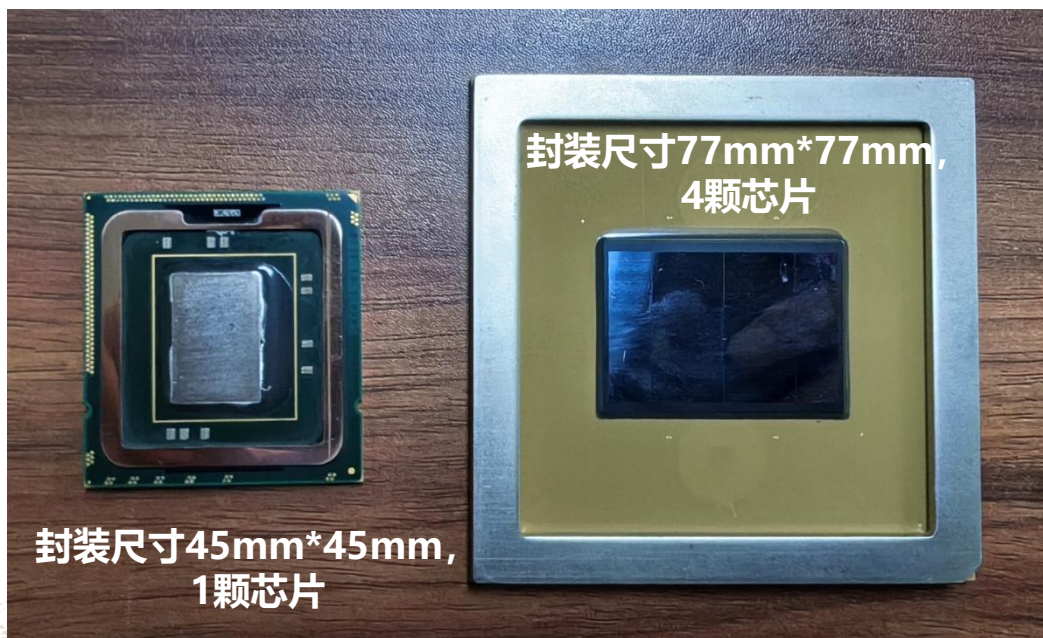


融合Hybrid Bonding, Nano-TSV, BEOL的3D IC架构

未来的先进封装？

■ 从集成电路到集成芯片 ——大尺寸封装

- 三维集成、异质集成



未来的先进封装？

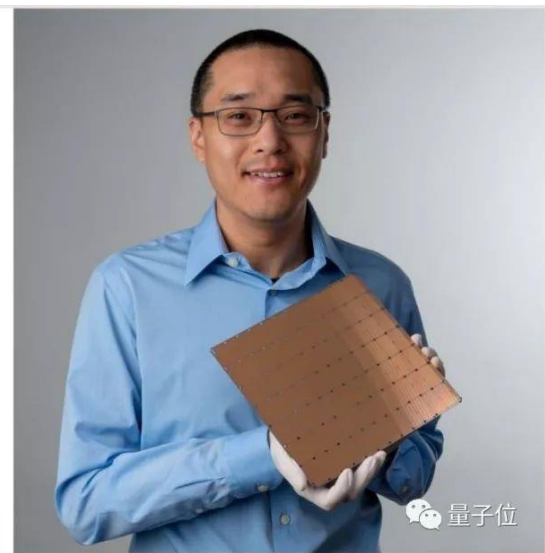
■ 大芯片封装

- “史上最大芯片诞生！1.2 万亿晶体管超级巨无霸，专为 AI 设计”——“晶圆级引擎” (Cerebras Wafer Scale Engine, WSE) ，2019年8月
- 其面积46225 平方毫米，拥有1.2 万亿个晶体管，400000 个核心，片上内存18G字节，内存带宽19PByte/s，fabric带宽100Pbit/s
- 是当时芯片面积最大的英伟达GPU的56.7倍

- 芯片功率达15千瓦，热膨胀系数失配严重
- 在芯片上方安装了一块“冷却板”，使用多个垂直安装的水管直接冷却芯片
- 由于芯片太大而无法放入任何传统封装中，设计了结合了连接PCB和晶圆两者的定制连接器以及冷却装置

Largest Chip Ever Built

- 46,225 mm² silicon
- 1.2 trillion transistors
- 400,000 AI optimized cores
- 18 Gigabytes of On-chip Memory
- 9 PByte/s memory bandwidth
- 100 Pbit/s fabric bandwidth
- TSMC 16nm process



量子位

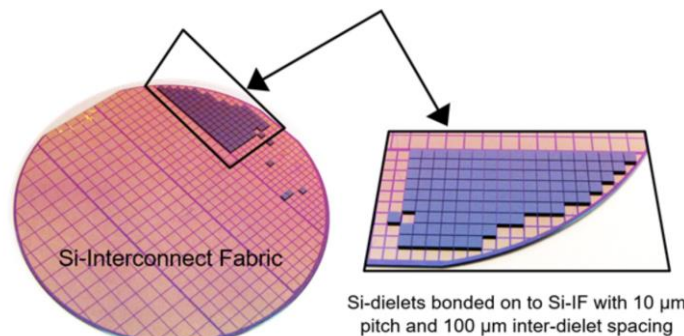
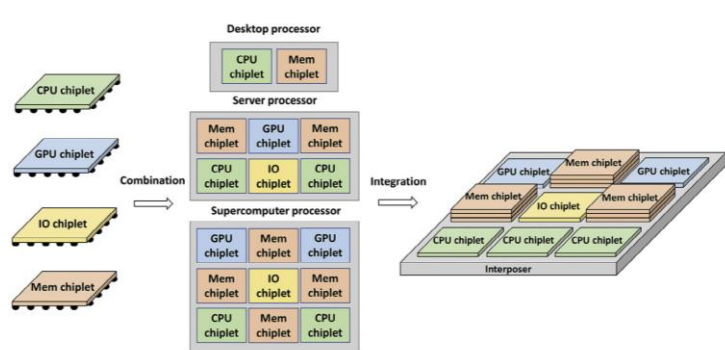
<https://tech.sina.com.cn/csj/2019-08-20/doc-ihytcern2173478.shtml>

未来的先进封装？



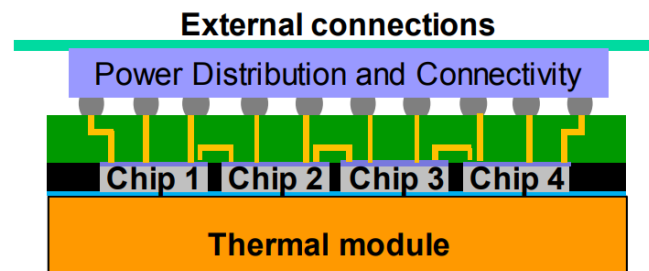
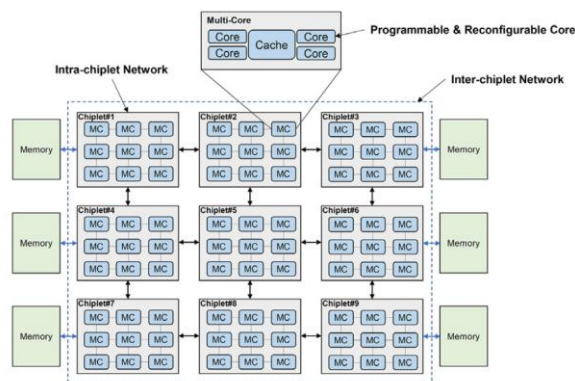
■ 晶圆级系统 (System-on-Wafer)

- 定制芯粒及晶圆级封装：实现高性能、可复用、高灵活度、短周期系统研制



UCLA: Silicon Interconnect Fabric

?



TSMC: InFO_SOW
晶圆级芯粒集成技术

晶圆级芯粒设计及网络拓扑优化



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